Semiconductor Primer Report

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Contents

01 Introduction to Semiconductors

1.1	The Wafer	05
1.2	Moore's Law	06
1.3	The Business Models	08
1.4	Process Nodes: What the Numbers Actually Mean	09
1.5	The Fabrication Process	10
1.6	Yield Economics	13
1.7	Transistor Architecture Evolution	14
1.8	Advanced Packaging	15
1.9	The Equipment Supply Chain – WFE	17
1.10	The Materials Supply Chain	19

02 Lithography

2.1	Glossary	22
2.2	What Lithography Actually Does	24
2.3	The Physics	25
2.4	History: The Birth of EUV & Rise of ASML	27
2.5	ASML: The Engineering GOAT	28
2.6	Inside an EUV Scanner	30
2.7	Multi-Patterning Before EUV	31
2.8	DUV vs EUV Comparison	32
2.9	Looking Forward: High-NA EUV	33
2.10	The Materials Supply Chain	35
2.11	ASML's Monopoly	37

03 Memory

3.1	Glossary	39
3.2	The Memory Hierarchy	42
3.3	AI Demand & Memory Wall	43
3.4	What is High-Bandwidth Memory (HBM)?	44
3.5	The 2.5D Package: Connecting HBM to the GPU	45
3.6	The HBM Technological Roadmap	46
3.7	Current Bonding Technology	47
3.8	Current HBM Players	48
3.9	Looking Ahead: Bonding Technology	50
3.10	The HBM Equipment Supply Chain	51
3.11	A Memorable Summary	52

04 Optics

4.1	Glossary	54
4.2	Why Light instead of Copper	62
4.3	Optical Fiber	63
4.4	Data Centre Networking Architecture	64
4.5	Transceivers	65
4.6	Lasers	67
4.7	Silicon Photonics	69
4.8	Digital Signal Processor (DSP)	70
4.9	Co-Packaged Optics	71
4.10	The External Laser Source	73
4.11	Epitaxy and MOCVD	75
4.12	Wavelength Division Multiplexing	76
4.13	Coherent Optics and Long-Haul Networking	77
4.14	Optical Circuit Switches	78
4.15	InfiniBand vs Ethernet	79
4.16	Key Supply Chain Players	80

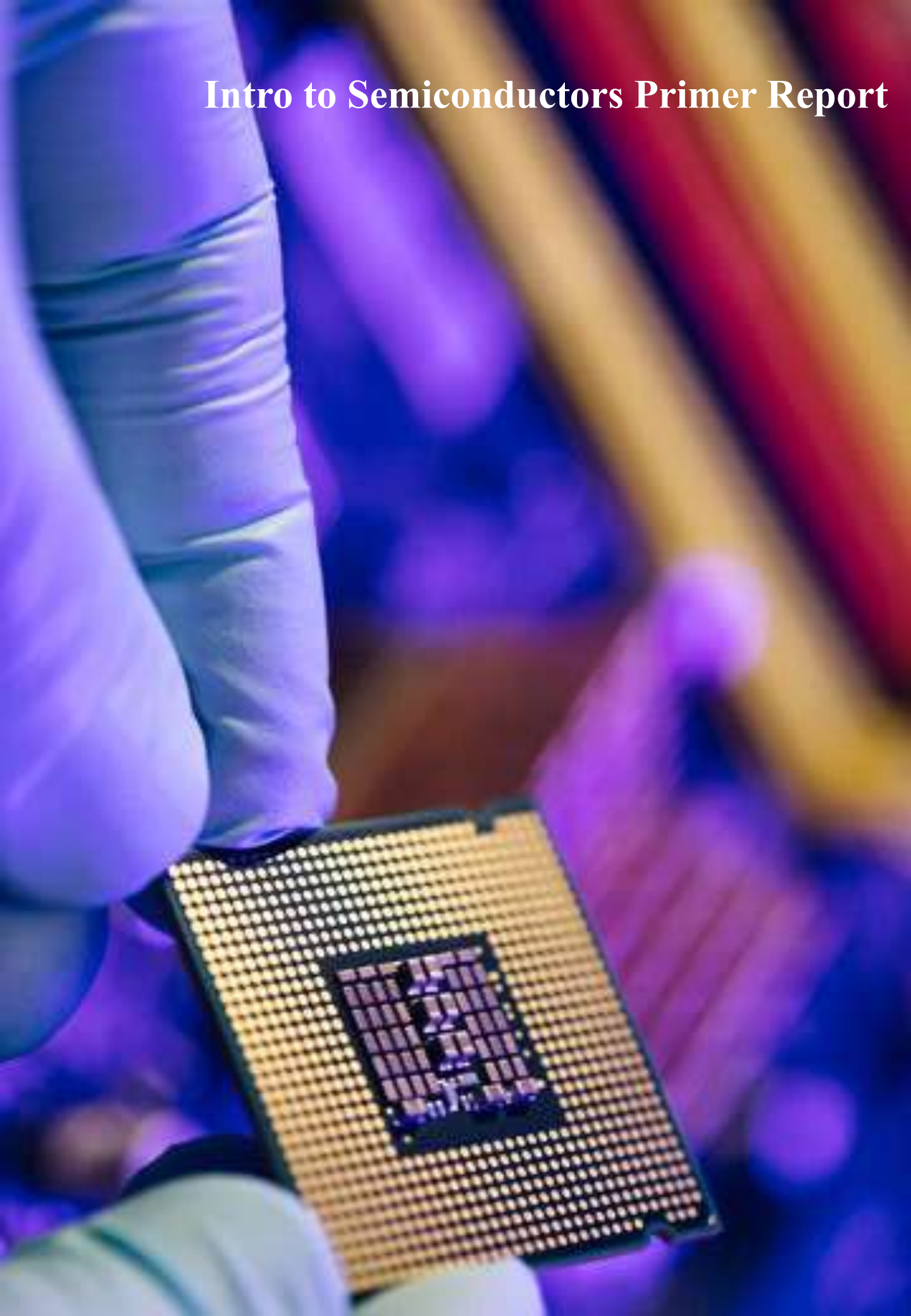
05 Power & Grid

5.1	Glossary	82
5.2	The Demand Shock	88
5.3	How the Grid Works	89
5.4	The Capacity Gap	90
5.5	Transformers	91
5.6	High-Voltage Equipment	92
5.7	The Data Centre Power Architecture	93
5.8	Battery Energy Storage Systems	94
5.9	Cooling	96
5.10	Nuclear Energy	97
5.11	Natural Gas	98
5.12	Renewable and Storage	99
5.13	Utilities as AI Infrastructure Investment Vehicles	100

06 Humanoid Robots

6.1	Glossary	102
6.2	Why Physical AI Now	107
6.3	Why Humanoid?	108
6.4	What is Inside a Humanoid Robot?	109
6.5	Vision-Language-Action Models	110
6.6	Actuation	112
6.7	Sensing and Perception	114
6.8	Compute and Motion Control	115
6.9	Power, Energy & Battery Constraints	116
6.10	Teleoperation	117
6.11	The Humanoid OEM Landscape	118
6.12	The 5 Tipping Point Conditions	120

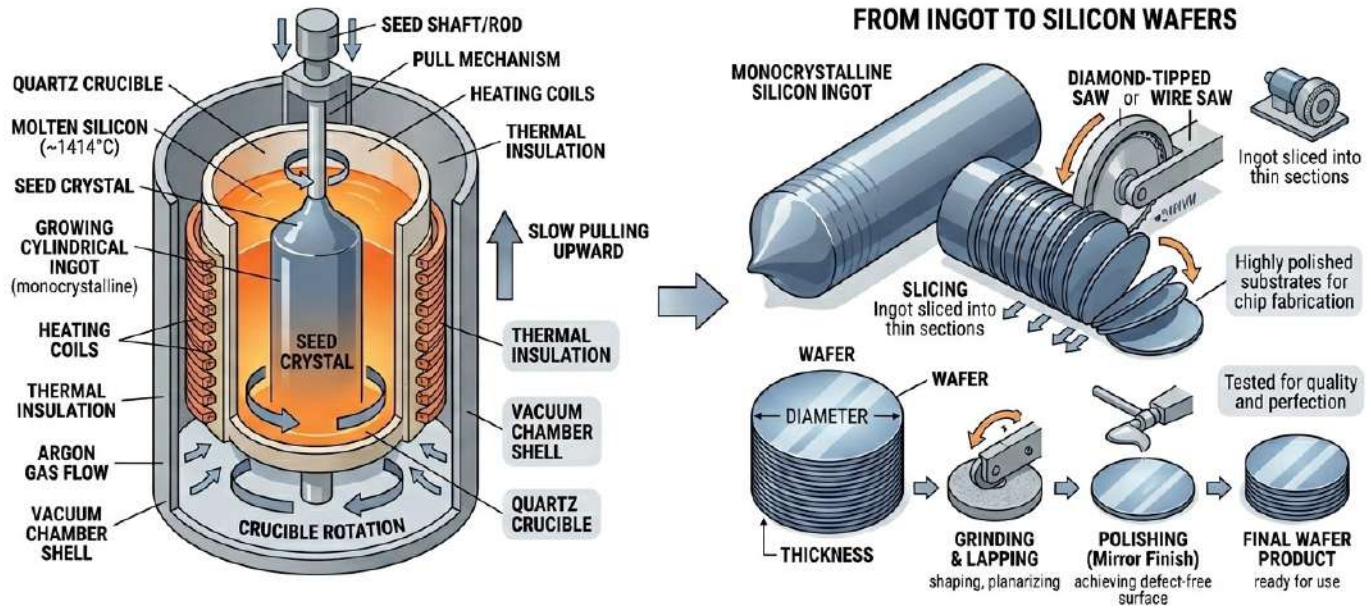
Intro to Semiconductors Primer Report



1. The Wafer

Everything begins with sand. Silicon, the second most abundant element in the Earth's crust, is refined to a purity of 99.9999999% (nine nines) and melted down in a quartz crucible at over 1,400 degrees Celsius. A seed crystal is dipped into the molten silicon and slowly pulled upward while rotating. This process, called the Czochralski method, produces a cylindrical silicon ingot weighing hundreds of kilograms and measuring up to 450 millimeter in diameter.

The Czochralski Process Diagram



That ingot is then sliced with diamond wire saws into thin circular wafers, typically 775 micrometer thick for the 300mm wafers used at all leading-edge fabs. Each wafer is then lapped and polished to a mirror finish using the same chemical mechanical planarization process used throughout chip manufacturing. This wafer is the blank canvas on which every transistor, logic gate, and memory cell will be built.

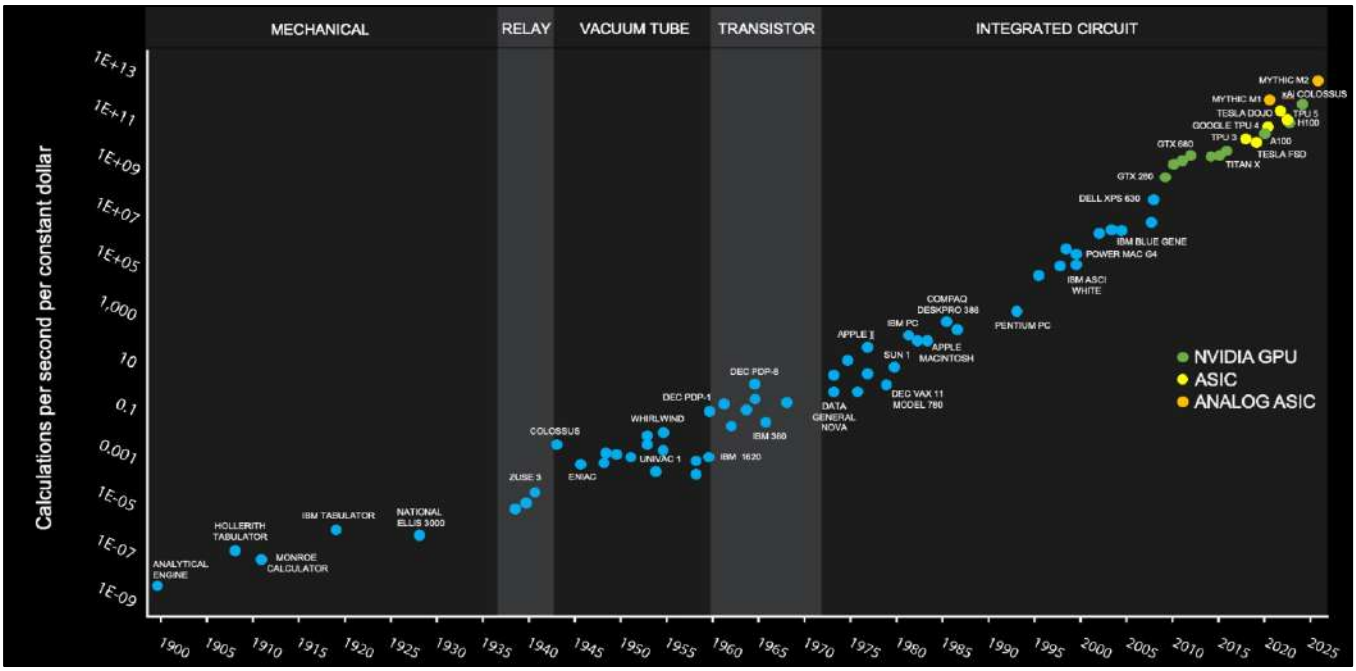
The economics of wafer size are straightforward and important. A 300mm wafer has 2.25 times the surface area of a 200mm wafer. Because the fixed cost of processing a wafer (the equipment time, the chemicals, the facility overhead) is roughly similar regardless of wafer size, more surface area means more chips per processing run and lower cost per chip. This is why the industry's transition from 200mm to 300mm wafers, completed in the early 2000s, was one of the most significant cost reduction events in semiconductor history. All leading-edge logic and memory fabs now run on 300mm wafers.

The Companies

- Shin-Etsu Chemical & Sumco (both Japanese companies), together control approximately 60% of global silicon wafer supply.
- SK Siltron (South Korea) and GlobalWafers (Taiwan) account for most of the remainder. The silicon wafer business benefits from high capital intensity, long customer qualification cycles, and a purity and surface quality standard that new entrants cannot quickly match.
- Shin-Etsu in particular holds the technology leadership position in the ultra-low-defect wafers required for leading-edge logic.

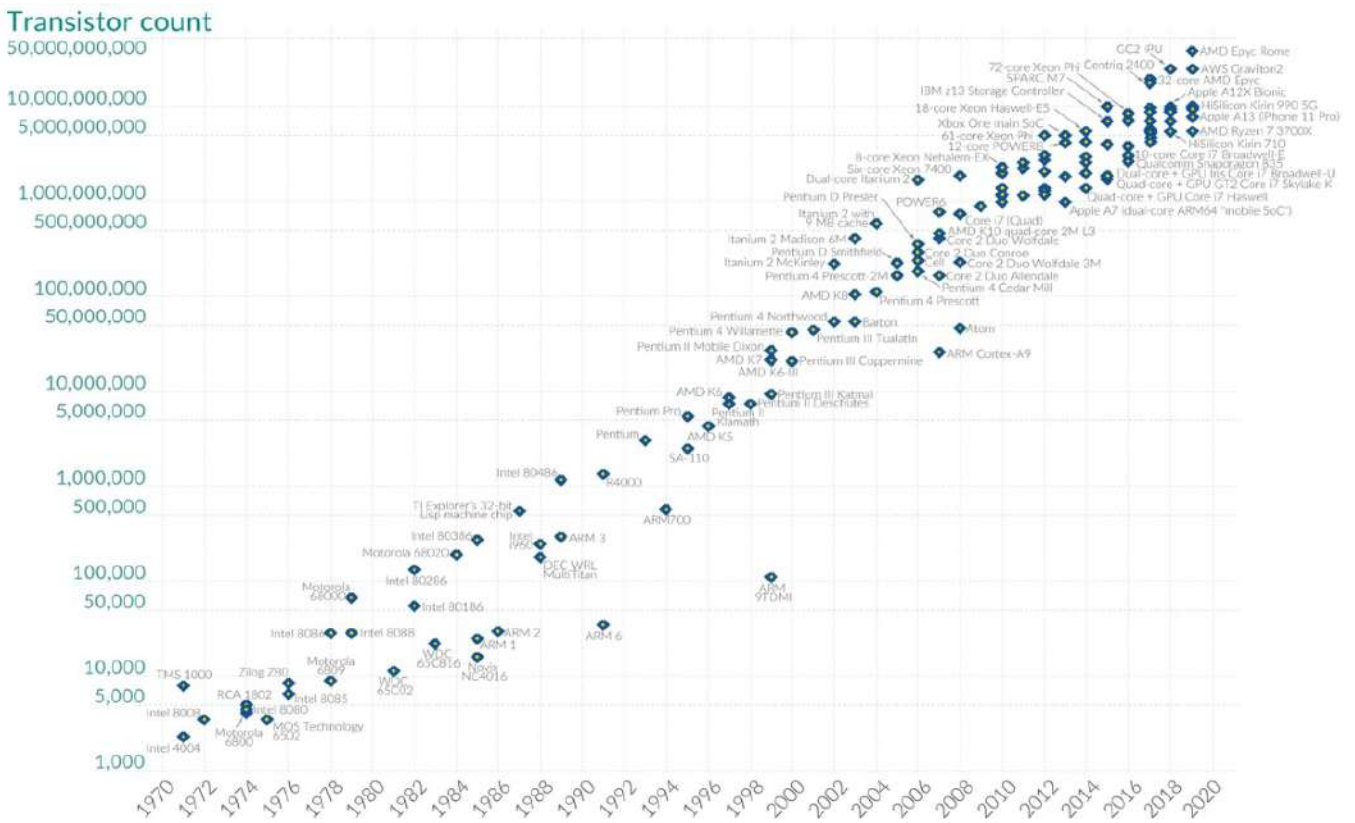
2. Moore's Law

In 1965, Gordon Moore, co-founder of Intel, observed that the number of transistors on an integrated circuit had been doubling approximately every year since the invention of the chip. He predicted this trend would continue. Revised to a doubling roughly every two years, **Moore's Law** became the central organizing principle of the semiconductor industry for five decades as a self-fulfilling engineering and economic roadmap that the industry committed to delivering.



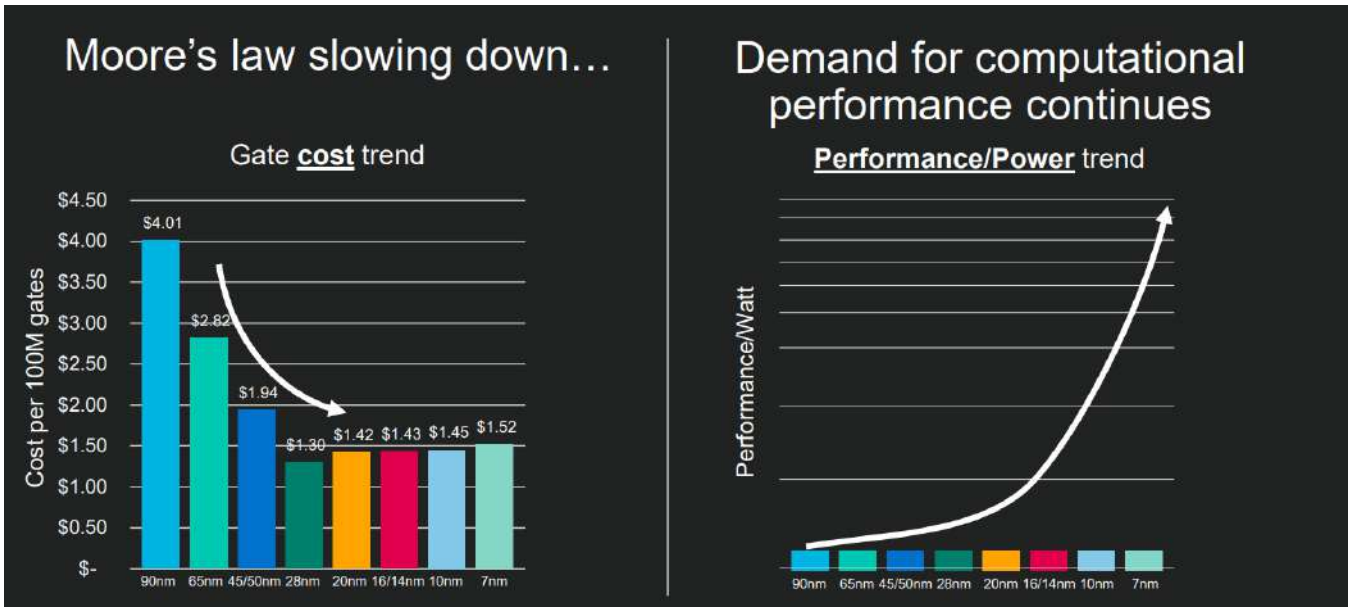
Why Moore's Law mattered economically

The profound implication of Moore's Law was that they got more powerful while simultaneously getting cheaper. When you pack twice as many transistors onto the same area of silicon, each transistor costs half as much to manufacture. When each transistor is also smaller and closer to its neighbors, it switches faster and consumes less power. Performance, cost, and power efficiency all improved simultaneously with each generation. This is a combination that no other technology in history has matched.



Why Moore's Law is slowing

The physics of transistor scaling has not broken down entirely, but the cost of continuing to scale has risen dramatically while the performance benefits per dollar spent have diminished. Three specific factors are responsible.



The Cost of Each New Node has Risen Sharply

Building a leading-edge fab capable of manufacturing at 3nm now costs $20 billion or more, versus $5 billion a decade ago. The equipment is more expensive, the process is more complex, and the yield is harder to achieve. The economic case for each successive node requires larger volumes of chips sold at sufficient price premiums to justify the investment.

Power density is a Physical Constraint.

Packing more transistors into the same area generates more heat. As data center chips are already thermally limited, adding more transistors does not help if the chip cannot dissipate the resulting heat. This is why clock speeds have been roughly flat since 2005 despite ongoing transistor count increases.

The Cost-per-Transistor Improvement is Slowing.

At the most advanced nodes, each transistor is so complex to manufacture, and process yields are so difficult to achieve, that the historic cost reduction per transistor at each new node is no longer reliably occurring. This is sometimes called the "end of Dennard scaling," referring to the set of physics relationships that previously ensured smaller transistors were proportionally cheaper and faster.

The industry's response to these limits is not to abandon scaling but to supplement it with new approaches such as new transistor architectures (FinFET, GAAFET, CFET), backside power delivery, chiplet-based integration, and advanced packaging. These are the subjects of later sections.

Source: Marvell 2020 Investor Day - Slide 43

The semiconductor industry divides into three distinct business model categories that determine how companies create and capture value. Understanding these models is essential for investment analysis because they have completely different financial profiles, capital requirements, and competitive dynamics.

The Integrated Device Manufacturer (IDM)

An IDM designs its own chips and manufactures them in its own fabs. Intel is the archetypal IDM. Samsung operates as an IDM for its own chip products while also running a separate foundry business. The IDM model historically provided advantages. It has tight integration between design and manufacturing teams could produce superior optimization, and owning the fab gave control over supply.

However, fab ownership requires enormous capital expenditure regardless of demand cycles, and the fixed costs of manufacturing create financial pressure during downturns. Intel's struggles to keep pace with TSMC's process technology from 2015 onward illustrate what happens when an IDM's manufacturing falls behind while its fixed cost structure remains unchanged.

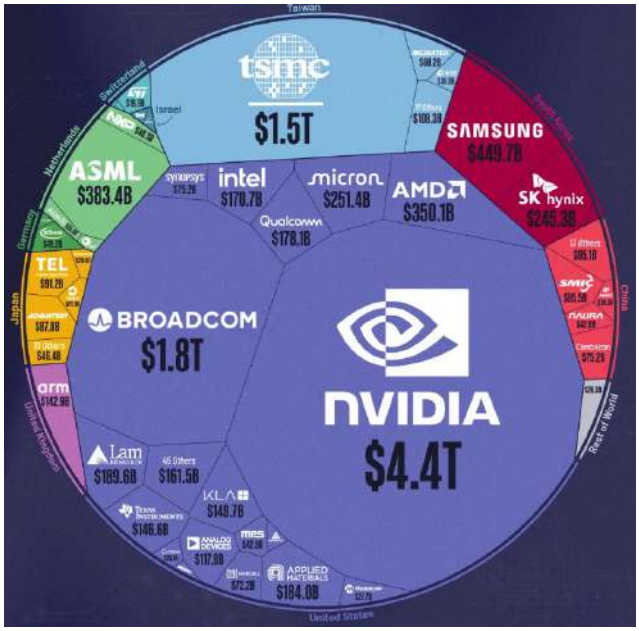
The Pure-Play Foundry

A pure-play foundry manufactures chips for external customers and designs nothing itself. TSMC, founded in 1987 by Morris Chang, invented this model. The insight was that the capital requirements of advanced manufacturing and the engineering requirements of chip design had grown so large that specialization was economically superior to vertical integration.

By serving multiple customers across multiple markets, a foundry can achieve utilization rates and process learning curves that no individual IDM could match. TSMC today serves Apple, Nvidia, AMD, Qualcomm, and hundreds of other customers from the same process platform, spreading its R&D and capex across an enormous revenue base.

The Fabless Company

A fabless semiconductor company designs chips and outsources all manufacturing to foundries. Nvidia, AMD, Qualcomm, Apple's chip division, and Broadcom are all fabless or predominantly fabless. The fabless model allows companies to focus entirely on design (the highest-value, highest-margin activity in the chip supply chain) without carrying the capital burden of fab ownership. A fabless company can spend its capital on R&D and engineering talent rather than on buildings and equipment that depreciate over decades.



The fabless-foundry division of labor has been the dominant organizational force in the semiconductor industry since the 1990s and is the primary reason TSMC has accumulated such a commanding market position. By serving the entire fabless universe, TSMC processes more wafers at leading-edge nodes than any IDM could justify building for its own products alone, giving it a process learning rate advantage that compounds over time.

Source: https://www.visualcapitalist.com/the-global-semiconductor-industry-in-one-giant-chart/

4. Process Nodes: What the Numbers Actually Mean

Investors in semiconductors regularly encounter references to "3nm," "5nm," "7nm," and similar process node designations. These numbers are almost entirely marketing labels and do not correspond to any physical dimension measured in nanometers on the actual chip. Understanding this distinction is important because it means that node comparisons across foundries (i.e "Samsung's 3nm vs TSMC's 3nm“) are not straightforward technical comparisons.

The Historical Meaning

In the 1990s and early 2000s, process node numbers did refer the gate length of the transistor, which is the distance between the source and drain in the channel. A "130nm process" genuinely had transistor gate lengths of approximately 130 nanometer. This made cross-foundry comparisons meaningful.

However, at approximately the 28nm node around 2011, the physical dimension that the node number referred to became ambiguous. Gate lengths were no longer shrinking proportionally to the node number because other techniques, such as high-k metal gate dielectrics and strained channels, were being used to improve performance without further shrinking the gate.

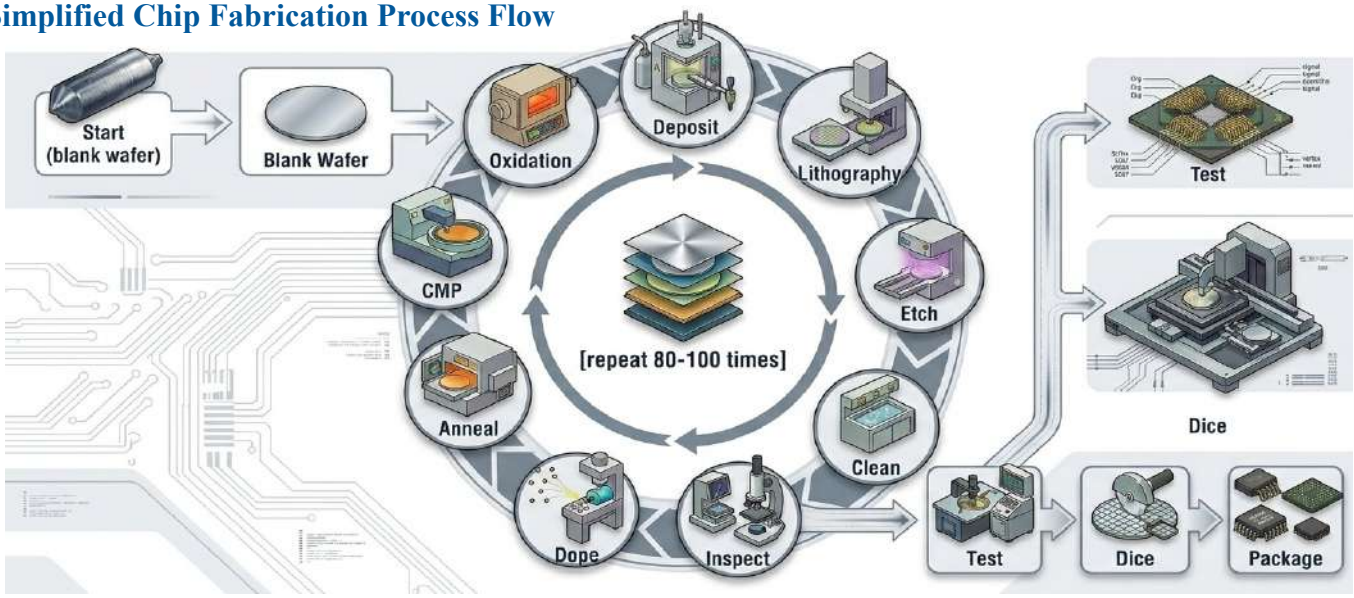
By the time the industry reached 10nm and below, the node numbers became largely arbitrary. Intel's "10nm" process, which struggled in production from 2016 to 2019, was approximately competitive with TSMC's "7nm" in transistor density. TSMC's "3nm" has a gate pitch and metal pitch that are not 3 nanometers in any physical measurement.

What Investors Should Use Instead

The meaningful metrics for comparing process nodes across foundries are transistor density (how many million transistors fit per square millimeter), gate pitch (the center-to-center distance between adjacent transistor gates), and metal pitch (the spacing of the finest interconnect wiring layer). These are published in foundry technology briefings, which performs physical teardown analysis of manufactured chips.

A finished chip is the product of 1,000 to 1,500 individual process steps performed in a sequence that takes two to four months from blank wafer to finished die. Each step must be executed with atomic-scale precision. A single particle of contamination as small as a few nanometers can create a killer defect that destroys a transistor and renders an entire chip nonfunctional. The process flow described below applies to leading-edge logic chips. Memory and other device types follow similar sequences with different specifics.

Simplified Chip Fabrication Process Flow



Step 1: Oxidation

A silicon dioxide layer is thermally grown on the wafer surface by exposing it to oxygen or steam at high temperatures of 900 to 1,200 degrees Celsius. This oxide layer serves simultaneously as an electrical insulator, a diffusion barrier preventing unwanted dopant migration, and a protective hard mask for subsequent process steps. Dry oxidation uses pure oxygen and is slower but produces a denser, higher-quality oxide preferred for gate dielectrics. Wet oxidation uses steam, grows oxide faster for applications where thickness matters more than density.

Step 2: Chemical Mechanical Planarization (CMP)

As layers are built up on the wafer, the surface accumulates topography, bumps, ridges, and uneven steps. Left uncorrected, these height variations cause catastrophic problems. Lithography tools have an extremely shallow depth of focus: at EUV nodes, measured in tens of nanometers. An uneven surface produces blurred, defective patterns. CMP flattens the wafer surface using a combination of chemical slurry and mechanical abrasion against a polishing pad. It is one of the most consumable-intensive steps in the fab, burning through large quantities of abrasive slurry and polishing pads. Entegris is the dominant supplier of CMP slurries and polishing pad conditioners, making it one of the most critical materials companies in advanced manufacturing.

Step 3: Lithography

The patterning step that determines the ultimate resolution of features on the chip. This step is covered in depth in the separate Lithography Primer. In brief, a light-sensitive photoresist is coated onto the wafer, a circuit pattern is projected through or reflected off a reticle and onto the resist using an ASML scanner, and a developer dissolves the exposed areas to leave a precise physical pattern for subsequent etching. At leading-edge nodes, this step uses EUV light at 13.5nm wavelength and is repeated 80 to 100 times across the full process flow.

Step 4: Cleaning

Cleaning is performed dozens or even hundreds of times throughout the manufacturing process. The standard sequence is the RCA Clean, developed at RCA Laboratories in 1965 and still in widespread use. SC-1 (ammonium hydroxide plus hydrogen peroxide and water) removes organic contaminants and particles. SC-2 (hydrochloric acid plus hydrogen peroxide and water) removes metallic ions. An HF dip removes native oxide that grows spontaneously on silicon exposed to air. Megasonic cleaning uses high-frequency acoustic waves to dislodge particles without scratching delicate surfaces.

Step 5: Etching

Etching selectively removes material from regions not protected by photoresist, transferring the lithographic pattern into the actual device layer underneath. Wet etching uses liquid chemical etchants and is isotropic where it etches equally in all directions, including laterally underneath the resist mask which limits its resolution and usefulness for critical dimensions.

Dry etching (Reactive Ion Etching) uses a plasma of reactive ions is generated in a low-pressure chamber and accelerated vertically toward the wafer surface. The combination of chemical reactivity and physical ion bombardment produces highly anisotropic etching, straight-down, with minimal lateral undercut, enabling the high-aspect-ratio structures critical for FinFETs, NAND flash cells, and interconnect trenches. Atomic Layer Etching removes material one atomic monolayer at a time, the removal equivalent of ALD deposition, for the most demanding precision steps. Lam Research and Applied Materials are the dominant suppliers of etch equipment.

Step 6: Stripping

After etching, the photoresist used as the protective mask must be completely removed. Wet stripping uses aggressive solvents or piranha solution to dissolve the resist. Dry stripping, increasingly preferred at advanced nodes, exposes the wafer to an oxygen plasma that oxidizes and volatilizes the organic photoresist into carbon dioxide and water vapor without generating the hazardous waste volumes of wet chemistry.

Step 7: Doping

Doping is the intentional introduction of impurity atoms into the silicon crystal to modify its electrical conductivity. Undoped silicon is a poor conductor. Adding the right impurities at the right concentrations and locations is what creates the p-n junctions that make transistors function.

N-type doping introduces atoms with one more valence electron than silicon, such as phosphorus, arsenic, antimony, creating free electron carriers (negative charge carriers). P-type doping introduces atoms with one fewer valence electron (primarily boron) creating holes (positive charge carriers). Ion implantation is the modern standard method. Dopant atoms are ionized, accelerated through a high-voltage column to a precisely controlled energy (which determines how deep into the silicon they penetrate), and directed at the wafer surface. Dose is controlled precisely by monitoring the ion current. Axcelis Technologies is the leading supplier of ion implantation equipment for advanced logic and memory manufacturing.

Step 8: Annealing

Ion implantation damages the silicon crystal lattice where incoming ions slam into lattice atoms, displacing them from ordered positions and creating a cascade of crystallographic defects. Simultaneously, implanted dopant atoms initially rest in interstitial positions where they do not electrically contribute free carriers. Annealing is a precisely controlled heat treatment that repairs this damage and activates the dopants. Rapid Thermal Annealing heats the wafer to very high temperatures for extremely short durations of seconds using high-power lamp banks, activating dopants with minimal diffusion. This will be critical at advanced nodes where even a few nanometer of unintended dopant migration would distort the transistor geometry. Flash and laser annealing go further by using millisecond or nanosecond pulses to heat only the wafer surface with essentially zero diffusion.

Step 9: Deposition

Deposition builds new material layers on the wafer surface that includes insulators, conductors, semiconductors, diffusion barriers, and more. Chemical Vapor Deposition introduces precursor gases into a heated chamber where they react on the wafer surface, depositing a solid film. Atomic Layer Deposition is the precision tool for the thinnest, most critical films, using alternating pulses of two precursor gases that each deposit exactly one atomic monolayer in a self-limiting reaction. ALD is essential for gate dielectrics of 1 to 2nm thickness that must be perfectly uniform and pinhole-free across the entire wafer.

Physical Vapor Deposition physically ejects atoms from a solid target material using ion bombardment, used primarily for metallic films such as titanium, tantalum, tungsten, and copper seed layers. Epitaxy grows a single-crystal layer atom by atom on the wafer surface, inheriting the crystal orientation of the substrate. Silicon-germanium epitaxy introduces germanium to strain the silicon crystal lattice in the transistor channel, dramatically increasing carrier mobility and therefore transistor speed.

Applied Materials and Lam Research are the dominant deposition equipment suppliers across most of these categories. ASM International is the key player in ALD specifically.

Step 10: Inspection and Metrology

Inspection runs throughout the entire process flow, not just at its end. A defect found at Step 10 costs far less to address than one discovered at Step 500. CD-SEM (Critical Dimension Scanning Electron Microscopy) measures the physical width of patterned features at nanometer scale.

At 3nm nodes, features may be only 15 to 20 atoms wide and deviations of even 1nm from specification cause parametric failures. Optical defect inspection uses high-speed laser scanners to compare each die's image against a reference and flag anomalies. Overlay metrology verifies that each successive layer is correctly aligned to the one below. A misregistration of a few nanometers between layers would cause device failure.

KLA Corporation is the undisputed leader in wafer inspection and metrology equipment, with dominant market share across most of these categories. As nodes shrink and defect budgets tighten, KLA's tools become more valuable, not less.

Step 11: Repeat

The sequence of deposition, lithography, etch, clean, and inspect is repeated for every layer of the chip. A leading-edge logic chip at 3nm has 12 to 18 metal interconnect layers plus all the device layers below them. The full process requires 1,000 to 1,500 individual steps and two to four months of continuous fab time. Every single step must be executed within specification. The accumulation of small errors across hundreds of steps is the primary driver of yield loss.

Yield is the single most important financial variable in semiconductor manufacturing and is systematically underappreciated by investors outside the industry. Yield is the percentage of dies on a processed wafer that pass all functional and parametric tests and are fit for sale.

Why Yield Matters so Much

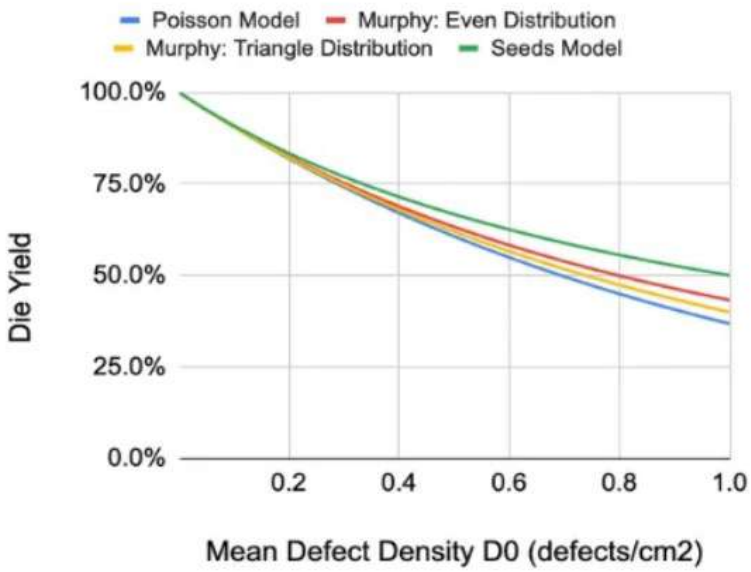
The cost of processing a wafer is essentially fixed regardless of how many good chips come off it. A wafer processed at TSMC's 3nm node costs approximately $17,000 to $20,000 to manufacture. If that wafer produces 100 dies and 80 pass testing, the cost per good chip is $200 to $250.

If process improvements raise yield from 80% to 90%, the cost per good chip falls from $200 to $178. That is an 11% cost reduction with no change in process flow, equipment, or facility investment. At the volumes leading-edge fabs run, this yield improvement directly becomes gross margin.

The Yield-Die Size Relationship

Yield is not just a function of process quality. It also depends directly on die size. Defects are distributed across the wafer surface at some average density determined by the process maturity. A large die has a higher probability of containing at least one killer defect than a small die processed on the same wafer.

This relationship is described by Murphy's yield model and is why very large chips like Nvidia's H100 have dramatically lower initial yields than smaller chips. This is also why TSMC charges enormous premiums for large-die products even at established nodes.



Yield Learning Curves

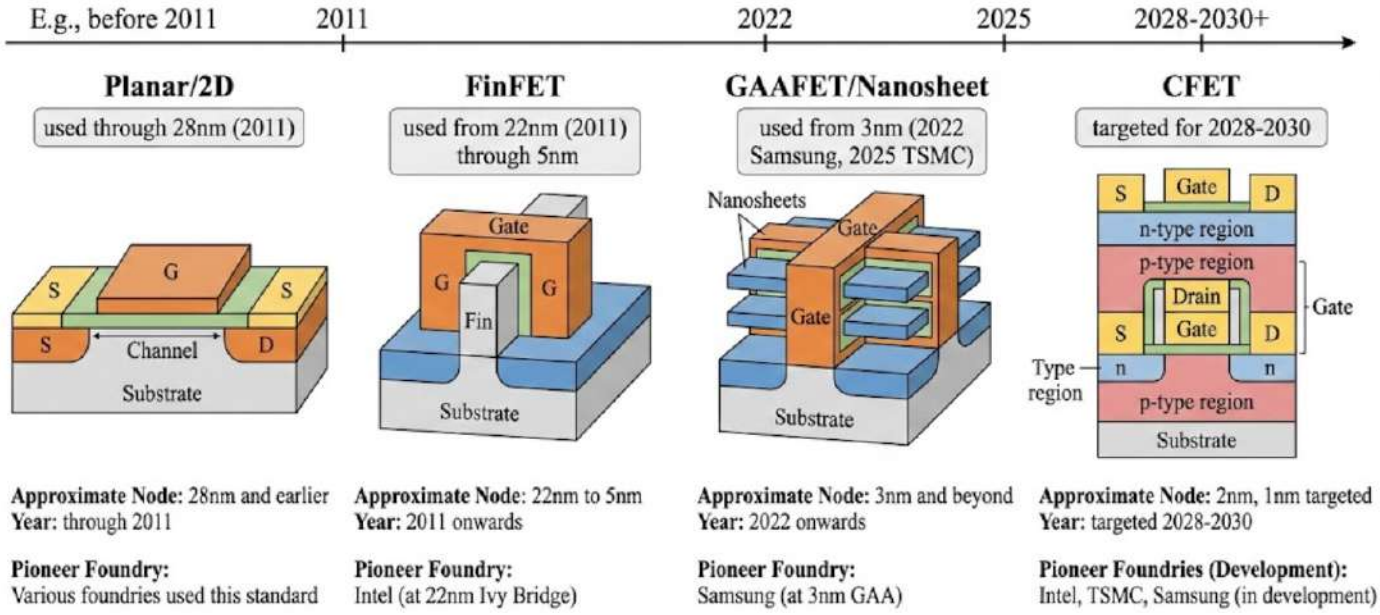
When a foundry begins production on a new process node, initial yields are low. Over the first 12 to 24 months of production, as process engineers identify and eliminate defect sources, yield improves along a learning curve. The pace of yield learning is one of the most important differentiators between foundries and between process generations. TSMC's reputation for fast yield learning is a key reason customers pay a premium to manufacture there. Faster yield learning means earlier access to high-volume, cost-competitive production. Samsung's recent struggles at 3nm and 4nm have been widely attributed to slower-than-expected yield learning, which directly translates to higher wafer cost per good chip for its customers.

Advanced Packaging & Yield

Rather than manufacturing one large die with a high defect probability, chiplets decompose the system into smaller, higher-yielding tiles that are then assembled at the package level. The trade-off is that packaging yield (the probability that a multi-chiplet assembly is functional) introduces a new yield term.

The history of transistor architecture is a story of one relentless engineering problem: leakage. To understand why architectures had to evolve, think of a transistor as a microscopic water valve. The Source is the water supply, the Drain is the sink, the Channel is the pipe between them, and the Gate is the valve controlling flow. As transistors shrank, the pipe became so short that the gate could no longer reliably shut off flow. This resulted in electrons leaking through even when the switch was supposed to be off.

Transistor Architecture Evolution Timeline



Planar (2D) Transistors: The Original Architecture

From the birth of the integrated circuit through approximately 2011 and the 28nm node, transistors were flat. The source and drain sat in the silicon substrate with the channel running flat between them along the surface.

The gate electrode sat directly on top of the thin gate oxide insulating it from the channel. The critical failure mode was the short-channel effect. When the source-to-drain distance shrank below approximately 28nm, the gate's electrostatic influence over the full channel depth weakened. Electrons could sneak through the deeper parts of the channel underneath the gate. This phenomenon is called subthreshold leakage where billions of transistors leaking femtoamps of current each sum to watts of wasted power, generating heat and draining battery life. Planar transistors hit a physical wall where further shrinking produced no useful benefit.

FinFET: The 3D Revolution

Intel introduced the first commercial FinFET at the 22nm node in 2011, a breakthrough that extended Moore's Law for a full decade. TSMC and Samsung followed at 16nm and 14nm respectively.

The core insight was to take the flat channel and stand it vertically, like a shark's dorsal fin. The gate electrode is then draped over the top and both sides of this fin, wrapping around three faces of the channel. This tri-gate geometry gives the gate dramatically better electrostatic control where there is no longer an exposed underside of the channel where electrons can leak away. The short-channel effect is suppressed, leakage plummets, and the transistor can operate at lower voltages while switching faster. FinFETs carried the industry from 22nm through 14nm, 10nm, 7nm, and 5nm generations.

Advanced packaging allows multiple chips, each manufactured on its optimal process node, to be integrated into a single system with performance approaching what a monolithic die would achieve. Therefore, achieving better economics and higher yield.

Economics of Chiplets

The economic reality of advanced nodes is that building ever-larger monolithic dies is becoming prohibitively expensive and low-yield. A 100 square millimetre die might yield 80% good units. An 800 square millimetre die on the same process might yield 5%.

The chiplet strategy breaks large systems into smaller, high-yielding tiles that are integrated at the package level. A chip's compute cores may need 3nm for maximum performance, but its I/O interfaces, analog circuits, and cache SRAM run efficiently on older, cheaper 7nm or 12nm processes.

Chiplets allow for a heterogeneous integration (matching each function to its optimal process node) which is the central idea behind AMD's Zen 3D architecture, Intel's Foveros platform, and Nvidia's use of CoWoS packaging for its H100 and H200.

The Interconnect Technology Spectrum

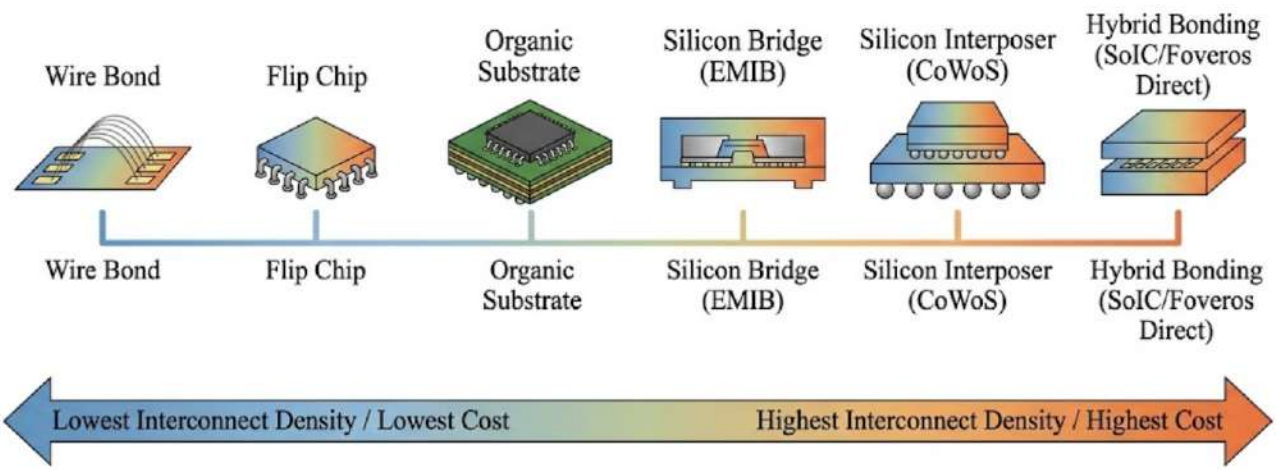
The performance of a chiplet-based system depends critically on how the individual chiplets communicate. The interconnect technology determines bandwidth, latency, and power consumption between chiplets, and it exists on a spectrum from “coarse but cheap” to “fine but expensive”.

At the lowest density end, standard flip-chip solder bumps connect chiplets to organic substrates at pitches of 100 micrometres or more. These are cheap and mature but the bandwidth between chiplets is limited by the sparse interconnect density.

Silicon interposers (large pieces of silicon patterned with high-density wiring) allow chiplets placed side by side to communicate at much higher density. TSMC's CoWoS (Chip-on-Wafer-on-Substrate) is the dominant implementation. The Nvidia H100 and H200 both use CoWoS, with the GPU die at the centre of the interposer surrounded by six to eight HBM memory stacks. Intel's EMIB (Embedded Multi-die Interconnect Bridge) embeds a small silicon bridge specifically at the chiplet-to-chiplet boundary rather than using a full interposer, reducing cost while achieving similar local interconnect density.

At the highest density, hybrid bonding directly fuses copper pads on opposing die faces at pitches of 1 to 10 micrometres. This enables chiplet-to-chiplet communication nearly as fast as within a single die. TSMC's SoIC and Intel's Foveros Direct use this approach. Hybrid bonding is the technology that enables stacking logic on logic, SRAM on logic, and eventually CFET architectures.

Advanced Packaging Technology Spectrum



The Advanced Packaging Competitive Landscape

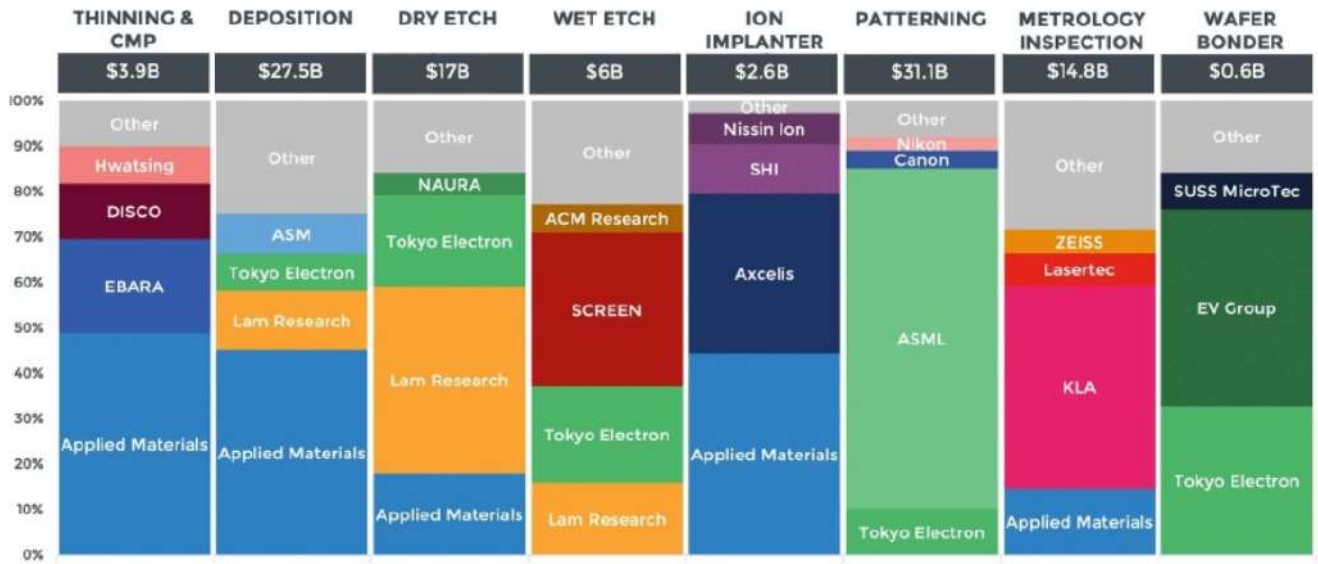
TSMC dominates advanced packaging through CoWoS for 2.5D interposer integration and SoIC for 3D hybrid bonding. Its packaging capacity has been a significant bottleneck for Nvidia AI GPU supply where expanding CoWoS capacity requires dedicated cleanroom space and specialised large-format lithography tools and cannot be done quickly.

Intel competes through its Foveros (3D stacking) and EMIB (2.5D bridge) platforms, with the strategic motivation of offering customers a one-stop-shop from design to advanced packaging. Samsung offers its own 2.5D and 3D packaging platforms but has been less successful than TSMC in winning leading-edge AI chip packaging business.

ASE Technology, Amkor, and JCET are the leading independent advanced packaging specialists (OSATs — Outsourced Semiconductor Assembly and Test) serving customers who want packaging services separate from foundry services.

Wafer Fabrication Equipment (WFE) is the collective term for all the machines used to manufacture semiconductor chips. WFE companies sell the tools that every foundry and IDM uses to process wafers, and their revenue is therefore a proxy for the entire semiconductor capital expenditure cycle. The WFE industry is highly concentrated by a handful of companies that hold near-monopoly or duopoly positions in specific equipment categories.

WFE Market Share by Equipment Category



ASML: The Lithography Monopoly

ASML manufactures every EUV lithography scanner in the world and is the sole source for the machines that print features at leading-edge nodes. Its competitive position is covered in depth in the Lithography Primer. From an investor perspective, the key points are that ASML's backlog consistently exceeds one year of revenue, its installed base generates growing recurring service and upgrade revenue, and every new leading-edge fab represents multiple billions in ASML tool demand. At approximately $380 million per High-NA EUV unit, each tool sale is a meaningful revenue event for a company with annual revenues of approximately $28 billion.

Applied Materials: The Broadest Platform

Applied Materials is the largest WFE company by revenue and has leading or strong positions across deposition, etch, CMP, ion implantation, and inspection. Its breadth gives it resilience across process steps and customer types. Applied Materials has also been building a growing position in advanced packaging equipment, particularly in the CMP, etch, and deposition steps required for hybrid bonding, positioning it to benefit from the advanced packaging growth cycle alongside traditional front-end WFE demand.

Lam Research: Etch and Deposition Specialist

Lam Research holds the leading position in plasma etch equipment and is a strong competitor in deposition, particularly for the high-aspect-ratio etch processes required for 3D NAND flash memory. Its installed base in memory manufacturing makes it significantly exposed to the NAND capital expenditure cycle, which is more volatile than logic WFE. Lam has been expanding into advanced packaging through its Syndion and VECTOR product lines.

KLA: Inspection and Metrology

KLA holds dominant positions in wafer inspection and metrology — the quality control layer that runs throughout the entire fabrication process. As node geometries shrink, defect budgets tighten and the value of KLA's tools increases proportionally. KLA's business is structurally attractive because its tools are needed regardless of which foundry or process technology wins the competitive battles in front-end manufacturing.

Source: https://siliconsemiconductor.net/article/120484/Global_Wafer_Fab_Equipment_revenue_poised_to_surge

ASM International: ALD Specialist

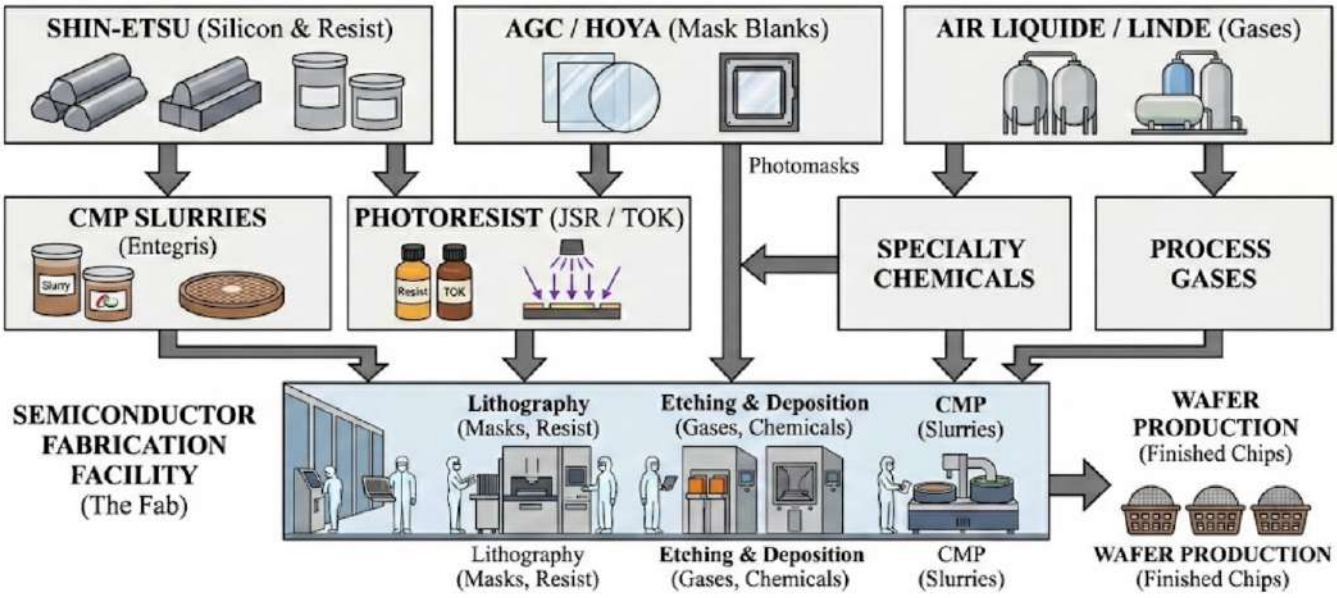
ASM International holds a strong position in atomic layer deposition equipment, a category that grows in strategic importance at every successive process node as the films being deposited become thinner and the uniformity requirements become tighter. ASM is smaller than the others but holds structural advantages in a category whose importance only increases with scaling.

Tokyo Electron: The Japanese Giant

Tokyo Electron (TEL) is the largest Japanese WFE company, with significant positions in etch, deposition, and cleaning equipment. It is a key supplier to both TSMC and Samsung. TEL's deep customer relationships in Japan and Korea give it structural advantages in serving those geographies.

Chemicals, gases, slurries, photomasks, photoresist, and the silicon wafers themselves represent a supply chain that is as concentrated and as strategically important.

Semiconductor Materials Supply Chain Map



Silicon Wafers

As discussed in Section 1, Shin-Etsu Chemical and Sumco together control approximately 60% of global silicon wafer supply. The barriers to entry are high as producing wafers at the purity, flatness, and defect density required for leading-edge manufacturing requires years of process learning. GlobalWafers (Taiwan) and SK Siltron (South Korea) are the other significant players.

Photoresist and Specialty Chemicals

Photoresist is the light-sensitive material coated onto wafers before each lithography exposure. For EUV specifically, the resist formulation must be compatible with high-energy EUV photons and their secondary electron chemistry. JSR Corporation, Shin-Etsu Chemical, and Tokyo Ohka Kogyo (TOK) are the dominant suppliers. JSR's acquisition of Inpria (the leading developer of metal-oxide resist for High-NA EUV) makes it the most strategically positioned resist company entering the next generation of lithography.

The Japanese government's 2023 facilitation of a state-backed buyout of JSR (taking it private through the Japan Investment Corporation) was an explicit recognition of the strategic importance of resist supply.

Photomask Blanks

Every lithography exposure requires a reticle — the patterned mask that carries the circuit design. EUV reticles require blank substrates coated with alternating silicon-molybdenum multilayer films that act as a mirror. AGC (formerly Asahi Glass) and Hoya, both Japanese companies, produce essentially all EUV mask blanks globally. This is a complete duopoly in a supply that is essential for every advanced chip manufactured.

CMP Consumables

Chemical mechanical planarization consumes enormous quantities of slurry (abrasive chemical mixtures) and polishing pads at every layer of every chip. Entegris dominates CMP slurry supply following its acquisition of CMC Materials. DuPont (Dow's former Electronic Materials division) is the leading polishing pad supplier.

Process Gases

Semiconductor manufacturing uses large volumes of specialty process gases such as silane, hydrogen chloride, ammonia, nitrogen trifluoride, and many others, These gases are at purities that industrial gas suppliers must maintain at parts-per-billion contamination levels.

Air Liquide (France), Linde (Germany-UK), and Air Products (United States) are the three dominant global specialty gas suppliers to semiconductor fabs. These are large industrial companies for which semiconductor gases are one segment among many, but their semiconductor businesses benefit from long-term supply agreements and high switching costs embedded in the qualification process.

The Japan concentration

A striking feature of the semiconductor materials supply chain is its concentration in Japan. JSR, Shin-Etsu, TOK, AGC, Hoya, and Sumco are all Japanese companies controlling critical supply categories.

Japan's semiconductor materials industry built its position over decades of close collaboration with Japanese chipmakers and then survived even as Japan's chip manufacturing industry declined. The geographic concentration of critical materials in Japan is a supply chain risk that the industry has been slow to address, and one that US and European governments have been increasingly flagging as a strategic vulnerability.

Lithography Primer Report



1. Glossary

Lithography Key Terms

Terms	Definitions
Directed Self-Assembly (DSA) Lithography	A next-generation patterning technique combining "top-down" conventional lithography with "bottom-up" chemical self-assembly of block copolymers (BCPs). It enhances resolution, reduces costs, and enables sub-10 nm feature sizes for semiconductor manufacturing by creating dense, ordered patterns via coating and annealing.
Deep Ultraviolet (DUV)	A photolithography technology using excimer laser light sources with wavelengths of 248nm or 193nm to print, or "print," the smallest features on a microchip. It is the industry standard for the majority of integrated circuit (IC) layers, offering high throughput and mature, reliable performance.
Extreme Ultraviolet (EUV)	An advanced lithography method utilizing an extremely short wavelength of 13.5nm, allowing for far higher resolution than DUV. Enables printing of **smaller,** more complex features required for advanced semiconductor manufacturing (7nm, 5nm, 3nm, and 2nm nodes).
G-Line (436nm)	A specific wavelength of light, commonly produced by mercury-vapor lamps, used in photolithography to pattern photoresists on semiconductor wafers. It is often used for non-critical, larger-resolution manufacturing, providing a cost-effective alternative to more advanced methods like i-line (365 nm) or EUV. Used for the backend manufacturing of chips where high precision is not required.
I-Line (365nm)	A cost-effective, mature photolithography technology using a 365 nm wavelength ultraviolet light from a mercury arc lamp to print circuit patterns on wafers. It is primarily used for manufacturing power semiconductors, IoT devices, and image sensors. While replaced by finer DUV (KrF/ArF) techniques for advanced chips, the i-line remains crucial for non-critical layers and, in many cases, offers high productivity for >250 nm features.
Krypton Fluoride (KrF)	Critical DUV (Deep Ultraviolet) photolithography technologies used to print circuit patterns on semiconductor wafers. **KrF (248nm)** handles mature, larger nodes (e.g., 90nm-130nm), while **ArF (193nm)** (dry or immersion) is vital for advanced nodes (sub-65nm to 7nm), often used in high-performance memory (DRAM/3D NAND).
Moore's Law	Moore's law predicted that transistor density would **double every 2 years,** delivering roughly **1.4x annual performance improvements.** However, AI training computers have outpaced Moore's law by 50-100x since 2012. Compute deployment doubling everyday 3.4 months. It is filled by massive parallelisation and unprecedented capital deployment

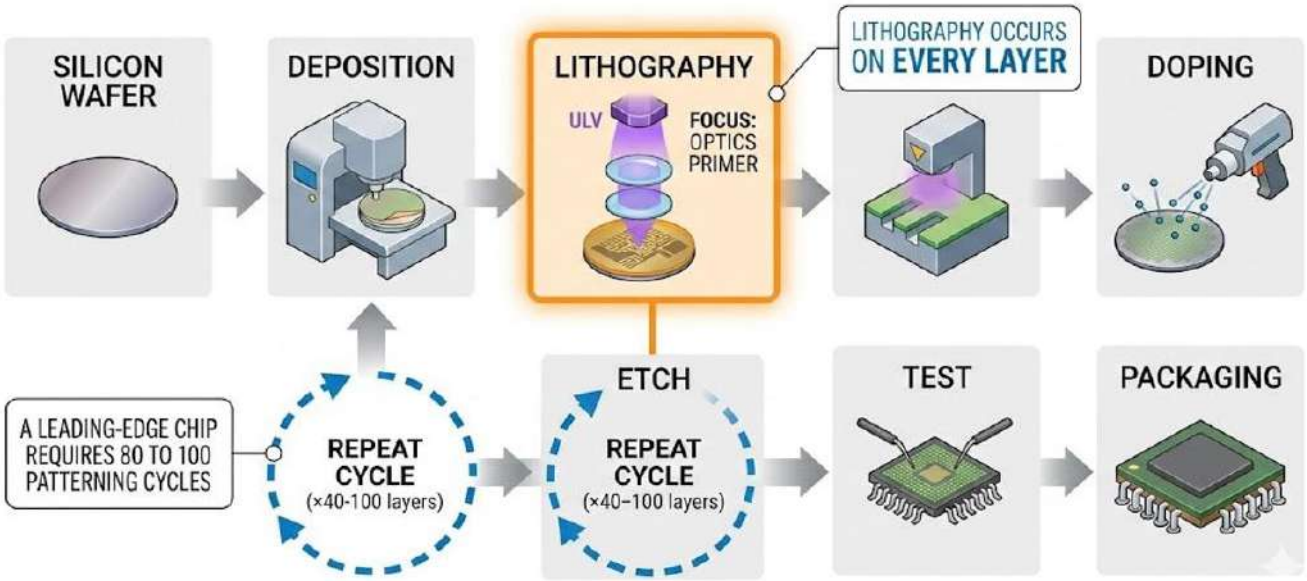
1. Glossary

Lithography Key Terms

Terms	Definitions
Nanoimprint Lithography	A cost-effective, high-resolution patterning technique that **stamps nanometer-scale structures directly** onto a substrate, offering a sustainable alternative to Extreme Ultraviolet (EUV) lithography. It achieves **sub-10nm resolution** with significantly lower energy consumption (approximately 10% of EUV) and reduced equipment costs.
Node	Defines the size, density, and performance capabilities of the transistors on a chip. The "node" number corresponded directly to the physical size of the transistor, specifically the gate length (the distance between source and drain) or the metal pitch (distance between wires).
Photolithography	The primary semiconductor fabrication method used to print intricate, nanometer-scale circuit patterns onto silicon wafers. It uses light, ranging from ultraviolet (UV) to extreme ultraviolet (EUV), through a mask to transfer designs onto a light-sensitive photoresist material, etched to create microchips.
Photoresist	A light-sensitive polymer used in semiconductor manufacturing to transfer circuit patterns onto wafers via photolithography. It acts as a sacrificial mask during etching or ion implantation, defining crucial micro-scale features. Includes positive (exposed areas = soluble) and negative (exposed areas harden).
Through Silicon Vias (TSVs)	They are microscopic copper pillars that **provide direct vertical electrical connections** through the silicon, enabling a significantly wider data bus than traditional RAM. This enables HBM by transmitting power and electrical signals vertically through the HBM stack at high speeds.
Transistors	A device used to amplify or switch electrical signals and power. Composed of semiconductor material, it acts as a tiny, solid-state gate controlling the flow of current within circuits. It is essential for modern electronics, with millions or billions embedded in integrated circuits to form processors and memory. It is made through lithography, a process that "prints" intricate circuit patterns onto silicon wafers. Light, or similar energy, projects a design mask onto a photoresist-coated wafer, defining where materials are removed or added to build the transistor structures.
Transistor Density	The number of transistors that can be fabricated within a specific, small surface area (usually per square millimeter) of a microchip, often exceeding 100 million per mm2 in modern chips. It is the primary metric for measuring semiconductor advancement, indicating how packed and efficient a processor is in terms of speed and power. It measures the concentration of switches on a chip. Higher density leads to faster, more powerful processors that consume less energy per operation.

2. What Lithography Actually Does

Semiconductor manufacturing is the process of selectively adding and removing materials from a silicon wafer in a precise geometric pattern, repeated across dozens of stacked layers, each one defining a different part of the circuit. Lithography is the step that defines the pattern for each layer. Without it there is no geometry, without geometry there are no transistors, and without transistors there is no chip.



The three-stage patterning sequence works as follows.

Coating

A light-sensitive material called photoresist is spin-coated onto the wafer surface. The wafer spins at high RPM while liquid resist is dispensed at the center, spreading into a perfectly uniform thin film across the surface. Spin speed directly controls thickness.

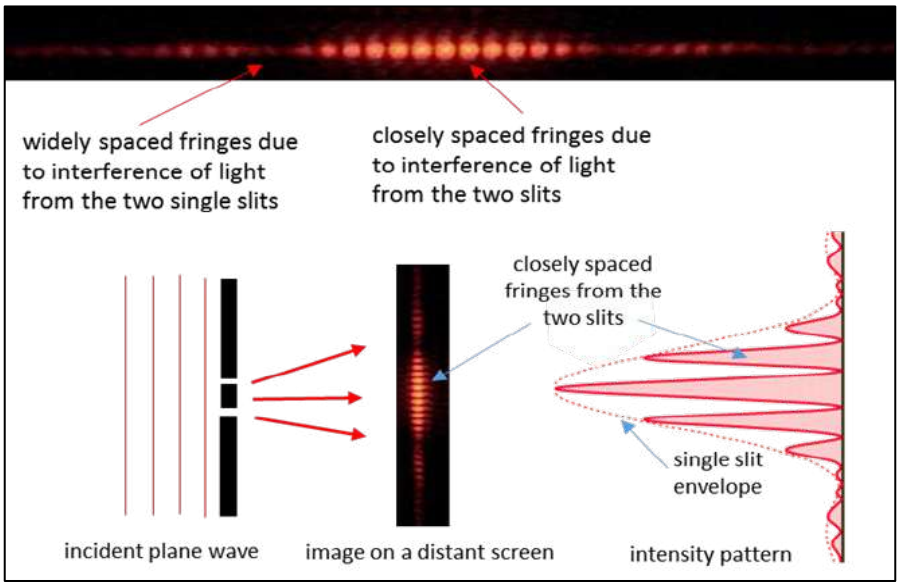
Exposure

A mask called a reticle, containing the circuit pattern for one layer, is loaded into the scanner. Light is projected through or reflected off the reticle, reduced in size by the projection optics, and focused onto the photoresist. Where the light strikes, a chemical change occurs. This is the most capital-intensive step in the entire fabrication process. A single EUV scanner from ASML costs approximately $180 million. A High-NA EUV scanner costs approximately $380 million.

Development

A chemical developer is washed over the wafer. In positive resist, the most common approach, exposed areas dissolve, leaving trenches and holes that expose the underlying material for the next process step. In negative resist, the exposed areas harden and the unexposed areas dissolve, leaving solid walls. Either way, the result is a precise physical pattern that will protect certain areas and expose others during etching or deposition.

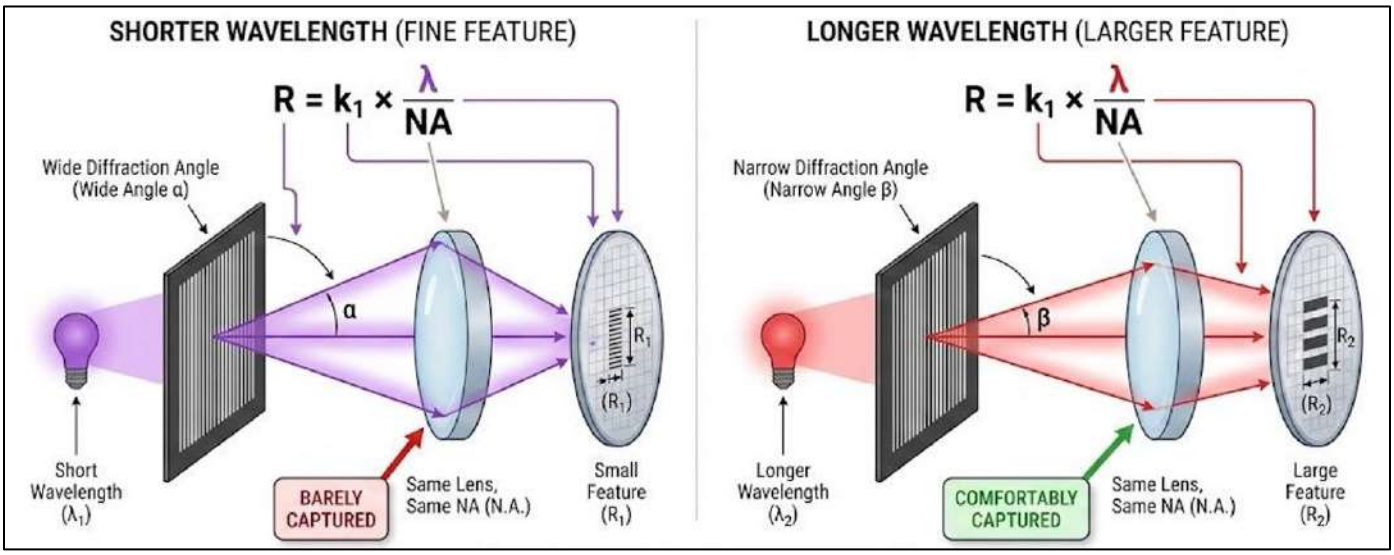
When light passes through the narrow gaps of a reticle, it does not travel in clean straight lines. Each gap acts as a new wave source, and the waves from adjacent gaps spread out and overlap. Where two wave peaks meet, they add into a bright spot. Where a peak meets a trough, they cancel into a dark spot. The result is that instead of a clean sharp image projected onto the wafer, you get a blurred, interference-distorted pattern.



The critical insight is that diffraction cannot be eliminated. Chip designers work backward from the pattern they want on the wafer and engineer the reticle geometry so that the diffraction patterns interfere constructively in exactly the right places. The patterning process is fundamentally an exercise in controlled wave manipulation.

The resolution limit of any lithography system is described by the Rayleigh equation: $\boldsymbol{R = k_1 \times \lambda / NA}$
where R is the minimum printable feature size, λ is the wavelength of light, and NA is the Numerical Aperture — a measure of how wide a cone of diffracted light the lens can collect.

The Rayleigh Equation Visualized

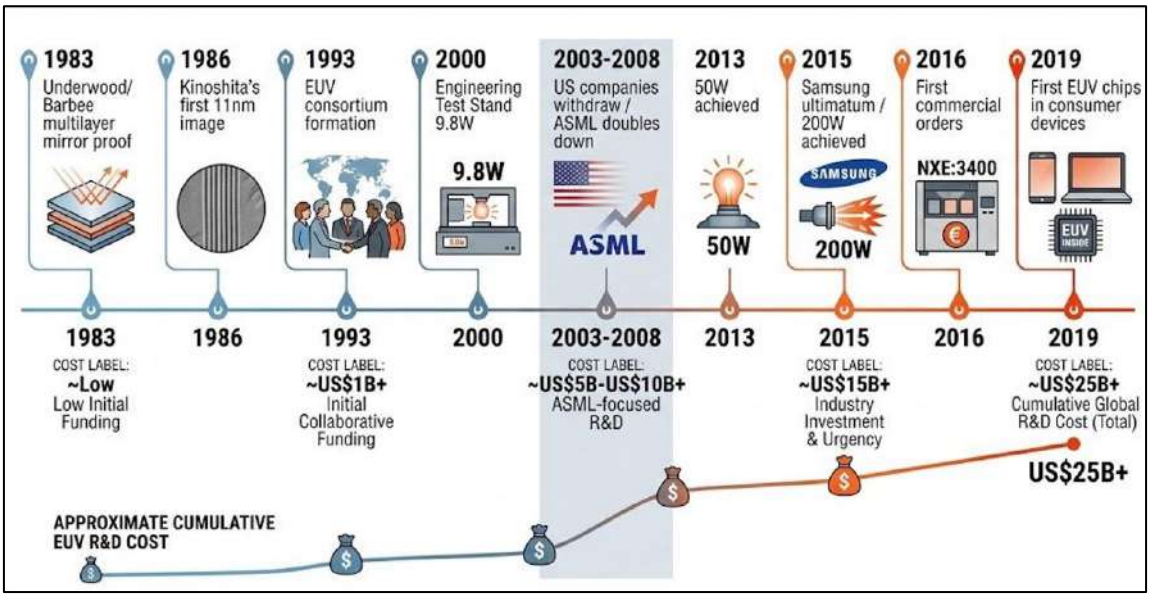


Physically, NA is the sine of the maximum half-angle of light the lens can gather. When a reticle diffracts light, the zero-order beam goes straight through, and the first-order beams angle out to either side. To resolve a feature, the lens must capture at least the zero-order and the first-order beams. The finer the features on the reticle, the wider that angle becomes, and the larger the lens system must be to capture them.

However, there is physical bottleneck on NA where the sine of any angle can never exceed 1. To avoid this problem, we require an infinitely large lens. Since NA cannot grow indefinitely, the only remaining path to smaller features is to shorten the wavelength of light.

Therefore, a shorter wavelength compresses the diffraction angles so the same lens can resolve finer patterns. This is precisely the dynamic that drove four decades of lithography wavelength reduction, from 436nm mercury lamps in the 1970s, progressively down to 365nm, 248nm, and finally 193nm deep ultraviolet light, which became the industry standard in the late 1990s and remained there for nearly two decades.

By around 2015, the semiconductor industry had extracted every last nanometer of resolution from 193nm DUV light. Multi-patterning techniques bought several more years of scaling at enormous cost and complexity.



In the early 1980s, Japanese scientist Hiroo Kinoshita proposed abandoning visible and near-UV light entirely and using X-ray wavelengths around 10 nanometer. In principle this would allow features far smaller than anything achievable with conventional UV sources. However, at these wavelengths, light is absorbed by essentially everything. The only way to make it work would be to operate in a vacuum, and without lenses (since any lens material would simply swallow the light), there was no obvious path to focusing the beam onto the wafer.

In 1983, Kinoshita discovered work by researchers Jim Underwood and Troy Barbee, who had demonstrated that specially engineered mirrors could reflect X-ray wavelengths by exploiting thin-film interference. The principle works as follows. A single layer of material reflects only a small fraction of X-ray light. But if you stack dozens of alternating layers of two materials with different refractive indices, each interface contributes a small reflected wave. By tuning each layer to exactly one quarter of the target wavelength in thickness, all those reflected waves arrive back in phase and add up constructively. Underwood and Barbee achieved around 6% reflectivity using tungsten and carbon layers, but it was proof that X-ray reflection using mirrors was physically possible.

Kinoshita used this principle to build three curved multilayer mirrors and produced an image of 4-micron lines using 11nm light. When he presented his findings to the Japanese Society of Applied Physics in 1986, the audience was almost entirely skeptical. There was nothing on Earth that naturally produced EUV light in usable quantities. Producing it in a lab meant building an artificial star.

Meanwhile at Lawrence Livermore National Laboratory, a team led by Andrew Hawryluk was studying X-ray emission from nuclear fusion reactions. Using multilayer mirrors to capture and analyze that emission, Hawryluk recognized the same lithography application. He presented his findings at a conference and was, like Kinoshita, dismissed by the audience. The scientific consensus was that making mirrors smooth enough, producing enough light, and doing all of it in a manufacturable system was simply not achievable.

Bell Labs became an early believer. With the Cold War winding down and the US government looking to commercialize national laboratory research, Intel, Motorola, AMD, and others eventually invested $250 million collectively into the effort. At the time, this represented the largest-ever private investment in a Department of Energy research project. By 1993 the technology had a formal name known as Extreme Ultraviolet Lithography (EUV).

In 2000, the consortium produced the Engineering Test Stand, the first fully functional EUV prototype. It produced 9.8 watts of 13.4nm EUV light, reflected by eight mirrors, and could print 70nm features. It proved EUV could work in principle. But it could only process about 10 wafers per hour. Commercial viability required hundreds of wafers per hour, run continuously, 24 hours a day, 365 days a year. The gap between prototype and product was enormous. American companies concluded it was unbridgeable and walked away.

ASML had spun out of Philips in the 1980s with little more than a shed, a barely functional wafer stepper, and a group of determined engineers. ASML partnered with German optics manufacturer Zeiss in a division of labor that would become history. Zeiss would take responsibility for the mirrors, and ASML would solve the light source.

The Mirror Problem

The mirror was arguably the hardest precision manufacturing problem ever attempted. At 13.5nm wavelength, any surface roughness causes light to scatter randomly rather than reflect cleanly.

A mirror for EUV light had to be smooth relative to 13.5nm, meaning surface roughness had to be held below 0.1nm, roughly 2 to 3 silicon atoms in height. A household mirror has average surface roughness of around 4,000 silicon atoms. EUV mirrors needed to be 1,000 times smoother.

Zeiss manufactured these mirrors using a sputtering process, depositing alternating layers of silicon and molybdenum. The pair was chosen over beryllium and molybdenum because beryllium is acutely toxic. The sputtering process is inherently imperfect, leaving microscopic bumps and gaps in each deposited layer. ASML engineers perfected an ion beam technique to flatten these irregularities. The ion beam gently agitates freshly deposited atoms, so they settle into surface voids before locking in place. This produces the atomically flat surfaces required.

The resulting silicon/molybdenum multilayer mirrors achieve a maximum reflectivity of around 70% per bounce. With six mirrors in the optical path plus the reflective reticle, only about 8% of generated EUV light ever reaches the wafer → A very low number!

To manage thermal drift, Zeiss built a real-time active control system directly into the optics. Robot-guided sensors continuously measure the exact position and angle of each mirror at picoradian precision. To contextualize, a pointing accuracy of one picoradian means that if you placed a laser on a mirror and aimed it at the moon, you could choose which side of a coin-sized target to hit. Heat from the plasma events shifts the mirrors by nanometers, and these active systems compensate in real time to maintain overlay accuracy of one nanometer or less across the entire wafer.

The Light Source Problem

Producing enough EUV light to drive a commercially viable tool was the hardest engineering challenge in the entire project and consumed most of the development time.

When a highly ionized atom recombines with a free electron, it releases the excess energy as a photon at a wavelength determined by the target material. For 13.5nm, tin is the material of choice. Its conversion efficiency is 5 to 10 times better than xenon, and its emission peak falls almost exactly at the target wavelength.

The challenge is that neutral tin atoms also absorb EUV light. If you simply vaporize solid tin, the surrounding neutral atoms reabsorb much of the light before it can escape. ASML's solution was to fire individual droplets of molten tin one at a time and vaporize each droplet completely and instantaneously, leaving no neutral tin cloud to reabsorb the emitted photons.

Executing this required solving four sequential problems

Droplet formation

Molten tin is pushed at high pressure through a microscopic nozzle vibrating at high frequency, producing a chaotic stream of droplets. The key discovery was that these irregular droplets naturally self-organize as they travel down the flight path, coalescing into regularly spaced, nearly identical droplets by the time they reach the interaction zone.

The Three-Pulse Strike Sequence

A low-energy pre-pulse first strikes the droplet and flattens it into a thin disc, maximizing surface area. A second pre-pulse rarefies the tin into a low-density gas cloud. Then the main pulse, delivered by a 20,000-watt CO2 laser four times more powerful than lasers used to cut steel, strikes the rarefied cloud and vaporizes it entirely into a plasma heated to over 220,000 Kelvin.

Droplet tracking

The droplets travel through hydrogen gas flowing at 360 km/h. Hitting a droplet the size of a white blood cell travelling at 250 km/h through this turbulent environment, three times in succession within microseconds, 50,000 times per second without missing, is not the easiest task.

ASML uses laser curtains along the flight path to detect each droplet's position and velocity in real time, then calculates exactly when to fire the main pulse accounting for the travel time of the laser pulse itself.

Mirror Contamination

The plasma events generate tin debris that would coat and destroy the collector mirror within hours. ASML filled the chamber with low-pressure hydrogen.

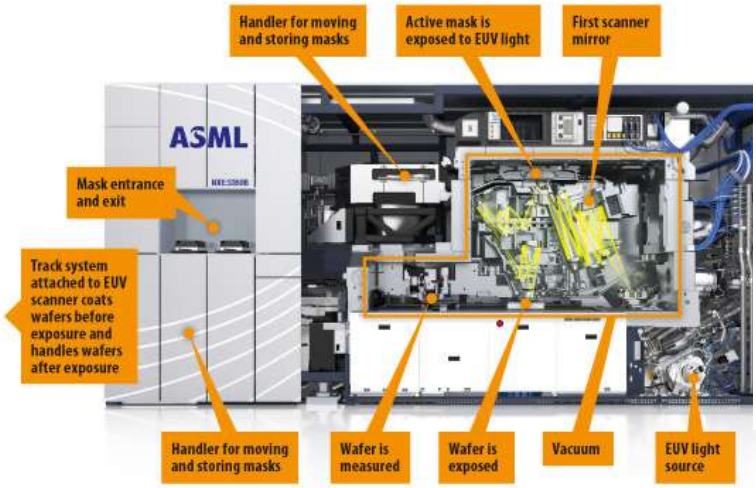
Hydrogen molecules slow the tin debris particles, and the small fraction that reaches the collector surface is removed continuously by chemical reaction → hydrogen combines with tin to form stannane gas, which is then pumped away. Managing hydrogen pressure is a precise balance since too little fails to protect the mirror, while too much absorbs EUV light and causes overheating from the 50,000 shockwaves per second propagating through the gas.

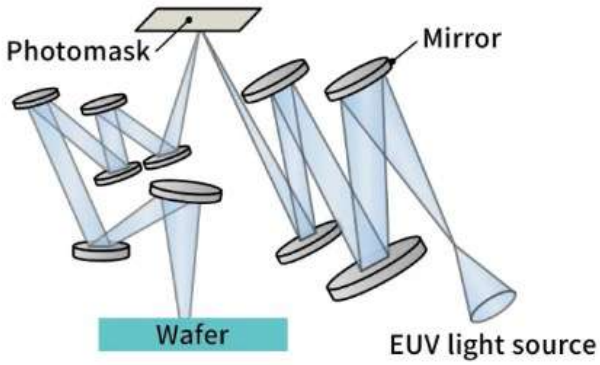
The Power Milestones

The engineering test stand in 2000 produced under 10 watts and the target for commercial viability was 100 watts. By 2013, ASML had reached 50 watts. Multi-patterning improvements extended the useful life of DUV tools, meaning EUV would only be cost-competitive at 200 watts and 125 wafers per hour.

The final obstacle was the mirror’s lifetime. The intense photon and ion bombardment degraded a protective coating on the collector mirror, requiring a shutdown every 10 hours for cleaning. It was only when engineers noticed that whenever the vacuum chamber was opened for maintenance, oxygen from the air contacted the mirror and the contamination improved. Controlled micro-dosing of oxygen into the vacuum environment turned out to continuously oxidize and remove the contaminating deposits, keeping the mirror clean indefinitely without shutdowns.

By 2016 commercial orders began arriving. By 2019, the first chips manufactured using EUV reached consumers. It had taken 30 years from Kinoshita's first images.





The Light Source

Generates EUV plasma using the tin droplet and CO2 laser system described in the previous section. The light source module is supplied by Cymer, a San Diego company ASML acquired in 2013.

The Illuminator

Conditions the collected EUV light into a precisely shaped beam. It homogenizes intensity, sets polarization, and controls the angular distribution of light hitting the reticle. By adjusting illumination mode, such as the dipole, quadrupole, annular, or freeform, engineers improve the contrast of specific pattern geometries on the wafer.

The Reticle

The reticle in EUV is reflective rather than transmissive. A flat substrate is coated with alternating silicon/molybdenum multilayer films acting as a mirror, with an absorber material patterned on top to block selected portions of the reflected light, encoding the circuit pattern.

Reticle blanks are supplied by AGC (formerly Asahi Glass, Japan) and Hoya (Japan) which produce essentially all EUV mask blanks.

The Projection Optics

Six ultra-smooth multilayer mirrors supplied entirely by Zeiss. In the standard EUV machine (*NA = 0.33*), the pattern is reduced 4x. In the High-NA machine (*NA = 0.55*), the projection optics reduce the pattern 8x in one direction and 4x in the other.

The High-NA mirrors are smoother still than those in the standard machine. If the standard EUV mirrors were scaled to the size of Germany, the tallest bump would be about 1mm. Scaled to the size of the Earth, the High-NA mirror's tallest bump would be no thicker than a playing card.

The Wafer Stage

Chip fields are printed one at a time, stepping across the wafer in a precise grid. To achieve 185 wafers per hour, the stage must scan back and forth at accelerations exceeding 20g. Overlay accuracy between successive layers must be held to within one nanometer.

The Full Machine

The complete EUV is built from components supplied by 5,000 companies worldwide. Each machine contains 100,000 parts, 3,000 cables, 40,000 bolts, and two kilometers of hosing. Shipping one machine requires 250 containers, 25 trucks, and seven Boeing 747s. It is assembled in a cleanroom which is 1,000 times cleaner than a hospital operating room.

It is important to note that before EUV reached commercial viability, fabs needed to print features at 10nm and 7nm using DUV tools that could only reliably resolve down to around 40nm in a single exposure.

The solution was to split dense patterns into multiple simpler passes, each printable with existing tools.

LELE (Litho-Etch-Litho-Etch)

The target pattern is decomposed into two interleaved, sparser sub-patterns. The first lithography pass prints every other line, spaced far enough apart to be printable, and those lines are etched into the wafer. A fresh coat of resist is applied, and the second pass prints the remaining lines exactly between the first set. The result is double the density at double the cost, double the cycle time, and with strict alignment requirements between the two passes. Any overlay error between pass one and pass two directly manifests as a pattern defect.

SADP (Self-Aligned Double Patterning)

A more elegant approach using chemistry to avoid the second lithography exposure. A set of lines is printed at achievable spacing. A conformal material is deposited over them, coating all surfaces uniformly. An anisotropic etch removes material from flat horizontal surfaces but leaves thin vertical deposits, called spacers, on the sides of the original lines. The original lines are then selectively etched away, leaving only the spacers. The spacers are now twice as dense as the original printed lines. Because spacer positions are defined by the geometry of the original lines rather than a second mask alignment, this method is self-aligned and avoids the overlay errors inherent to LELE.

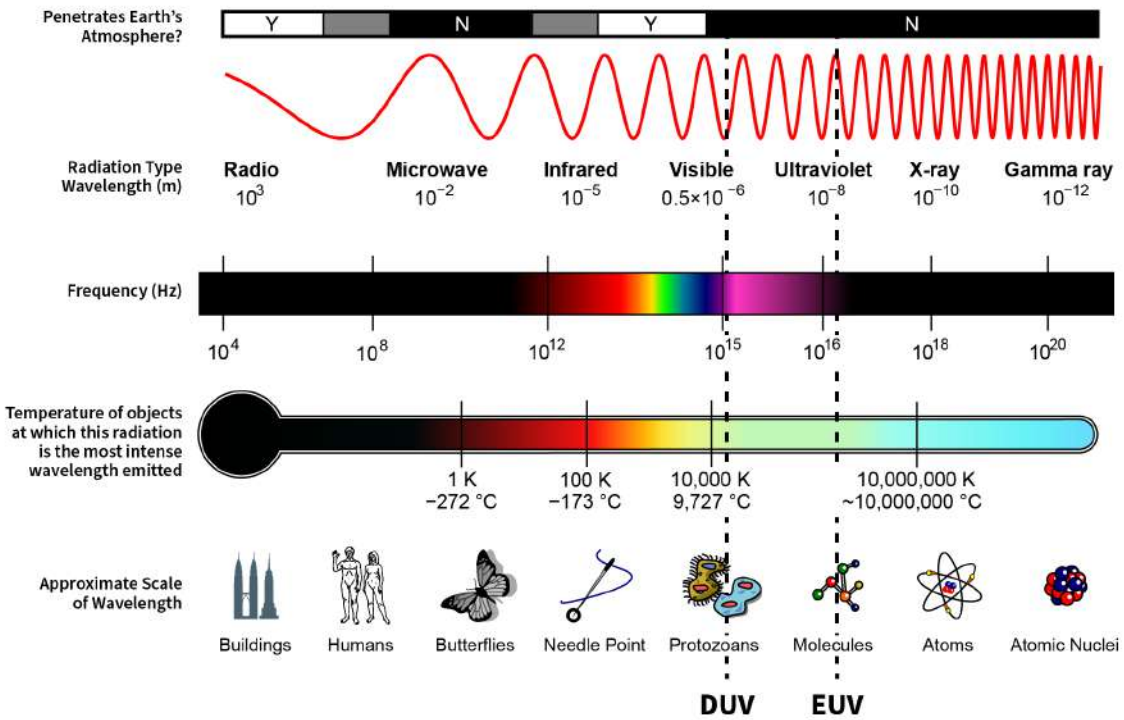
The reason EUV adoption was so economically significant is precisely that it eliminates most multi-patterning steps on critical layers. A single EUV exposure can replace four or more DUV exposures for the same layer, dramatically reducing cycle time, consumable usage, overlay error accumulation, and cost per wafer. Each EUV tool in practice displaces four to five DUV tools for the layers it covers, which is why the industry's capital expenditure shifted so dramatically toward ASML once EUV reached commercial viability.

Source: https://www.rapidus.inc/en/tech/te0005/, https://bits-chips.com/article/what-makes-an-euv-scanner-tick/

8. DUV vs EUV Comparison

Feature	DUV (Deep Ultraviolet)	EUV (Extreme Ultraviolet)	High-NA EUV
Wavelength	193nm (ArF excimer laser)	13.5nm (plasma-generated)	13.5nm (plasma-generated)
Numerical Aperture	1.35 (immersion)	0.33	0.55
Optics	Refractive (glass lenses)	Reflective (multilayer mirrors)	Reflective (multilayer mirrors)
Environment	Water immersion	High vacuum	High vacuum
Photomask	Transmissive	Reflective	Reflective (anamorphic)
Approximate tool cost	~$70M	~$180M	~$380M+
Key supplier	ASML, Nikon, Canon	ASML (monopoly)	ASML (monopoly)
Min. feature (single exposure)	~40nm	~13nm	~8nm
Wafers per hour	275+	185	~90–100 (early ramp)

Wavelength Comparison DUV vs EUV



Note that DUV is not yet obsolete. The vast majority of layers in even the most advanced chips are still printed using DUV, because most layers do not require the resolution that only EUV can deliver.

The economic case for using EUV applies only to the five to fifteen most critical pattern layers per chip. ASML still sells and services a large installed base of DUV tools (Nikon and Canon remain competitors in DUV) but EUV is entirely ASML's domain.

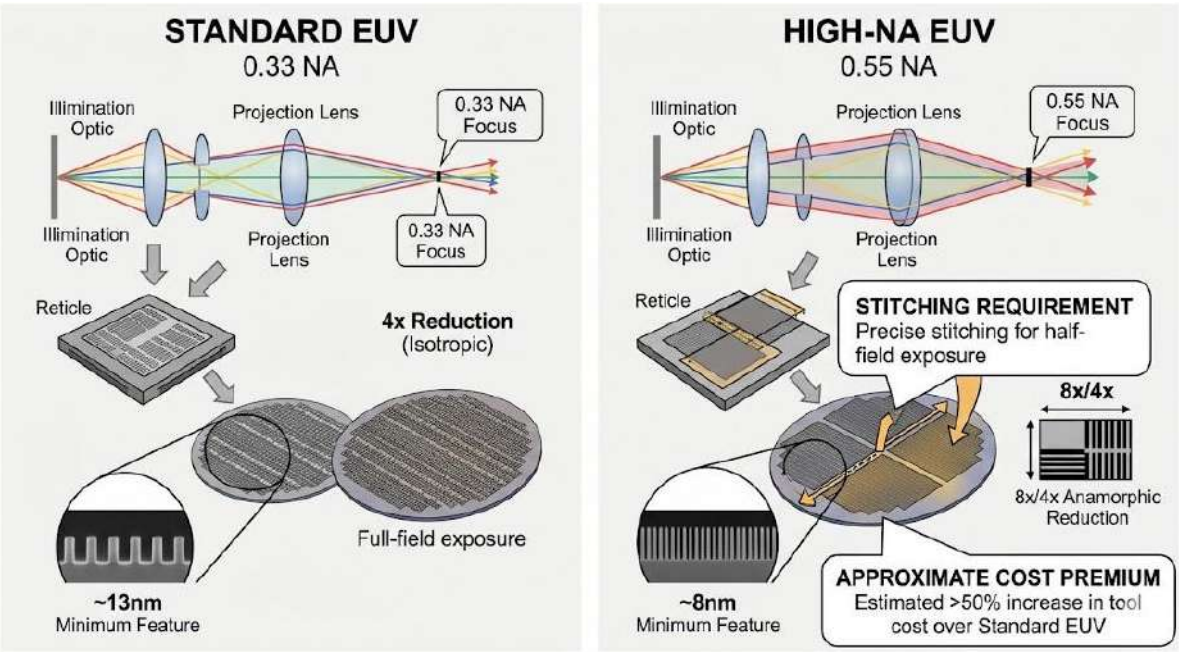
As mentioned in previous sections, the standard EUV machine runs at a Numerical Aperture of 0.33. ASML's next platform, the High-NA EUV scanner, raises the NA to 0.55. This enables single-exposure feature resolution down to approximately 8nm.

This is essential for spearheading the next advanced chips that are designed at <= 2nm which increasingly require features that even standard EUV cannot print in a single pass without returning to multi-patterning workarounds.

0.55 Requires a Fundamental Redesign

Simply increasing the aperture of the existing optics to 0.55 would require a lens diameter so large that the machine would be physically impossible to build. ASML solved this with an anamorphic optical design, meaning the projection optics reduce the reticle pattern at different ratios in the horizontal and vertical directions, specifically a 8x reduction vertically and 4x reduction horizontally. This asymmetric compression allows the mirrors to remain physically buildable while achieving the effective resolution of a much higher NA system.

However, the trade-off is that because the reduction ratios differ across axes, the machine can only print half the chip area in a single exposure compared to the standard EUV machine. This constraint is known as the half-field limitation. Critical layers with the finest features must be printed in two overlapping exposures with stitching at the boundary. Managing the stitching seam without introducing defects is a non-trivial process engineering challenge that Intel, the first High-NA customer, is actively working through.



The Photoresist Challenge

High-NA EUV creates a photoresist problem that the current generation of chemical resist cannot easily solve. Standard EUV photoresist is designed around the stochastic interaction of 13.5nm photons with organic chemical amplification.

At High-NA, the number of photons per exposure is lower (because smaller features collect fewer photons per unit area), and the resolution requirements are tighter. As a result, the statistical variation in where chemical reactions occur starts to matter significantly. Random variation at the molecular scale, known as stochastic noise, becomes a primary source of defects.

The leading solution being developed is metal-oxide resist, specifically tin-based organometallic compounds. These materials react more reliably and with higher sensitivity to EUV photons than conventional organic resists, and their inorganic metal content provides higher etch resistance. JSR, Inpria (acquired by JSR), and Merck KGaA are among the companies furthest along in metal-oxide resist development. The technical and supply chain implications of this resist transition are discussed further in Section 9.

The Customer Landscape

Intel was the first recipient of High-NA EUV tools, taking delivery in early 2024 for use in its Intel 14A node, expected to enter production in 2026. Intel's aggressive adoption of High-NA is part of its strategy to close the technology gap with TSMC. TSMC and Samsung are expected to follow with High-NA deployment for their respective sub-2nm process generations, likely from 2026 to 2027.

At approximately $380 million per tool, High-NA EUV is more than double the cost of standard EUV. The fabs that deploy it will require fundamentally different economic justifications. The yield improvement and multi-patterning avoidance at leading nodes must offset the tool cost premium over a smaller addressable layer count than standard EUV. Early industry estimates suggest each High-NA tool will still effectively displace two to three standard EUV tools on the layers it covers, keeping the economic case positive at volume.

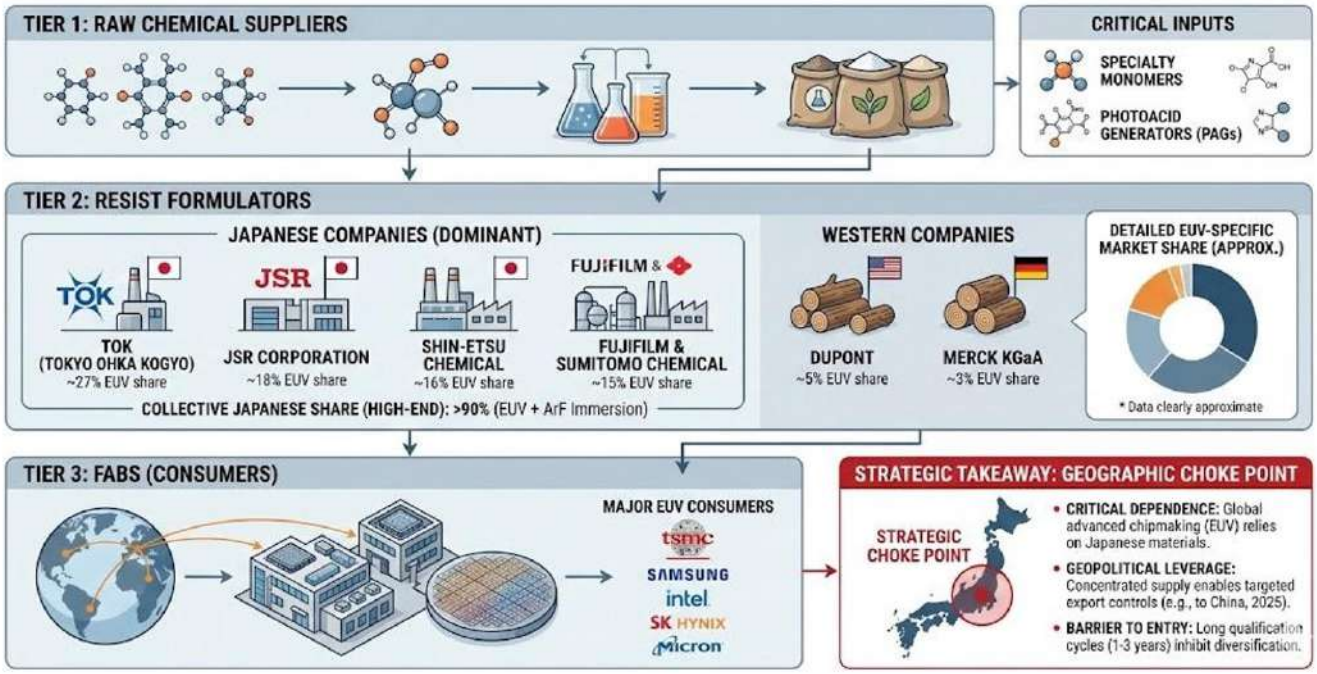
With the lithography scanner getting most of the attention, the materials has an equally important role in determining whether a pattern on the wafer can be reproduced accurately. Photoresist and its associated materials are a specialized supply chain that has consolidated into a small number of primarily Japanese and European companies.

Photoresist:

Photoresist is a light-sensitive polymer applied to the wafer before each exposure. Its job is to undergo a precise, localized chemical change exactly where light strikes, then respond predictably to the developer chemical, producing clean, sharp pattern edges.

For DUV, chemically amplified resists use a photoacid generator where a small number of photons creates acid molecules that then catalyze a much larger chemical amplification reaction, improving sensitivity and allowing practical exposure times.

EUV resist is more challenging because EUV photons are far more energetic than DUV photons and interact with resist differently. EUV photons generate secondary electrons that do most of the actual chemistry. Managing where those secondary electrons travel and therefore where the chemical reaction actually occurs is a primary determinant of the achievable resolution and edge roughness of the printed pattern.



The Key Suppliers

JSR Corporation is the largest dedicated EUV photoresist supplier. The company's strategic importance was implicitly recognized by the Japanese government, which in 2023 facilitated a buyout of JSR by the Japan Investment Corporation, a state-backed fund, taking it private. The stated rationale was supply-chain security and R&D investment continuity. JSR also acquired Inpria, the leading developer of metal-oxide resist for High-NA EUV, making it the most broadly positioned resist company entering the High-NA era.

Shin-Etsu Chemical is the world's largest semiconductor photoresist producer by volume, with dominant positions in DUV resist and a growing EUV portfolio. It also dominates the silicon wafer market, making it one of the most vertically exposed companies to overall semiconductor volume growth.

Tokyo Ohka Kogyo (TOK) is another major Japanese resist supplier with strong positions in both DUV and EUV applications.

Merck KGaA, through its semiconductor materials division, supplies specialty resist formulations, anti-reflective coatings, and developer chemicals. It has been actively investing in EUV-compatible materials including spin-on glass and carbon hard mask materials that are used in conjunction with resist in the multi-layer stack.

Pellicles

A pellicle is a thin membrane, stretched over a frame, that sits a few millimeters above the reflective surface of the EUV reticle. Its purpose is to keep particles from landing on the reticle surface. In DUV, particles on the reticle print as defects on the wafer and pellicles keep the reticle surface clean.

In EUV, developing a pellicle is dramatically harder because the membrane must transmit EUV light (already scarce, since only 8% reaches the wafer after all mirror reflections) while surviving the intense photon bombardment of the plasma source without degrading.

Early EUV fabs operated without pellicles, relying instead on faster reticle inspection and replacement cycles. Developing a reliable EUV pellicle has been an industry priority for a decade. The leading materials are polysilicon and carbon nanotube membranes, each with different trade-offs in transmission, durability, and manufacturability. ASML, Mitsui Chemicals, and IMEC have been the primary development partners. Achieving greater than 90% EUV transmission through the pellicle while maintaining a usable lifetime is an unsolved problem at High-NA, where the photon flux is even more demanding.

Why the Monopoly Cannot be Replicated Easily

The two closest competitors in lithography, Nikon and Canon of Japan, both participated in the EUV development programs during the 1990s and both withdrew before commercial viability was achieved. Neither company has since developed EUV capability. Nikon has openly stated that it does not plan to enter EUV and is pursuing next-generation patterning alternatives (nanoimprint lithography, discussed below). Canon is similarly focused on DUV extensions.

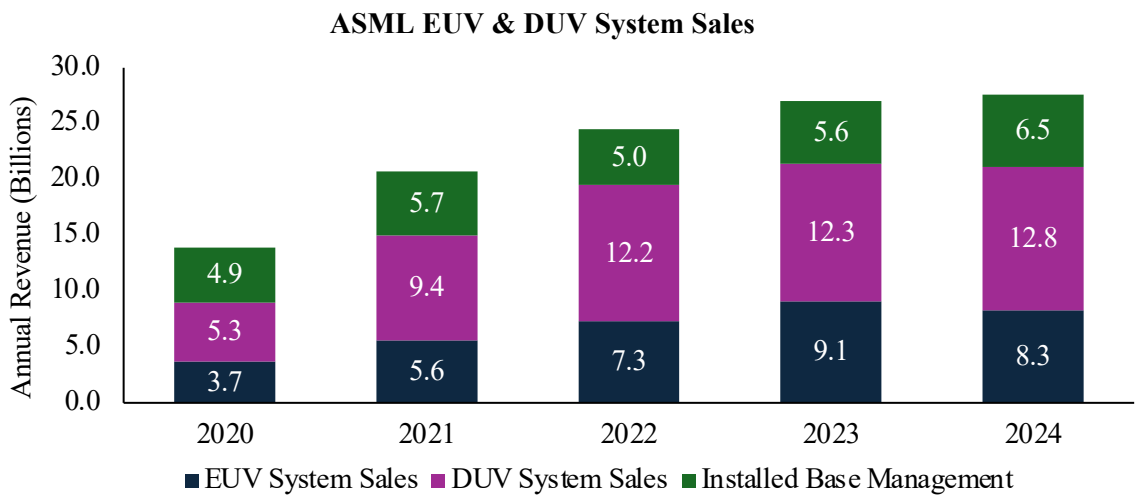
The barrier to entry is the 30-year accumulation of physical knowledge embedded in every subsystem of the machine. The tin droplet light source, the Zeiss mirror polishing process, the active thermal compensation in the optics, the hydrogen management system that were mentioned earlier. Each of these required decades of trial and error that cannot be shortcut by capital or reverse engineering. A competitor starting today would need to solve all these problems simultaneously, with no guarantee of success. They are up against a customer base that has already committed its process roadmaps and engineering teams.

ASML estimates its own installed base of EUV tools will require continuous service, upgrades, and consumables throughout the tool lifetime of 10 to 15 years, creating a growing recurring revenue base independent of new tool shipments.

Financial Scale

ASML generates revenue from three streams → New tool sales, installed base management (service, upgrades, and parts), and lithography systems for DUV, which still represents a large portion of the overall wafer fabrication market.

ASML operates in a market where its three primary EUV customers, TSMC, Samsung, and Intel, are all increasing capital expenditure simultaneously to build next-generation fabs. Every advanced logic chip manufactured at 3nm and below today requires ASML's EUV tools. Every new fab at those nodes requires multiple EUV scanners. With High-NA tools priced at $380 million each and a leading-edge fab requiring 10 to 20 of them, each new greenfield fab represents $4 to $8 billion in ASML tool revenue alone, before DUV tools are counted.



One Structural Risk

ASML's monopoly in EUV is durable, but it is not immune to longer-term disruption. Two alternative patterning technologies are being pursued.

Directed Self-Assembly (DSA) uses block copolymers that organize themselves at nanometer scale under controlled conditions, potentially bypassing the need for high-resolution optical exposure on certain layers.

Nanoimprint Lithography (NIL), championed by Canon, physically presses a patterned template into resist rather than using light. Canon has been commercializing NIL for NAND flash applications and claims a cost advantage at certain layers. Neither technology threatens EUV at leading logic nodes in the foreseeable future, but both represent long-term optionality that the industry is monitoring.



Memory Primer Report

Memory Key Terms

Terms	Definitions
Bit Output	The massive data-generating capability of modern devices, driven by memory density (DRAM/NAND) and high-speed processing. As of 2025, the industry is transitioning to AI-driven HBM with extremely high data rates.
Cache Levels	Specialised, high-speed static RAM (SRAM) memory integrated into processors to bridge the speed gap between the CPU and slower DRAM. They are organized hierarchically: L1 is smallest / fastest per core, L2 is larger / slower, and L3 is largest / slowest, typically shared across cores to minimize data latency.
Compute Express Link (CXL) KV Cache	A specialised, high-performance storage architecture for LLM inference that uses CXL technology to offload the KV cache from limited GPU HBM to external, capacity-optimized CXL-attached DRAM. It enables long-context LLM inference (e.g., 32K to 1M+ tokens) by addressing the "memory wall" where the KV cache grows to exceed GPU capacity.
Duel-In-Line Memory (DIMM)	A standard circuit board containing RAM chips used in computers, offering a 64-bit data path for faster, more efficient data transfer compared to older technologies. DIMMs use separate, independent electrical contacts on both sides, typically for desktops, while smaller SO-DIMMs are used in laptops.
Dynamic Random Access Memory (DRAM)	Uses a single transistor and capacitor per bit, requiring constant, power-consuming, and slow refreshing, but offers high density, low cost, and high capacity suitable for main system memory. DRAM stores bits as electric charges within capacitors. DRAM must be refreshed periodically (hundreds of times per second) to maintain data, whereas SRAM retains data as long as power is supplied. DRAM is much cheaper and denser, allowing for higher storage capacity in a smaller physical space.
High Bandwidth Memory	High-performance, 3D-stacked DRAM memory technology designed to provide massive bandwidth, low latency, and high energy efficiency. By vertically stacking DRAM memory chips (dies) and placing them directly alongside a processor (like a GPU) via an interposer, it allows for significantly faster data transfer than traditional DDR or GDDR memory. Must be very precise to the nanometers/ Supply of HBM is not increasing as fast as demand for HBM.
Hybrid Bonding / Direct Bonding Interconnect (DBI)	An advanced semiconductor packaging technology that connects stacked chips directly using copper-to-copper, rather than solder bumps. Copper pads on the face of each die are polished to atomic flatness, chemically activated, precisely aligned and placed in direct contact. Copper pads fuse through solid-state diffusion during a low temperature annealing. It enables ultra-fine pitch connections with no underfill material or gaps in dies and higher interconnect density

Memory Key Terms

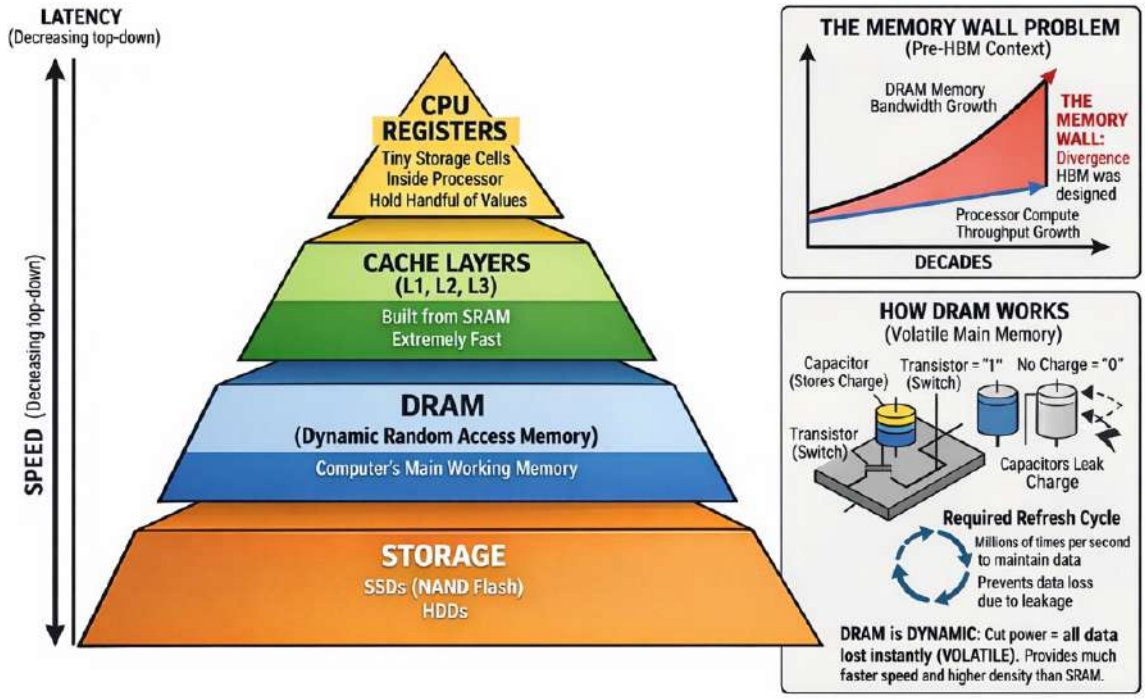
Terms	Definitions
Interposer	An electrical interface routing component, often a silicon wafer, which is placed between a semiconductor chip and a substrate or another chip. Acting as an intermediary, it enables advanced 2.5D packaging by providing high-density, short-path connectivity (using TSVs and RDLs), allowing multiple chiplets to function as a single, compact system with lower power requirements.
Key-Value (KV) Cache	An optimisation technique in transformer models that stores intermediate attention computations for previous tokens during text generation, avoiding redundant calculations. It significantly speeds up LLM inference from quadratic to linear time, allowing faster responses by reusing past keys and values.
Latency	The time delay between an **input** (request for data) and the corresponding **output** (response). It measures the duration for data to move between components or systems, usually in nanoseconds or microseconds. Low latency is crucial for real-time responsiveness in devices and AI.
Logic-To-Memory Interconnect	Advanced, high-density connection technologies that directly link distinct semiconductor dies, specifically computational logic (processors like GPUs) and memory (HBM or DRAM), within a single package. Unlike traditional methods where chips are connected via a printed circuit board (PCB), these interconnects enable 2.5D or 3D stacking, bringing memory physically closer to logic to maximise data speed and energy efficiency.
Mass Reflow Molded Underfill (MR-MUF)	An advanced semiconductor packaging technique that combines chip bonding (mass reflow) and underfill encapsulation (molding) into one process. It involves simultaneously soldering stacked chips and filling the gaps with liquid Epoxy Molding Compound (EMC) to enhance durability, heat dissipation, and production efficiency, notably in HBM3/3E production. It is SK Hynix's proprietary approach, developed and patented beginning in 2019.
Microbumps	Essential in HBM for enabling 3D vertical stacking of DRAM dies, creating high-density, low-latency interconnects between memory layers and processors. By providing thousands of connections, they achieve superior bandwidth while aiding in thermal management and heat dissipation within tightly packed, high-performance computing systems. They also allow for pre-testing of DRAM dies before final bonding.
Memory	A small, integrated circuit semiconductor device that stores digital data in electronic equipment, such as computers, phones, and USB drives. These chips act as the "brain's" storage, holding data either temporarily for quick access by the CPU (Volatile/RAM) or permanently (Non-volatile/Flash/ROM). They are responsible for storing data, applications, and operating system instructions, with speed and capacity defining device performance.

Memory Key Terms

Terms	Definitions
Nand Flash	a non-volatile storage technology that retains data without power, used in SSDs, USB drives, smartphones, and memory cards. It offers high storage density, fast read/write speeds, and durability due to lack of moving parts, making it ideal for cost-effective, high-capacity, long-term data storage. It is a substitute for HBM.
Static Random Access Memory (SRAM)	SRAM uses 4–6 transistors per bit and needs no refresh, providing high-speed, low-latency performance ideal for CPU cache SRAM uses flip-flop circuits to hold bits, and is much more expensive.
Thermal Compression Bonding (TCB)	a high-precision semiconductor packaging process that joins chips to substrates or other chips using simultaneous heat, pressure, and time. It is critical for advanced, high-density packaging, such as 2.5D/3D ICs and AI chiplets, enabling fine-pitch interconnections below 10 nm. Hanmi Semiconductor and Hanwha Precision Machinery produce TCBs and supply them to micron.
Thermal Compression with Non-Conductive Film (TC-NCF)	a dielectric adhesive material used in semiconductor packaging, **applied as a film to wafers before dicing.** It acts as an insulator, protecting microbumps and providing underfill protection. It is laminated, then cured using thermocompression bonding (TCB) to connect chips to substrates, optimizing fine-pitch assembly. **NCF** is applied to the wafer diced, and then, during assembly, is heated and pressed (thermocompression) to melt, fill gaps, and cure. This eliminates the need for separate capillary underfill steps and flux, reducing processing time and costs, and prevents filler entrapment. Made of thermoplastic resins.
Video Random Access Memory (VRAM)	is a specialised, high-speed type of RAM physically located on a computer's graphics card or integrated into the GPU's processor package. It acts as a dedicated buffer between the GPU and the display, storing graphical data, textures, and frame buffers required to render images on a screen.
Von Neumann Bottleneck	a performance limitation in computer architecture caused by the speed difference between the **CPU** and **memory.** Because data and instructions share a single bus, the CPU frequently waits for data transfer, resulting in idle time and restricted overall system speed.

2. The Memory Hierarchy

Memory inside a computer is not a single uniform technology. It is a pyramid of tradeoffs between speed, cost, and capacity. Understanding this pyramid is the starting point for understanding why HBM was invented and why it has become one of the most important components in AI infrastructure.



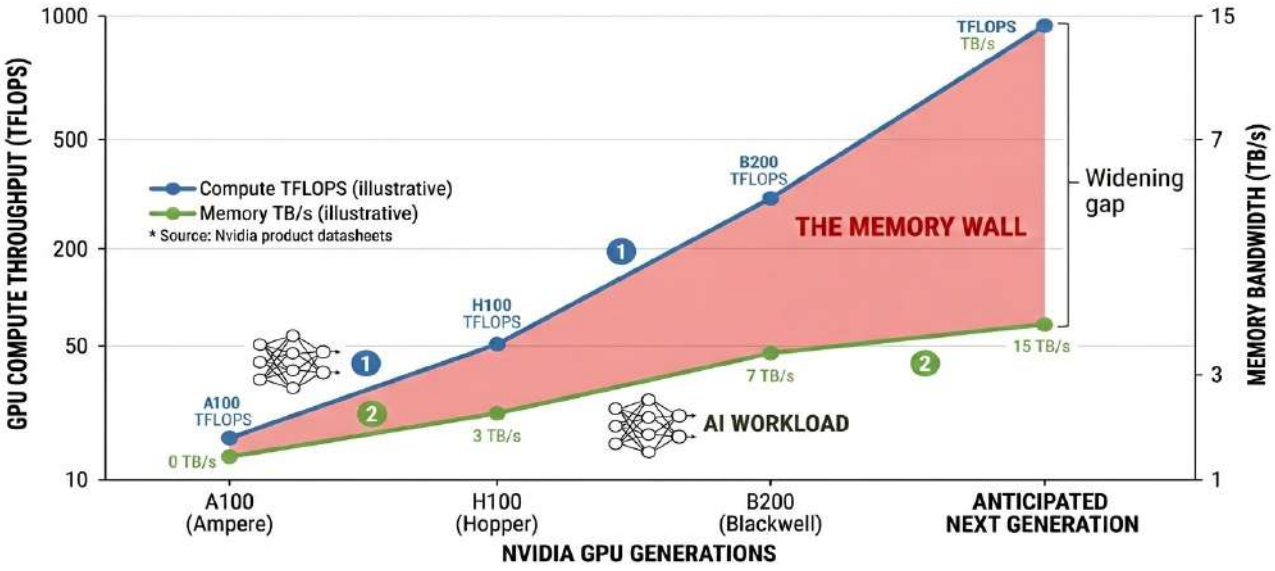
At the very top sit CPU registers, tiny storage cells inside the processor that operate at full processor clock speed but hold only a handful of values at any time. Just below are the cache layers (L1, L2, L3), built from SRAM (Static Random Access Memory), which are extremely fast but physically large and expensive per bit stored. Below that is DRAM (Dynamic Random Access Memory), the computer's main working memory. DRAM is slower than cache but far cheaper and denser, making it practical as the primary pool of memory a processor draws from during computation. Below DRAM sits storage which are SSDs built from NAND flash, and hard disk drives (HDDs). Each layer of the pyramid is roughly ten times slower than the layer above it.

DRAM is the critical working memory tier for all computation. Every instruction a processor executes requires feeding data from DRAM into the processor's registers. For decades, the rate at which processors could execute instructions grew faster than the rate at which DRAM could supply data. This divergence is called the memory wall, and it is the fundamental problem HBM was designed to solve.

The gap between processor compute speed and memory bandwidth has existed for decades and was manageable for most conventional workloads. AI inference and training transformed it into a critical infrastructure constraint.

The core operation in modern AI is the transformer attention mechanism, introduced in the landmark 2017 paper "Attention Is All You Need". The key takeaway is that attention mechanism must calculate relationships between every token (word or word fragment) in a sequence and every other token simultaneously. Memory requirements scale quadratically with sequence length, meaning doubling the context window quadruples the memory needed to process it.

During inference, an additional mechanism called the KV cache stores the context from all previous tokens, so the model does not recompute it from scratch with every new word generated. This KV cache grows linearly with each token produced throughout a generation.
Thus, modern large language model inference is almost never limited by the GPU's raw arithmetic speed. The processor's compute units sit idle for a vast majority of time and waits for data to arrive from memory. Adding more compute cores without adding proportionally more memory bandwidth produces essentially no improvement in AI throughput. As such, memory bandwidth becomes the binding constraint.



Before HBM, the standard solution for GPU memory was GDDR (Graphics Double Data Rate) memory, chips mounted on the printed circuit board surrounding the GPU and connected via a parallel electrical bus. A high-end GDDR6X configuration on a gaming GPU might deliver a 384-bit bus and approximately 1 TB/s of aggregate bandwidth. For AI training and inference at scale, this was insufficient by an order of magnitude.

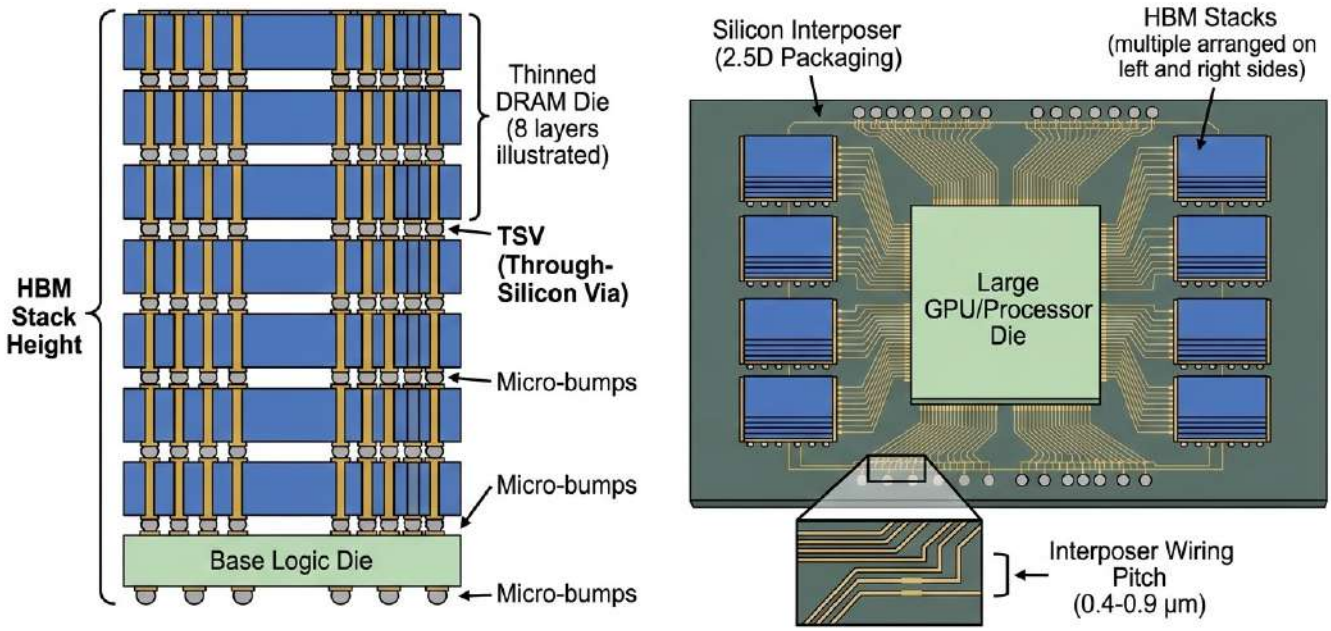
Standard Memory Bandwidth Comparison

Memory Type	Bus Width	Aggregate Bandwidth	Primary Use
DDR5 (CPU main memory)	64-bit	~90 GB/s	General compute
LPDDR5 (mobile)	128-bit	~68 GB/s	Smartphones
GDDR6X (gaming GPU)	384-bit	~1 TB/s	Consumer graphics
HBM2e	1024-bit x 4 stacks	~1.6 TB/s	Nvidia A100
HBM3	1024-bit x 6-8 stacks	~3.35 TB/s	Nvidia H100
HBM3e	1024-bit x 8 stacks	~4.8 TB/s	Nvidia H200/B200
HBM4 (expected)	2048-bit x 8 stacks	~10+ TB/s	Next-gen AI accelerators

High Bandwidth Memory solves the memory wall by fundamentally rethinking the physical relationship between memory and processor. Rather than placing memory chips on a circuit board some distance away from the GPU and connecting them with a narrow electrical bus, HBM stacks memory chips vertically and places the entire stack directly beside the processor on a shared silicon platform.

The Physical Stack

Individual DRAM dies are thinned from the standard 775 micrometer wafer thickness down to roughly 50 micrometers before stacking, producing dies thin enough to stack like floors in a building while fitting within standard server packaging height constraints. A typical HBM3e stack is 8 to 16 dies tall.



Through-Silicon Vias (TSVs)

The enabling technology of HBM is the Through-Silicon Via (TSV), a microscopic copper-filled cylindrical hole etched vertically through each silicon die. TSVs create direct electrical highways connecting every layer of the stack. A single HBM stack contains thousands of TSVs. This dense vertical interconnect produces the 1024-bit wide data bus, meaning 1024 parallel copper conductors running simultaneously through the stack. For context, a standard DDR5 memory module uses a 64-bit bus, making HBM's interface 16 times wider per stack. TSV fabrication requires precise deep reactive ion etching to drill each via hole, deposition of an insulating dielectric liner, and copper electroplating to fill the via without voids or seam defects.

The Base Logic Die

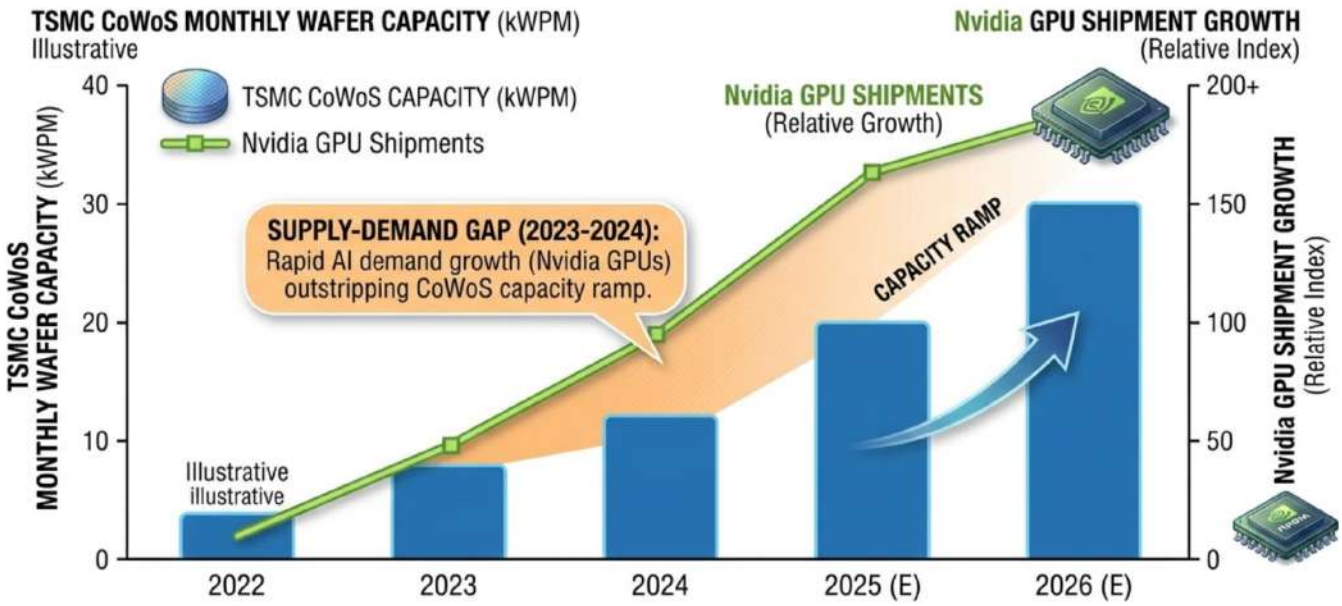
At the bottom of each HBM stack sits a logic die rather than a memory die. This base die acts as the control manager for the entire stack, handling all read and write requests, managing refresh operations for the DRAM layers above, performing error correction (ECC), and routing data between the TSV columns and the external interface. The sophistication of the base die has grown with each HBM generation and is becoming a primary competitive differentiator in HBM4, discussed later.

Micro-bumps

The DRAM dies in the stack are connected to each other and to the base die through micro-bumps, tiny solder joints at pitches of 40 to 55 micrometers. The technique used to form these connections is one of the most consequential manufacturing battlegrounds in HBM, discussed in detail in Section 5.

A HBM stack alone solves nothing. Its bandwidth advantage is only realized when it is connected to the processor at extremely high density through an interconnect platform far more capable than a conventional printed circuit board.

The solution is the silicon interposer, a large piece of silicon that serves as a shared mounting surface for both the GPU and the HBM stacks. Because the interposer is fabricated using the same lithography tools used to make chips, it can be patterned with extremely fine wiring at pitches of 0.4 to 0.9 micrometers, connecting tens of thousands of parallel signal lines between the GPU and each HBM stack. The HBM stacks sit only millimeters from the GPU die. The entire assembly, consisting of the GPU die, HBM stacks, interposer, and organic substrate, is called a 2.5D package because it is not full 3D stacking (dies are side by side rather than on top of each other) but far denser than conventional 2D circuit board assembly.

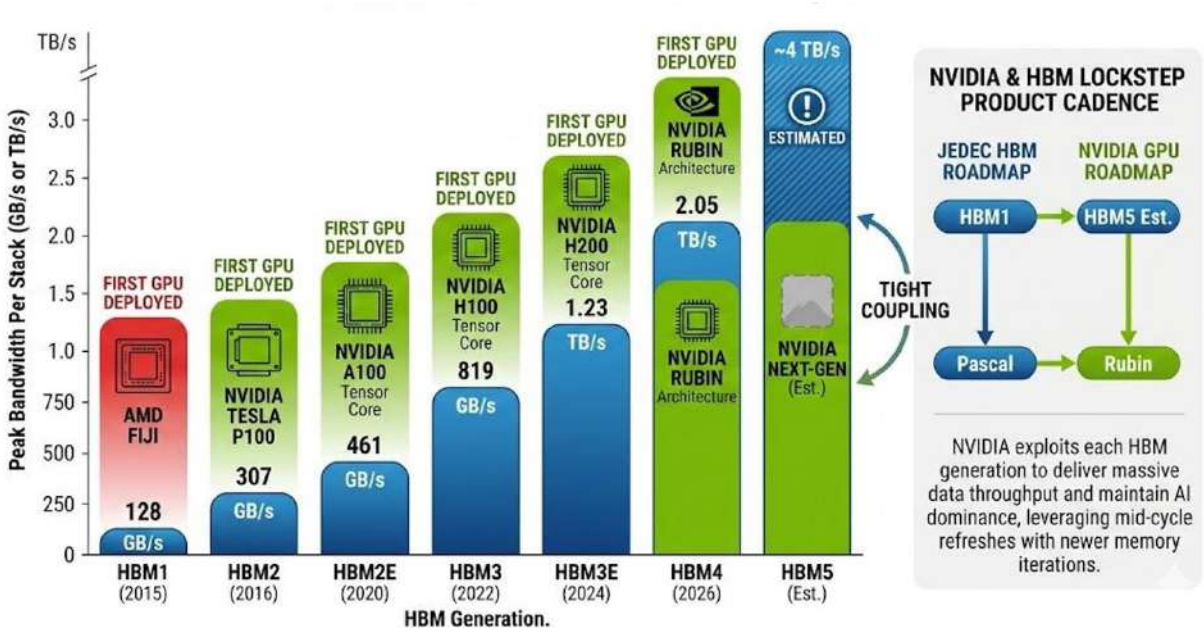


CoWos Implementation

TSMC's CoWoS (Chip-on-Wafer-on-Substrate) is the dominant implementation of this architecture. The Nvidia H100 and H200 both use CoWoS, with the GPU die at the center of the interposer and six to eight HBM stacks arranged around its perimeter. TSMC's CoWoS capacity has been one of the most significant physical bottlenecks in the AI infrastructure buildout. Demand from Nvidia, AMD, and custom silicon programs at the major hyperscalers substantially exceeded available CoWoS capacity throughout 2023 and into 2024, contributing directly to GPU supply constraints and extended lead times. As a result, expanding CoWoS capacity requires building dedicated cleanroom space and procuring large-format interposer lithography tools. It cannot be done quickly.

6. The HBM Technological Roadmap

HBM has gone through five commercial generations since 2015, with each generation delivering substantially higher bandwidth per stack through a combination of faster signaling speeds, wider interfaces, and taller die stacks.



The HBM upgrade cycle is tightly coupled to the Nvidia GPU product cycle. Nvidia follows a pattern where the primary upgrade between successive generations of the same chip architecture is often an increase in HBM capacity. The H200 carried 50% more HBM capacity than the H100 (141 GB vs 80 GB), achieved by upgrading from HBM3 to HBM3e and increasing stack height from 12-Hi to 16-Hi. The B200 to B300 transition is expected to be similarly driven by an HBM capacity increase, and the Rubin to Rubin-Ultra transition is projected to quadruple HBM content.

The HBM market is forecast to grow at approximately 50% CAGR from 2024 to 2028. From a financial structure perspective, HBM is meaningfully less cyclical than conventional DRAM because contract pricing is negotiated approximately one year in advance, providing revenue visibility that commodity DRAM spot pricing does not. The price per bit of HBM is currently 5 to 10 times higher than DDR5, reflecting both manufacturing complexity and sustained demand.

Generation	Bus Width	Max Stack Height	Peak Bandwidth per Stack	First Deployment
HBM1	1024-bit	4-Hi	128 GB/s	2015 (AMD Fiji)
HBM2	1024-bit	8-Hi	256 GB/s	2016 (Nvidia P100)
HBM2e	1024-bit	8-Hi	410 GB/s	2020 (Nvidia A100)
HBM3	1024-bit	12-Hi	819 GB/s	2023 (Nvidia H100)
HBM3e	1024-bit	16-Hi	1.2 TB/s	2024 (Nvidia H200)
HBM4	2048-bit	16-Hi	~2.4 TB/s	2025-2026 (est.)
HBM5	2048-bit+	20-Hi	TBD	2028-2029 (est.)

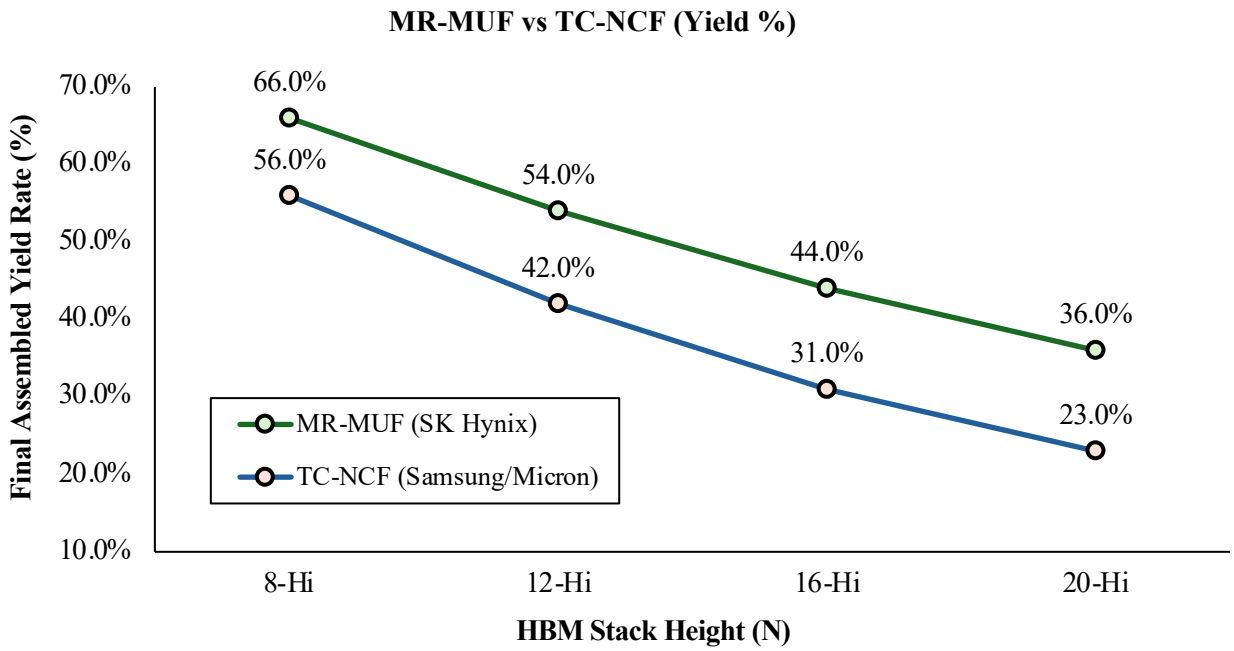
The hardest part of HBM manufacturing is the die stacking and bonding process. This is where SK Hynix built the advantage that has defined the current competitive landscape.

TC-NCF (Thermal Compression with Non-Conductive Film)

This is the bonding approach currently used by Samsung and Micron. Each DRAM die is pressed onto the stack at approximately 300 degrees Celsius with significant compressive force. A non-conductive film provides the insulating underfill between die layers. The high temperature creates risk of warpage, particularly as stack heights increase and thermal gradients across the stack become steeper. The compressive force creates risk of mechanical damage to individual dies. Alignment tolerances are tight and become harder to maintain reliably as more layers are added. TC-NCF yield rates are structurally challenging to scale with increasing stack height.

MR-MUF (Mass Reflow Molded Underfill)

SK Hynix's proprietary approach, developed and patented beginning in 2019. Rather than bonding each die individually under high temperature and force, MR-MUF first places all dies into the stack at room temperature using low-force micro-bump connections. The entire stack then undergoes a single mass reflow step, and a liquid epoxy underfill is simultaneously injected to encapsulate all gap spaces under molding. This approach carries several structural advantages over TC-NCF.



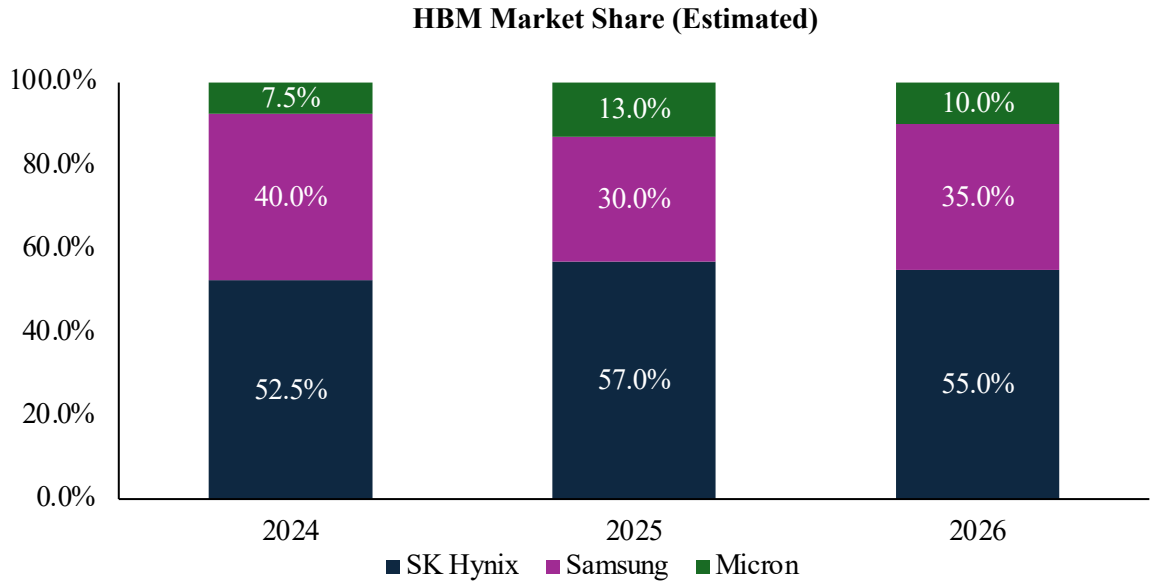
The Power of SK Hynix

The process operates at room temperature and low force, essentially eliminating warpage and mechanical damage risk. In addition, it uses approximately three times as many thermal bumps as TC-NCF, dramatically improving heat dissipation out of the stack. This is a critical advantage as stack heights increase from 8-Hi to 12-Hi to 16-Hi and thermal density rises accordingly.

The yield advantage of MR-MUF is approximately 20 percentage points higher average yield versus TC-NCF. This advantage compounds with stack height in a mathematically significant way. For example, if a single die has a 95% yield rate, the final assembled yield for an 8-Hi stack will be 66% (*0.95 to the 8th power*) and for a 16-Hi stack, that same die yield produces a final stack yield of only 44%. A 20-point yield improvement from MR-MUF therefore has an outsized effect on profitability at high stack heights.

SK Hynix's MR-MUF advantage is further reinforced by an exclusive supply arrangement. The epoxy molding compound enabling MR-MUF's uniform gap filling is supplied exclusively to SK Hynix by Namics Corporation, a Japanese specialty chemicals company. Samsung and Micron cannot replicate MR-MUF by copying the process steps alone as they lack access to the specific chemistry. Furthermore, SK Hynix holds extensive patents on the process architecture, enhancing its structural MOAT.

Three companies control the HBM market: SK Hynix (South Korea), Samsung (South Korea), and Micron (United States).



SK Hynix

SK Hynix has been the clear winner of the HBM era due to a decade of continuous, uninterrupted investment in a technology that most of its competitors abandoned. The company co-developed the first generation of HBM with AMD in 2013, introduced HBM2 in 2017, unveiled its MR-MUF bonding process in 2019, and delivered HBM3 in 2021 and HBM3e in 2023. This decade-plus of sustained investment created a technological learning curve and a supply chain ecosystem which includes the exclusive Namics epoxy arrangement that competitors cannot replicate on a short timeline.

SK Hynix serves as Nvidia's primary HBM supplier and supplied essentially all of the HBM3e used in the H100 and H200 ramp. Nvidia accounted for over 60% of total global HBM consumption in 2024 and is estimated to account for over 70% in 2025. As such, SK Hynix holds more than 60% of the total HBM market by revenue.

In Q1 2025, SK Hynix surpassed Samsung to become the largest DRAM company by revenue for the first time in decades. This milestone was achieved through the price premium of HBM which is valuated at 5 to 10 times the price per bit of commodity DDR5. Even a modest mix shift toward HBM will dramatically increases revenue per unit of manufacturing capacity.

Samsung Electronics

Samsung's HBM story over the past five years is one of the most consequential strategic miscalculations in recent semiconductor history. In 2019, Samsung dismantled its dedicated HBM development team, citing limited market potential for the technology. The team was dissolved approximately three years before ChatGPT-3 triggered the AI infrastructure boom and drove explosive demand for precisely the HBM3 and HBM3e products Samsung was no longer adequately staffed to develop. The decision severed the talent pipeline and stalled technical progress at exactly the wrong moment.

Samsung's difficulties are compounded by front-end process challenges. Since the 1a nanometer DRAM node, Samsung has experienced yield rates that trail SK Hynix. In the context of HBM manufacturing, the stacking process makes post-assembly repair of individual dies essentially impossible as any die with a latent defect that enters the stack becomes a permanent defect in the finished product.

Samsung's yield disadvantage has been a material factor in its failure to qualify for Nvidia's most demanding HBM programs on schedule. Samsung is now targeting 1c nanometer DRAM and a 4nm logic process for its HBM4 base die in an attempt to close the gap, but reported results at 1c nm have been unsatisfactory while SK Hynix has already achieved strong yields at the same node.

Source: *Statista*

Micron

Micron is the most positively developing player in the HBM landscape. As the only US-headquartered HBM supplier, Micron benefits from geopolitical tailwinds in the form of US government interest in maintaining a domestic source for a strategically critical component.

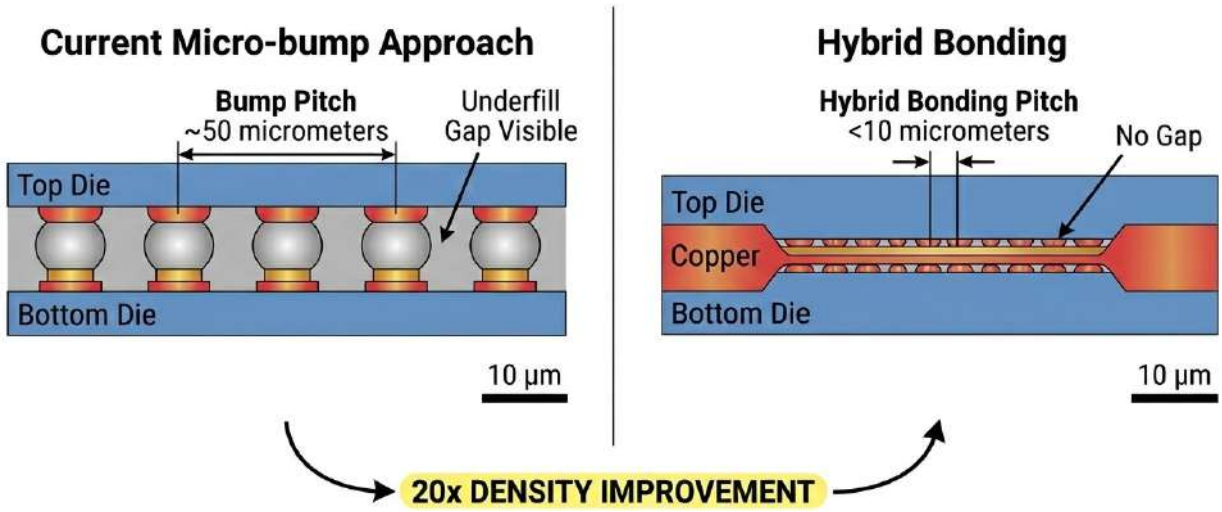
Micron has qualified its HBM3e product with Nvidia and is actively ramping production. The company has historically excelled at capital allocation where it has acquired capacity cheaply during downturns and managing its cost structure with discipline. In 2025, Micron broke ground on a $7 billion HBM advanced packaging facility in Singapore, targeting mass production readiness in 2027. Micron is unlikely to displace SK Hynix at the technology frontier in the near term, but it is a credible and growing second source that customers value for supply chain diversification.

The current bonding technology landscape, SK Hynix's MR-MUF versus Samsung and Micron's TC-NCF, will be disrupted when the industry transitions to **hybrid bonding** for HBM5 and beyond

Why Hybrid Bonding?

Hybrid bonding eliminates micro-bumps entirely. Instead, copper pads on the face of each die are polished to atomic flatness, chemically activated, precisely aligned, and placed in direct contact. The copper pads fuse through solid-state diffusion during a low-temperature annealing step below 400 degrees Celsius. The result is a direct copper-to-copper bond with no solder, no underfill material, and no physical gap between dies.

The performance advantages are substantial. Hybrid bonding supports interconnect pitches below 10 micrometers, compared to 40 to 55 micrometers for current micro-bumps which results in roughly 20 times more connections per unit area. Eliminating the bump gap and underfill material also reduces the total stack height, allowing more dies to be stacked within the 720 to 775 micrometer height envelope specified by JEDEC standards. Thermal conductivity improves because there is no low-conductivity underfill between dies impeding heat flow out of the stack.



The Yield Advantage Is as Important as the Performance Advantage

Hybrid bonding does not only improve bandwidth density, it also eliminates one of the primary yield failure modes in current HBM manufacturing. Micro-bumps fail through misalignment during placement, voiding during reflow, and solder bridging when adjacent bumps are at fine pitches. These failure modes become harder to control at every additional die layer and every generation of pitch reduction. By replacing solder joints with a direct copper diffusion bond, it removes this entire category of failure. For HBM5 at 20-Hi stack heights, this yield benefit may matter as much as the interconnect density improvement.

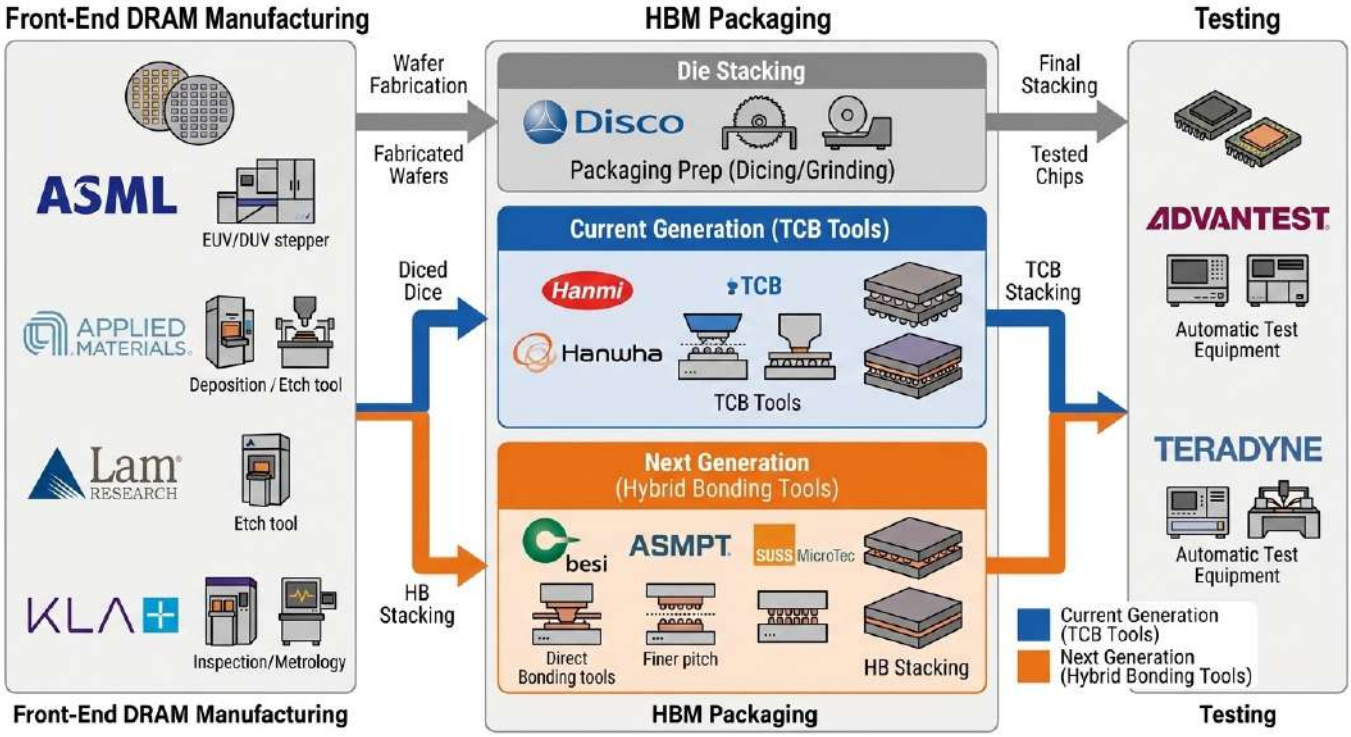
The Current Bottleneck

The reason hybrid bonding is not yet in production is cost. A hybrid bonding tool costs approximately $3 million, roughly double a thermal compression bonder. Beyond the bonder itself, hybrid bonding requires additional CMP (chemical mechanical planarization) steps to achieve atomic-scale surface smoothness across the full die face where any surface variation larger than a few nanometers prevents successful copper fusion. The entire process must also meet front-end fabrication cleanroom standards (Class 10 or better) rather than the less stringent back-end packaging standards used for conventional HBM. As such, this requires different facility specifications, contamination controls, and process discipline than what today's packaging operations have historically maintained.

The JEDEC height standard relaxation for HBM4, from 720 to 775 micrometers, was specifically designed to allow MR-MUF and TC-NCF to remain viable for one additional generation, deliberately delaying the forced industry transition to hybrid bonding until HBM5.

The current industry consensus places hybrid bonding mass adoption at HBM5 in 2028 to 2029. Micron's $7 billion Singapore packaging facility targets mass production in 2027 and SK Hynix's $3.87 billion Indiana advanced packaging plant targets production start in 2028, both timelines consistent with HBM5 ramp preparation.

For investors seeking exposure to HBM growth without directly investing in the memory companies themselves, the equipment supply chain offers several well-positioned opportunities.



Thermal Compression Bonders

Hanmi Semiconductor and Hanwha Precision Machinery are the dominant Korean suppliers of TCB tools for HBM packaging. Hanmi has been SK Hynix's primary TCB supplier and is also the main supplier to Micron. ASMPT and Kulicke & Soffa (KLIC) are active in TCB but historically stronger in logic packaging, where higher precision requirements justify their premium pricing.

Next Generation: Hybrid Bonders

Besi is the current leader in chip-to-wafer hybrid bonders for HBM applications, with its current generation achieving 100 nm placement accuracy and a second generation targeting 50 nm accuracy and sub-4 micrometer bump pitch. Besi has partnered with Applied Materials, which contributes CMP, etch, deposition, and cleaning process expertise to the combined packaging solution.

ASMPT has partnered with EV Group, the established leader in wafer-to-wafer permanent bonding. Korean TCB incumbents Hanmi and Hanwha are developing hybrid bonding extensions of their existing platforms but currently lack the precision and low-temperature process control required at the leading edge.

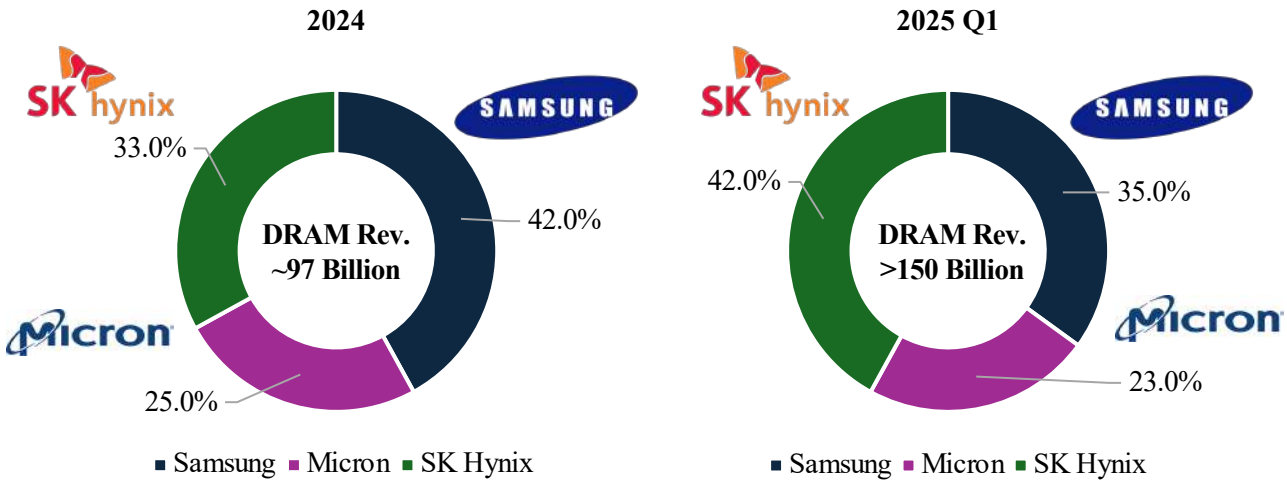
Supporting Equipment Categories

Applied Materials and Lam Research dominate TSV etch, deposition, and CMP processes. KLA and Camtek lead in metrology and inspection tools for HBM stacks. Disco is the dominant supplier of wafer grinding equipment used to thin each DRAM die before stacking. SUSS MicroTec and Tokyo Electron supply temporary bonding and debonding systems used during the thinning and handling process. Advantest and Teradyne supply the automated test equipment (ATE) used for HBM testing.

The TCB-to-hybrid bonding transition is the **most consequential equipment market shift anticipated** in HBM manufacturing over the next five years. The outcome of the Besi-versus-ASMPT competition for hybrid bonding leadership will significantly influence which companies capture the capital expenditure flowing from the HBM5 generation build-out from 2027 onward.

DRAM: A Well-Consolidated Oligopoly

Samsung, SK Hynix, and Micron control approximately 95% of global DRAM revenue. The final major consolidation was Micron's acquisition of Elpida, Japan's last DRAM company, in 2013. Since that consolidation, the DRAM industry's return on invested capital has improved substantially. The three survivors now typically manage production cuts and capacity discipline during downturns rather than flooding the market with supply, producing a more rational pricing environment than the fragmented era that preceded it.



HBM has improved DRAM economics at the margin in two ways. First, HBM contract pricing is negotiated approximately one year in advance, providing revenue visibility that commodity DRAM spot pricing does not offer. Second, the 5 to 10 times price premium per bit relative to DDR5 means that converting manufacturing capacity from commodity DRAM to HBM production dramatically increases revenue yield from the same physical fab space.

NAND: Structurally Weaker

The NAND flash market has five active competitors — Samsung, Kioxia, SanDisk (now spun off from Western Digital), SK Hynix, and Micron. This makes it less consolidated than DRAM and therefore less profitable through the cycle.

Kioxia and SanDisk operate through a unique joint venture where Kioxia controls 100% of fab capacity and splits wafer output 50/50 with SanDisk. Both companies lack the profitable DRAM businesses that Samsung, SK Hynix, and Micron use to absorb losses during NAND downturns, making them structurally the most vulnerable players in the market.

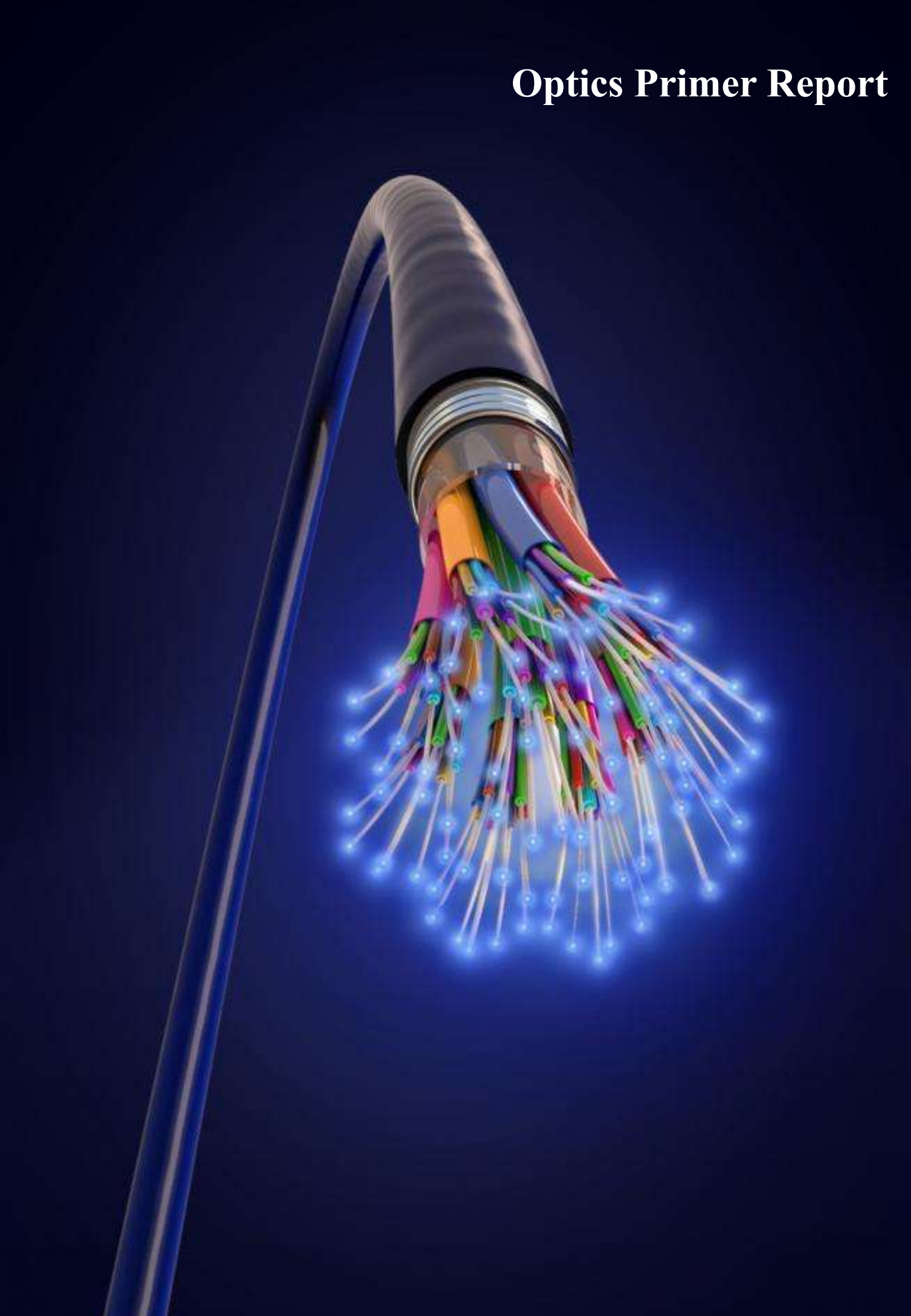
SK Hynix's 2021 acquisition of Intel's NAND business (now operating as Solidigm) established it as the second-largest NAND company and further reduced the number of independent players. Furthermore, NAND consolidation is widely expected and would structurally improve margins for surviving participants.

AI as a Structural Demand Tailwind

DRAM demand is undergoing a structural shift toward data center applications driven by AI. The data center segment, which was previously roughly equal in size to the PC and mobile segments, is projected to account for approximately two thirds of total DRAM bit demand by 2029. HBM is the highest-value segment within that data center demand, growing at approximately 50% CAGR from 2024 to 2028.

This growth is highly visible, contracted in advance, and tied to capital expenditure commitments by hyperscalers and AI accelerator companies that are among the most financially robust customers in the global technology economy.

Optics Primer Report



Optics Key Terms

Terms	Definitions
Analog Semiconductors	electronic components that process, amplify, or convert continuous **real-world signals,** such as sound, light, temperature, and pressure, into **electrical signals.** Unlike digital chips that use binary (0s and 1s), these components manage voltage and current that vary continuously over time.
Application Specific Integrated Circuits (ASICs)	a custom-designed microchip optimised for a single, specialised task rather than general-purpose use. ASICs provide superior performance, high power efficiency, and smaller physical footprints by implementing dedicated hardware for specific workloads, making them ideal for high-volume, performance-driven applications like cryptocurrency mining, AI processing.
Attenuation	the reduction in strength (amplitude or power) of a signal or wave as it travels through the material. It is caused by scattering, absorption, or defects within the crystal lattice. In high-frequency acoustic waves in silicon or germanium, this is caused by phonon-phonon interactions (Akhieser losses), which are heavily dependent on temp.
Bend Sensitivity	The degree to which a semiconductor material or device's electrical conductivity or structure changes due to mechanical bending. For flexible and printed electronics, low bend sensitivity is preferred, meaning the device maintains stable performance (e.g., low resistance change) even when bent. Bending induces tensile or compressive strain, which alters the band gap and electron mobility within the material.
Catastrophic Optic Damage	an irreversible, power-limiting failure mechanism in semiconductor laser diodes, characterised by sudden, localised melting of the semiconductor material at the facet, resulting in immediate device death. It is driven by thermal runaway due to high optical absorption at the facet.
Chromatic Dispersion	a fundamental phenomenon where the phase velocity and group velocity of light propagating through the medium are dependent on the optical frequency or wavelength. Because light sources (like lasers) have a finite spectral width, different spectral components within a single pulse travel at different velocities. As a result, the light pulse spreads out or broadens in time as it travels.
Coarse Wavelength Diversion Multiplexing (CWDM)	Transceivers and multiplexing components that transmit multiple data signals over a single fiber strand using **18 distinct,** wide-spaced wavelengths (1270–1610 nm). **Spaced 20 nm apart,** this enables lower-cost, uncooled lasers for short-to-medium range networks (up to 80 km), optimised for metro access and enterprise connectivity.

Optics Key Terms

Terms	Definitions
Complementary Metal-Oxide Semiconductor (CMOS) Fabrication	The multi-step manufacturing process used to create integrated circuits (ICs) by integrating both N-type (NMOS) and P-type (PMOS) field-effect transistors on a single silicon substrate. It is the dominant technology for Very Large Scale Integration (VLSI) chips, such as microprocessors, microcontrollers, and memory (RAM/ROM), due to its low power dissipation, high noise immunity, and high packing density.
Continuous Wavelength (CW) Lasers	A device that emits a steady, uninterrupted, and constant beam of light, with a power output that remains stable over time. CW lasers are lower in complexity and cost than EMLs. Unlike pulsed lasers that fire in short bursts, CW lasers are continuously pumped, providing continuous energy, and are widely used in applications needing constant power, such as welding, cutting, printing, and medical procedures. They are supplied primarily to silicon photonics customers who use a separate modulator on their own silicon chip to encode data onto the light.
Co-Package Optics (CPO)	An advanced technology that integrates optical components (lasers/transceivers) directly onto the same package as high-speed switching silicon (ASICs) or processors, rather than using separate pluggable modules. This proximity drastically reduces power consumption (by 73% per optical link, compared to DSPs) and increases bandwidth density for data centers and AI. CPOs require precision fiber coupling and thermal isolation that conventional electronics manufacturing is currently not designed for. The pluggable market allows for buyers to pit multiple transceiver manufacturers against each other, while CPOs consolidate optical functions into the switch package, reducing procurement flexibility. Once CPO is in mainstream production, optical transceivers will no longer be required, but DSPs will still be needed to be mounted onto the chips. A standard **144-port CPO-enabled switch** requires approximately 18 ELS modules, so ELS is essential in the CPO supply chain.
Dense Wavelength Division Multiplexing (DWDM)	A high-capacity fiber-optic transmission technique that multiplexes multiple data signals onto a single fiber by using different wavelengths (colours) of laser light. It increases bandwidth by supporting 40, 80, or over 160 channels, enabling long-haul, high-speed transmission.
Digital Signal Processor (DSP)	A specialised, high-speed microprocessor designed to manipulate analog signals, such as audio, video, or sensor data, in real-time after they have been converted into digital format. DSPs use optimised architectures to quickly perform mathematical calculations like compression, filtering, and noise cancellation. 800 DSP-based transceivers consume **16-17 watts** and a DSP chip consumes roughly ½ of the total, DSP power consumption is a large fraction of total facility power drawn.

Optics Key Terms

Terms	Definitions
Distributed Feedback Laser (DFB)	A semiconductor laser diode that uses a periodic grating structure integrated into its active region to provide optical feedback, ensuring single-frequency operation with a very narrow spectral linewidth. Unlike standard lasers, DFB lasers **do not rely** on end mirrors, but rather on Bragg scattering from this grating, resulting in high wavelength stability for telecommunications and sensing. It is essential for **DWDM applications** where many closely spaced wavelengths must be packed onto a single fiber without interference, used for long-haul systems.
Electro-Absorption Modulator (EAM)	A semiconductor device that controls the intensity of a laser beam by altering its absorption coefficient through an applied voltage, often using the Quantum-Confined Stark Effect (QCSE). Key features include a small footprint, low power consumption, and high-speed modulation for optical fiber comms.
Electro-Absorption Modulated Lasers (EML)	A high-speed semiconductor laser device that monolithically integrates a Continuous-Wave (CW) laser source (typically a Distributed Feedback or DFB laser) with an Electro-Absorption Modulator (EAM) on a single chip. It works by generating a constant light beam that is then intensity-modulated by the EAM section via an applied voltage, offering high-speed, long-distance performance with low dispersion. 200G EML commands ~2x the price of a 100G EML, but occupies the same wafer footprint, making it favourable for EML manufacturers. **EMLs** remain dominant in WDM apps, while **SiPho** remains in parallel fiber apps. The wavelength of an EML laser is determined by thickness and composition of specific quantum well layers within the device stack. Each layer must be grown with atom-level uniformity across the entire wafer.
Epitaxy	The substrate acts as a seed crystal, forcing the deposited film to adopt the same crystallographic structure and orientation as the underlying substrate. Epitaxy requires each deposited atom to lock into a specific position extending the perfect crystal structure of the layer beneath it. Each atom must fall exactly into the correct grid position, or the crystal develops a defect that propagates upward through subsequent layers and destroys the device performance.
Epitaxy Growth Process	A precise, high-temperature deposition process where a thin layer of crystalline material (the epitaxial layer or "epi-layer") is grown on a crystalline base (the substrate). The key characteristic is that the deposited layer replicates the crystallographic structure, orientation, and lattice registry of the substrate, acting as an extension of the underlying crystal.

Optics Key Terms

Terms	Definitions
Erbium-Doped Fiber Amplifiers (EDFAs)	An optical repeater device that amplifies weak optical signals directly in fiber-optic communication systems, typically in the **1530-1565 nm C-band,** without requiring O-E-O (optical-electrical-optical) conversion. It works by using a pump laser (980 nm or 1480 nm) to excite erbium ions (Er3+) doped into the fiber core. **DWDM** requires precisely temperature-controlled lasers and sophisticated optical amplifiers to compensate for accumulated signal loss.
External Laser Source (ELS)	A disaggregated, pluggable optical module that provides Continuous Wave (CW) light, separate from the primary silicon photonics chips in CPO architectures. By moving heat-generating laser sources out of the immediate switch or ASIC package into a manageable, field-replaceable module, ELS addresses thermal challenges and simplifies maintenance in high-bandwidth data center applications. A separate, air-cooled module is mounted onto the GPU or switch package, but the laser itself remains remote. A **single ELS module** must supply light to multiple optical engines simultaneously.
Indium Phosphide (InP)	A semiconductor laser diode that utilises Indium Phosphide as its base material to generate, amplify, and emit coherent, high-intensity light. These lasers are central to modern optoelectronics due to their ability to emit in the infrared range, making them essential for high-speed fiber-optic telecommunications, especially in the **1550 nm wavelength** window, supporting the lowest attenuation wavelengths for fiber-optic communication. They offer high electron mobility and high-frequency switching, essential for transmitting large volumes of data. The laser's wavelength is determined by the thickness of specific crystal layers grown atom by atom; any deviation from specifications will ruin the device.
Laser-Induced Damage Threshold (LIDT)	The maximum laser intensity or fluence (energy per unit area) an optical component can withstand without suffering permanent, irreversible damage. According to ISO 21254, it is the highest incident radiation level where the extrapolated probability of damage is zero. It is a critical parameter for defining the power handling limits of optics.
Linear Pluggable Optics (LPO)	A high-speed, low-power optical transceiver technology for data centers that removes the DSP from the transceiver module. By using linear analog components rather than DSP-driven retiming, LPO reduces power usage cost, and latency, making it ideal for short-reach AI and machine learning networks. LPO saves substantial power and lowers latency, but can only go up to 2km.

Optics Key Terms

Terms	Definitions
Metal Organic Chemical Vapour Deposition (MOCVD) Reactors	Specialised chemical vapour deposition systems used for growing high-purity crystalline semiconductor thin films and multilayer structures, primarily for compound semiconductors (III-V and II-VI materials). These reactors operate by introducing precise amounts of metal-organic compounds (precursors) and hydrides, carried by a gas, into a heated chamber, where they decompose on a heated substrate to deposit atoms in a controlled epitaxy process. Aixtron SE (German company) holds **90% of market share** in MOCVD reactors for optoelectronics. Its proprietary **Planetary Reactor Design** has a gas flow dynamic that achieves near-perfect layer uniformity across the entire wafer.
Multi-Mode Fiber (MMF)	A type of optical fiber primarily used for communication over short distances, such as within buildings or on campus networks (LANs). Its definition hinges on its ability to support multiple, simultaneous light paths (modes) due to a relatively large core diameter. The large core allows multiple light modes (rays) to propagate through the fiber simultaneously, usually at 850 or 1300nm wavelengths.
Optical Circuit Switches (OCS)	A high-performance network device that establishes direct, physical, and dedicated optical paths between endpoints **without converting** the signal into electricity. Unlike traditional electronic packet switches that parse, buffer, and route data at every hop, OCS creates a "transparent" pipe that allows light (data) to travel from **input** to **output** with near-zero latency, negligible signal loss, and no rate dependency. OCS is increasingly used as a foundational technology in hyperscale data centers, AI training clusters, and high-performance computing (HPC) for its ability to handle immense bandwidth while dramatically reducing power consumption (often by **40-70%** vs traditional switches).
Optical Micro-Electric Mechanical Systems (MEMS)	A specialised class of MEMS that integrate electrical, mechanical, and optical components on a micrometer scale. They function by using tiny mechanical actuators, such as mirrors, gratings, or shutters, to sense, manipulate, or modulate optical signals (light). These devices are fabricated using batch micromachining techniques, similar to semiconductor manufacturing (photolithography, etching, deposition), commonly using materials like silicon and silicon dioxide (SiO2).
Optical Modulator	A device that changes the properties of a light beam, such as amplitude, phase, polarisation, or direction, using an external signal, typically electrical. A key component in fiber-optic communications to encode high-speed data, as well as in laser systems for switching and pulse shaping. To turn a continuous-wave laser into an information-carrying signal, enabling, for example, high-speed telecommunications. Modulators often use materials that change their optical properties when an electrical field is applied, such as lithium niobate $LiNbO3$).

Optics Key Terms

Terms	Definitions
Optical Semiconductors	A device that combines semiconductor technology with optics to convert electrical energy into light (emission) or light into electrical energy (detection). These components, often made from compound materials like gallium arsenide (GaAs), manipulate photons and electrons to operate, including LEDs, lasers, photo sensors, and solar cells.
Photonics	The science and technology of generating, controlling, and detecting photons (particles of light), essentially acting as the optical counterpart to electronics. It utilises light, spanning ultraviolet to infrared, for applications like fiber-optic communication, laser processing, medical imaging, and consumer electronics, offering immense speed and precision.
Scale Across	Extending the high-performance network fabric across multiple, geographically separated data centers or campuses to treat them as a single logical supercluster. **Optics Focus:** Long-haul connectivity (metro or regional distances, >10km to hundreds of km) requiring ultra-high bandwidth, high reliability, and low latency to manage data movement between sites. **Technology:** Coherent Optical Transceivers (e.g., ZR/ZR+) and Data Center Interconnect (DCI), often leveraging Optical Circuit Switches (OCS). This layer allows for capacity pooling and distributed training.
Scale Out	Networking multiple scale-up units (nodes/racks) across a data center floor to form a massive compute cluster, often using a Spine-Leaf (Clos) architecture. **Optics Focus:** High scalability and medium-to-long reach (tens to hundreds of meters). It enables "any-to-any" connectivity between thousands of GPUs. **Technology:** Primarily 800G/1.6T Pluggable Optics (OSFP/QSFP) utilizing DSPs (Digital Signal Processors) for reliable error correction (FEC) over long distances (10km+), as LPO/NPO are less stable over these distances.
Scale Up	Tightly coupling accelerators (GPUs/TPUs) within a single server, rack, or adjacent racks to make them behave as one single, large processor with a unified memory pool. **Optics Focus:** Short-range, ultra-low latency, and extremely high bandwidth. It connects GPUs to NVLink Switches to form a "supernode" or "GPU island". **Technology:** Historically copper, but shifting to Linear Direct-Drive Optics (LPO) and Near-Packaged Optics (NPO) to extend reach slightly while maintaining low latency. Future adoption of Co-Packaged Optics (CPO) is expected by 2026 to integrate optics inside the GPU package.

Optics Key Terms

Terms	Definitions
Silicon Photonics (SiPh)	A technology that uses silicon as an optical medium to generate, process, and transmit data using light (photons) rather than electricity, fabricated on specialised silicon chips using standard **CMOS fabrication processes.** It combines high-index waveguides for tight light confinement with microelectronics, enabling compact, low-power, and high-speed optical devices. SiPho yields thousands of chips per run using the **same** defect control discipline as semiconductor manufacturing.
SiPh Transceivers	The next-generation optical modules that leverage silicon-based materials and CMOS manufacturing techniques to integrate photonic components, such as modulators, detectors, and waveguides, onto a single, compact chip. These transceivers convert electrical signals to optical signals (and vice versa) for high-speed data transmission, with key components fabricated directly on silicon substrates to achieve lower costs, smaller sizes, and higher power efficiency, particularly for 400G, 800G, and 1.6T data rates. SiPho transceivers require half the number of lasers as an EML product and benefit from EOS of silicon wafer production. However, SiPho cannot efficiently generate laser light because silicon has an indirect electronic bandgap that does not support efficient photon emission.
Single-Mode Fibers	An optical fiber with a very small core diameter, typically **8–10 micrometers**, designed to carry only a single ray or transverse mode of light, which is the fundamental mode (HE11). By restricting light to a single propagation path, single-mode fibers eliminate modal dispersion, the distortion caused by multiple paths, which allows for significantly higher bandwidth and much longer transmission distances compared to multimode fibers.
Spine-Leaf Architecture	A 2-tier data center network topology consisting of leaf switches (access layer) and spine switches (backbone layer), designed to provide high-bandwidth, low-latency, and highly scalable connectivity, typically relying on optical fiber for inter-switch communication. In this architecture, every leaf switch is connected to every spine switch, creating a non-blocking, full-mesh fabric.
Transceivers	It serves as the physical interface between networking equipment (switches, routers) and optical fiber cables, converting "electronic language" into high-speed light impulses, typically using lasers. And a **receiver** (a photodetector converts incoming light back to electric signals). Data from the GPU must travel as an electrical signal across the PCB to reach the transceiver. Since signals degrade before optical conversion, every transceiver requires a **Digital signal processor (DSP)** for error correction and cleanup.

Optics Key Terms

Terms	Definitions
Ultra-High Power Laser (UHP)	specialised optical components designed to manage, direct, and manipulate laser beams with very high power densities (typically >10 kW Continuous Wave or high-energy pulsed systems) without suffering from structural failure, damage, or significant performance degradation. These optics are essential for high-power laser material processing, defense, and research applications, often employing advanced Ion Beam Sputtered (IBS) coatings to maximise the Laser-Induced Damage Threshold (LIDT). UHP lasers require dies **2-3x larger** than standard lasers, reducing chips per wafer. They require more complex and thicker epitaxial stacks that increase reactor time per wafer.
Vertical Cavity Surface Emitting Lasers (VCSEL)	a semiconductor laser diode that emits light vertically from the top surface of a chip, rather than from the edge like traditional lasers. It uses high-reflectivity mirrors (Distributed Bragg Reflectors) to form a short vertical cavity, enabling high-speed data communication, 3D sensing, facial recognition, and low-power, cost-effective manufacturing. VCSELs can be manufactured in arrays at very low cost, and are hence used as the dominant light source for intra-building data centre connections.
Wavelength Multiplexing (WDM)	a passive fiber-optic technology that increases data capacity by **multiplexing** multiple optical signals, each using a **distinct light wavelength** (colour), onto a single fiber. It works by using a multiplexer at the transmitter to combine signals and a demultiplexer at the receiver to separate them. WDMs allow multiple independent data streams to share a single optical fibre simultaneously, each with a different wavelength of light. At the **transmit end,** a multiplexer combines multiple laser signals at different wavelengths onto 1 fibre. At the **receiver end,** a demultiplexer separates them back out.

Electrical signals travelling through copper wires suffer from several fundamental physical limitations that become crippling at high data rates over distance.

Attenuation

The signal weakens as it travels. At 400 Gbps and above, copper cables can only carry signals reliably over one to two meters before requiring expensive active signal conditioning. Think of it as shouting across a dinner table versus shouting across three football fields. Effective at short range, physically impossible at long range no matter how hard you try. As engineers push more data through copper at higher speeds, the electrons generate heat from resistance and the signal collapses within meters. Compensating with thicker, heavier cables only extends this slightly.

Bandwidth-Distance Product

The maximum data rate achievable over copper falls sharply with distance. At the 800 Gbps and 1.6 Tbps speeds demanded by next-generation AI clusters, copper physically cannot bridge the 50, 100, or 500 meters separating server racks in a warehouse-scale data center.

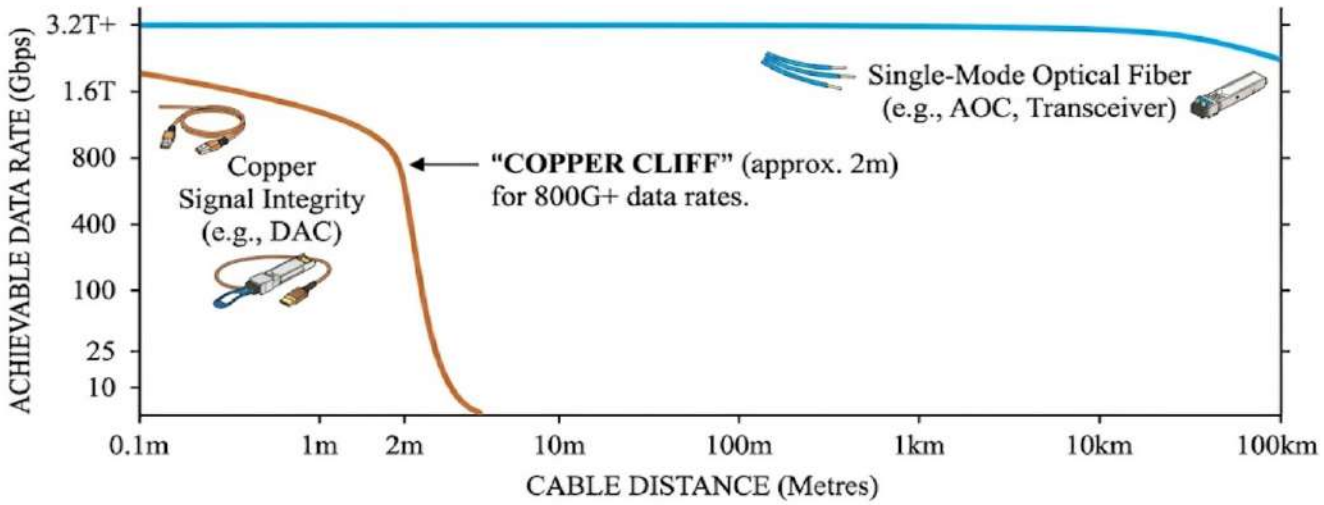
Electromagnetic Interference

Copper cables both emit and absorb electromagnetic noise, requiring shielding that adds weight and cost.

Power Consumption

Driving high-frequency electrical signals through copper at high data rates requires significant power in the transmit and receive circuitry, a problem that compounds severely at scale when a single cluster contains millions of connections.

Copper vs Optical Fiber – Signal Integrity against Distance



Light travelling through optical fiber suffers almost none of these limitations. Photons have no mass and experience no resistance. A single fiber strand can carry many independent wavelengths of light simultaneously through wavelength division multiplexing, multiplying capacity without adding physical strands. Power consumed per bit transmitted is dramatically lower than copper.

Modern AI models require stitching together 10,000, 20,000, or 100,000 GPUs into a single coherent supercluster. These chips must communicate constantly at extraordinary speeds. Since that many GPUs cannot fit in a single server rack, they are spread across a warehouse-scale hall. Optics is the only technology that can connect them reliably at the speeds AI demands.

Source: https://www.inneos.com/the-copper-cliff/

Single-Mode Fiber (SMF)

Has a very narrow core of 8 to 10 micro-meters that constrains light to travel in a single propagation mode, essentially a straight line along the fiber axis. This eliminates modal dispersion — the blurring of pulses caused by different light rays taking slightly different paths — enabling reach of tens to hundreds of kilometer with very high bandwidth. Single-mode fiber is used for all long-distance telecommunications, data center interconnects between buildings, and increasingly for the highest-speed intra-data-center connections.

Multi-Mode Fiber (MMF)

Has a wider core of 50 to 62.5 micro-meters that allows multiple propagation modes simultaneously. It is cheaper to manufacture and easier to connect with lower-cost light source. However, it is limited in reach to typically 100 to 500 meters before dispersion degrades signal quality. It is used for short intra-data-center connections where cost efficiency matters more than reach.

Key metrics to Understand:

Attenuation

As mentioned previously, attenuation measures how much signal is lost per kilometer, expressed in decibels per kilometer. Standard SMF achieves approximately 0.2 dB/km.

Chromatic Dispersion

Describes how different wavelengths of light travel at slightly different speeds through fiber, spreading pulses and limiting bandwidth over distance.

Bend sensitivity

Practically important for data center deployments. Traditional fiber loses signal if bent too tightly, and bend-insensitive formulations are critical for installations where cables route through tight angles in cable trays.

The fiber cable and connectivity supply chain is dominated by a small number of large manufacturers. Corning (United States) is the world's largest optical fiber producer and holds a commanding position in SMF manufacturing. Prysmian (Italy) and Sumitomo Electric (Japan) are the leading competitors. Corning's technology leadership in ultra-low-loss fiber formulations, particularly its TXF and SMF-28 Ultra product lines optimized for hyperscale data center deployments, gives it a materials differentiation that generic fiber suppliers cannot easily replicate. The data center fiber buildout for AI infrastructure is a direct revenue driver for Corning, which has guided to significant hyperscaler-driven demand growth.

Before diving into transceivers and photonics, it is useful to understand how data center networking is structured, because different parts of the network have different requirements and different technologies serve them. This architecture is the map that explains which optical technologies go where and why.

Scale up

Refers to the lowest level of networking. It connects individual GPU chips together inside a single rack or tightly coupled cluster, so they act as one giant unified processor. Scale up has historically been done entirely with copper.

The dominant protocol is Nvidia's NVLink. A single NVLink-connected rack today packs 72 GPUs, 36 CPUs, and 18 NVSwitch chips interconnected by over 5,000 copper cables totaling approximately two miles of wiring. Scale up handles extraordinary bandwidth because signals travel very short distances. The architecture forces all GPUs to be physically proximate which is a fundamental constraint that co-packaged optics will dissolve.

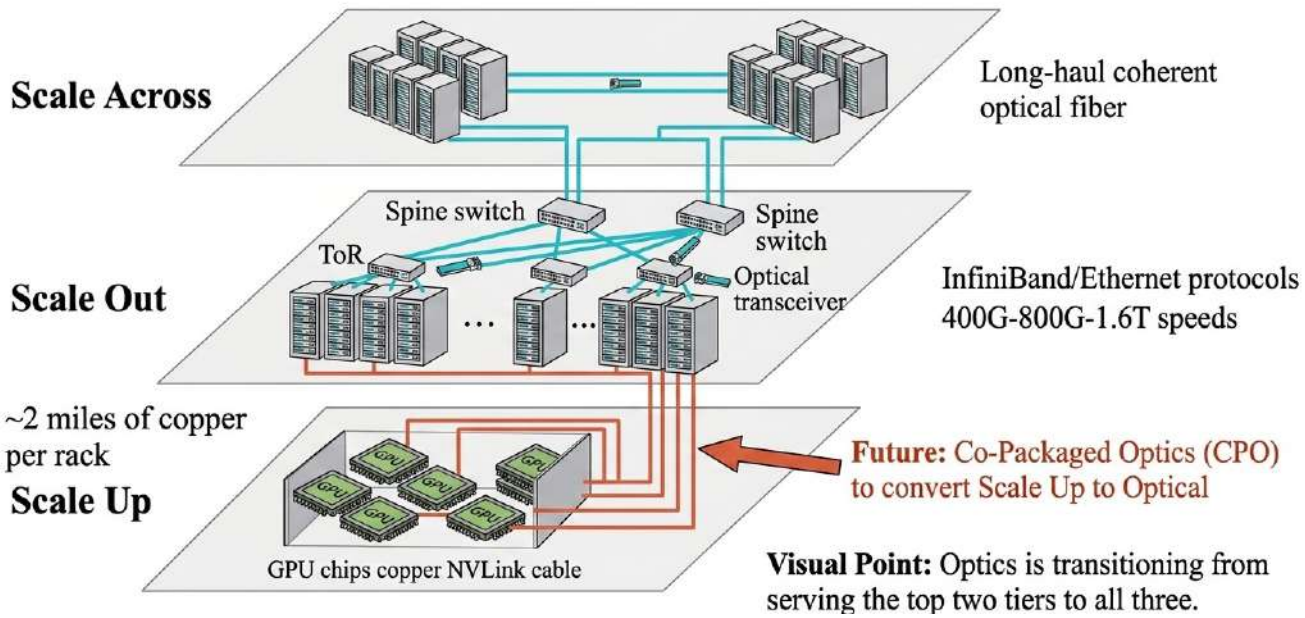
Scale out

Refers to connecting many scale-up pods together through a central switching fabric. Signals must travel much farther, so they require significant signal conditioning at both transmit and receive ends. Scale out has already adopted optical fiber extensively through InfiniBand and Ethernet protocols.

Scale across

Refers to connecting separate data center sites when a single facility reaches its power or space limits. This requires long-haul coherent optical networking at its most demanding and is a significant growth driver for coherent optical systems suppliers.

Data Center Networking Architecture Diagram

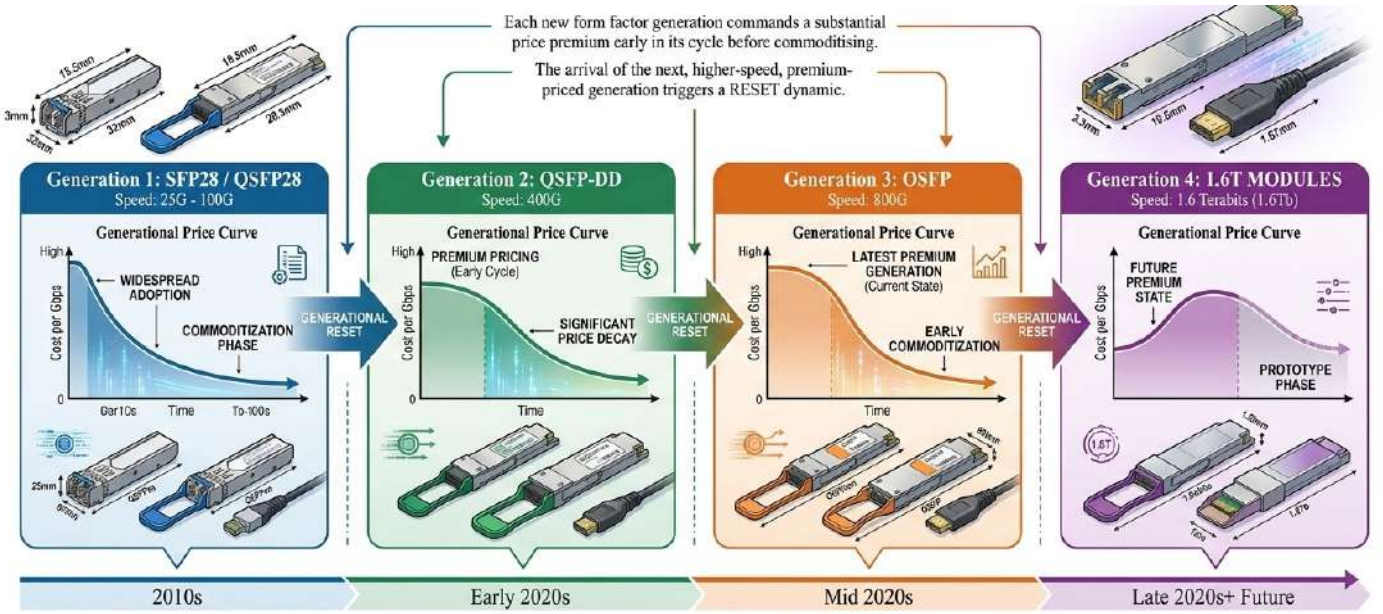


Scale out has been the domain of optical transceivers for years. Scale up is the now the new frontier, and it represents a far larger total addressable market because there are far more chip-to-chip connections inside a cluster than between clusters. The ratio of scale-up to scale-out optical content in a next-generation Nvidia rack is estimated at approximately 4.5 to 1.

At every point where data enters or exits a fiber, a transceiver converts electrical signals to optical and back. Transceivers are the critical interface components at the edge of every switch, router, and server network interface card.

A transceiver contains a transmitter (a laser diode or VCSEL that converts electrical signals to light pulses) and a receiver (a photodetector that converts incoming light back to electrical signals). The form factor, speed, and reach have evolved rapidly to track the bandwidth demands of AI infrastructure.

Transceiver Form Factor Evolution and Speed Progression



Form Factor	Max Speed	Typical Reach	Use Case
SFP28	25 Gbps	10km SMF	Legacy top-of-rack
QSFP28	100 Gbps	500m MMF / 10km SMF	Prior-gen server NICs
QSFP56	200 Gbps	100m MMF / 2km SMF	Current fabric switches
QSFP-DD	400 Gbps	100m MMF / 2km SMF	AI cluster fabric
OSFP	400-800 Gbps	100m MMF / 2km SMF	AI cluster, 800G networks
1.6T modules	1.6 Tbps	Short reach SMF	Next-gen AI networking (2025-2026)

The Generational Reset Dynamic

When data center speeds double from 400G to 800G to 1.6T, the new generation does not simply cost twice as much as the old one. It often commands a premium of more than two times because the engineering challenges involved in each speed doubling are not linear.

Signal integrity, thermal management, and precision manufacturing all become exponentially harder. Early in a new generation cycle, manufacturing yields are low, and few suppliers can build units reliably. This allows companies with the relevant technology to command strong pricing. Over time, as engineering challenges are solved and more competitors enter, volumes ramp and margins compress until the next generational reset arrives.

The transceiver module business is fundamentally a manufacturing assembly operation as it relies on assembly rather than deep IP. As a result, it trends toward commoditization over time. Chinese manufacturers, particularly Innolight and Eoptolink, excel at high-volume production and continuous price reduction. Western suppliers must focus on differentiated segments where new generations early in the cycle where Chinese manufacturers have not yet caught up. This includes specialized architectures such as LPO, or the underlying optical components used inside those modules.

Fabrinet

Occupies a unique position as the dominant contract manufacturer for optical components and modules. Rather than competing in final product markets, Fabrinet manufactures optical assemblies and components for companies including Lumentum, Coherent, and others. Its revenue is therefore a diversified proxy for overall optical component demand, with less exposure to the competitive pricing pressure that affects module assemblers.

6. Lasers

Every optical connection requires a laser. The type of laser determines the performance, reach, and cost of the link. Four laser architectures serve the data center market, each optimized for different requirements.

Dimension	VCSEL	DML / DFB-DML	EML	CW Laser
What it is	Directly modulated surface-emitting laser	Directly modulated edge-emitting laser, often DFB-based	DFB laser & monolithic electro-absorption modulator (EAM)	Continuous-wave light source feeding an external modulator / SiPh engine
Typical fiber / reach	MMF, short reach	SMF, short- to medium-reach	SMF, medium-reach datacom	Reach depends on the external modulator & optical engine
Modulation	Direct	Direct	Integrated external modulation	External modulation required
Best-fit applications	SR modules, AOCs, short intra-DC links	Cost-sensitive SMF links, CWDM/LR-style links	Higher-performance pluggables, 400G/800G/1.6T class optics	Silicon photonics transceivers, CPO, external laser source architectures
CPO relevance	Low	Low to moderate	Moderate	Highest / primary
General positioning	Lowest-cost / shortest-reach option	Middle ground between VCSEL and EML	Higher-performance, tighter optical control	Best when optics are disaggregated from modulation
Representative vendors	Broadcom, Lumentum, Coherent	Lumentum, Coherent	Broadcom, Lumentum, Coherent	Lumentum, Coherent, Broadcom

VCSELs (Vertical Cavity Surface Emitting Lasers)

Emit light perpendicular to the chip surface and can be manufactured in arrays at very low cost. They can be tested at the wafer level before dicing, making them cost-effective for short-reach applications on multi-mode fiber under 500 meters. VCSELs are the dominant light source for intra-building data center connections and consume approximately 1 pJ/bit, making them highly power-efficient at short reach.

EMLs (Electro-absorption Modulated Lasers)

Integrate a laser and a modulator in a single device. The laser produces a continuous wave of light, and the modulator switches it on and off at extremely high speed. EMLs can operate at 100 Gbps and now 200 Gbps per lane, making them the standard for high-performance longer-reach single-mode applications. The transition from 400G to 800G to 1.6T transceivers drives a corresponding shift from 100G to 200G per lane EMLs. A 200G EML commands approximately double the price of a 100G EML but occupies roughly the same wafer footprint, making the mix shift from 100G to 200G per lane highly favorable for EML manufacturers.

DFB lasers (Distributed Feedback Lasers)

Single-frequency lasers with very narrow spectral linewidth, essential for DWDM applications where many closely spaced wavelengths must be packed onto a single fiber without interference. Their precise wavelength stability makes them the standard light source for long-haul and metro coherent optical systems.

Continuous Wave (CW) Lasers

Emit a constant, unmodulated stream of light. They are supplied primarily to silicon photonics customers who use a separate modulator on their own silicon chip to encode data onto the light. CW lasers are lower in complexity and cost than EMLs, but the requirements for CW lasers are evolving dramatically as CPO architectures emerge, discussed in detail in Section 9.

Indium Phosphide (InP)

The dominant substrate material for high-performance lasers. InP laser manufacturing uses small wafers, historically 3-inch and now transitioning to 4-inch, using specialised MOCVD reactors. These reactors require extremely precise epitaxial growth: the laser's wavelength is determined by the thickness of specific crystal layers grown atom by atom, and any deviation from specifications ruins the device. InP manufacturing is capital-intensive, capacity-constrained, and requires years of process learning to achieve competitive yields. The supply chain for InP laser wafers is therefore a genuine bottleneck in the optics scaling story.

Silicon photonics represents the most important platform transition in the transceiver industry. Silicon photonic transceivers integrate optical functions, includes waveguides, modulators, and photodetectors, directly into a silicon chip using standard CMOS fabrication processes compatible with conventional logic chip fabs.

Traditional InP manufacturing uses 3-inch or 4-inch wafers, yielding hundreds of laser dies per wafer with high touch labor and sensitivity to process variation. Tower Semiconductor, the leading silicon photonics foundry, manufactures SiPho devices on 200mm silicon wafers. A 200mm wafer has 16 times the surface area of a 3-inch InP wafer, so instead of hundreds of chips per run, a SiPho wafer yields thousands of chips per run using the same defect control disciplines as mainstream semiconductor manufacturing.

At 1.6T data rates, the industry is beginning a significant shift from EML-based transceivers toward SiPho-based transceivers for certain applications. SiPho transceivers typically require half the number of lasers as an equivalent EML product and benefit from the scale economics of silicon wafer manufacturing. However, EMLs are not disappearing: EML architecture remains dominant in WDM applications while SiPho takes share in parallel fiber applications. The two architectures are coexisting and serving different portions of the market.

InP vs Silicon Photonics

Dimension	InP PIC Platform	Silicon photonics Platform
Representative wafer size	Historically 3–4 inch common in InP photonics / electronics	200mm is common in commercial SiPh foundries; 300mm also exists
Example used for graphic	3-inch wafer	200mm wafer
Wafer diameter	76.2 mm	200 mm
Raw wafer area	~4,560 mm²	~31,416 mm²
Area multiple vs 3-inch InP	1.0x	6.9x
Manufacturing ecosystem	Smaller-wafer III-V / compound semiconductor flow	CMOS-compatible, larger-wafer, higher-volume ecosystem
Strengths	Native lasers / SOAs / active photonics integration	Wafer-scale economics, dense passive integration, stronger foundry scaling
Constraint to note	Smaller wafers and more specialized processing can limit scale economics	Often still needs external / heterogeneous III-V light source integration
Foundry annotation	SMART Photonics historically 3-inch, moved to 4-inch in 2024	Tower Semiconductor PH18 is an open 200mm SiPh foundry

The fundamental limitation of silicon photonics is that silicon cannot efficiently generate laser light because silicon has an indirect electronic bandgap that does not support efficient photon emission. Every SiPho chip therefore requires an external III-V laser source coupled into the silicon waveguide. This coupling is technically challenging and represents the primary engineering frontier for the platform. Several approaches are under development to integrate the laser source more tightly with the SiPho chip, including wafer bonding of InP laser material directly onto silicon.

Tower Semiconductor

Holds a near-monopoly position in SiPho foundry services. Its proprietary process design kit (PDK) is the standard that most transceiver and CPO designers have built their products around. Switching foundry partners requires redesigning the optical waveguide geometry and re-qualifying the entire process. This creates a switching cost moat that protects Tower's position even as GlobalFoundries and TSMC pursue secondary SiPho capabilities.

In a conventional pluggable transceiver architecture, optical transceivers sit in cages on the faceplate of a switch or server network interface card. Data from the GPU must travel as an electrical signal across the printed circuit board to reach the transceiver.

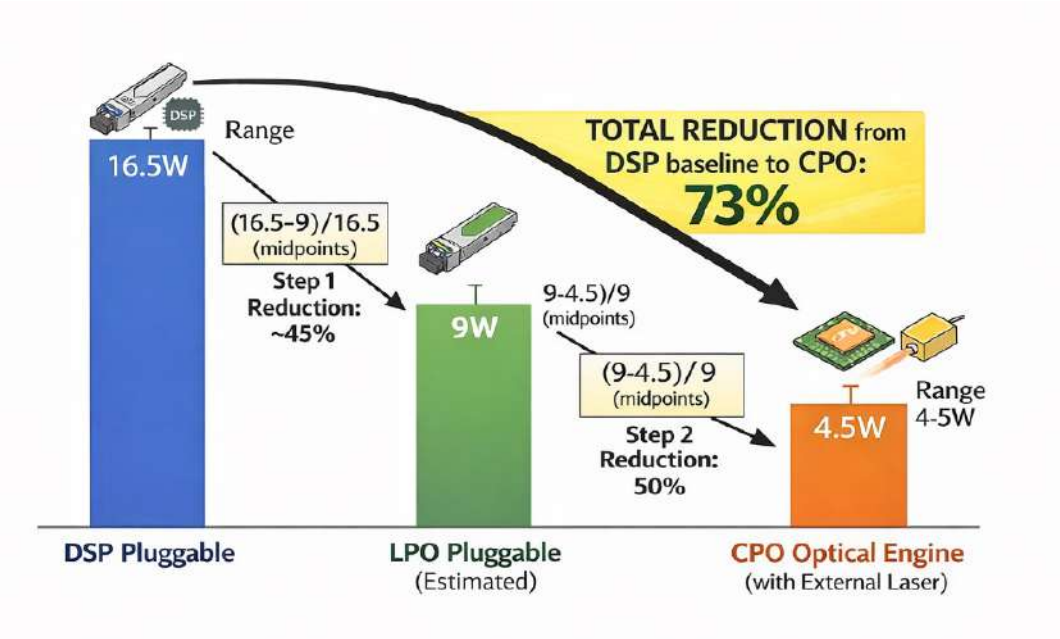
This PCB journey, despite being only a few inches, degrades the signal significantly. To clean up the degraded signal before optical conversion, every traditional transceiver requires a Digital Signal Processor (DSP) chip responsible for equalization, forward error correction, and signal cleanup.

The DSP is the single largest consumer of power in a traditional transceiver. An 800G DSP-based transceiver consumes 16 to 17 watts. The DSP chip inside it consumes roughly half of that total. At the scale of a 100,000-GPU cluster with tens of thousands of transceivers, DSP power consumption becomes a meaningful fraction of total facility power draw.

Linear Pluggable Optics (LPO)

Represents an intermediate step. LPO removes the DSP entirely, placing the signal conditioning burden on higher-performance analog driver and transimpedance amplifier components. LPO saves substantial power and reduces latency. However, it demands more sophisticated analog components that can perform complex equalization functions previously handled in the digital domain. LPO is an important transition architecture bridging the pluggable era and the co-packaged era.

Power Consumption Comparison Across Transceiver Architectures



Marvell

The leading DSP chip supplier for high-speed data center optical transceivers, having built this position through the 2021 acquisition of Inphi. Coherent DSPs for long-haul systems are supplied by Marvell, Cisco (through its Acacia acquisition), and Coherent. As LPO and CPO reduce or eliminate the DSP requirement in data center transceivers, Marvell's data center DSP business faces long-term headwinds from architectural transition, partially offset by its growing position in coherent DSPs for scale-across applications where DSPs remain essential.

Traditional pluggable transceivers will hit a fundamental physical wall as port speeds approach 1.6T and beyond. The electrical channel loss across the PCB path between the switch ASIC and the pluggable module becomes too large to manage efficiently. Power consumption and signal integrity both degrade beyond acceptable limits.

Co-Packaged Optics (CPO)

Resolves this by moving the optical engines directly inside the switch package, co-located with the switch ASIC itself. Instead of optical conversion happening at the faceplate of the switch, it happens millimeters from the compute silicon. The result is a 73% reduction in power consumption per optical link compared to traditional DSP-based pluggables. An 800G DSP transceiver consumes 16 to 17 watts. An equivalent CPO optical engine consumes only 4 to 5 watts.

CPO is architecturally significant for both scale-out and scale-up networking, but for different reasons.

Scale Out

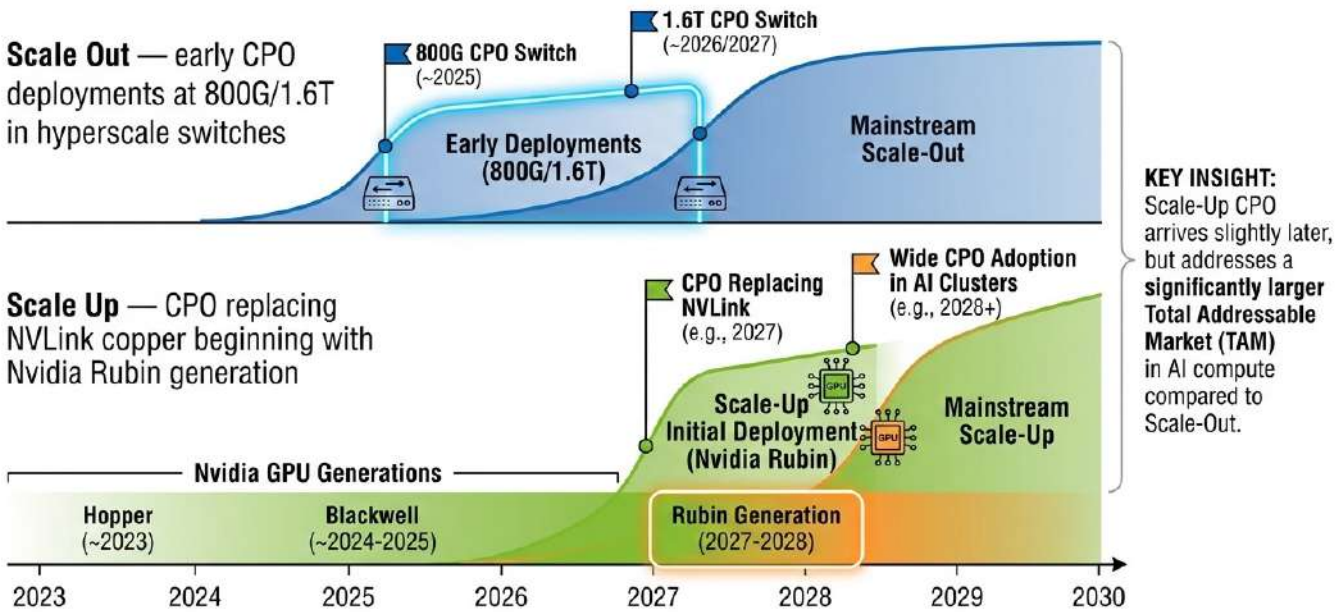
The primary benefit is total cost of ownership. Lower power consumption per port, higher port density on the switch faceplate, and better signal integrity at high speeds.

Scale Up

At the speeds demanded by next-generation GPU interconnects, 14.4 Tbit/s per GPU in Nvidia's Rubin generation, copper cannot function reliably even at the very short distances inside a rack. CPO is the only architecture that can deliver optical bandwidth at these speeds and density levels.

Once NVLink is replaced with optical connections via CPO, the physical constraint that forces GPUs to be packed tightly together disappears. GPUs can be spread across two racks, four racks, or an entire data center row while remaining in a single coherent NVLink memory domain with full bandwidth. The scale-up domain that previously comprised 72 GPUs in one rack can expand arbitrarily. Every NVLink connection that was previously a copper cable at zero optical cost becomes an optical link.

CPO Adoption by Network Tier



Current Barriers to CPO Adoption:

Serviceability is the most cited concern. In a pluggable setup, replacing a failed laser is a simple hot-swap. However, in a CPO architecture, a component failure can affect 64 or more ports simultaneously.

Vendor lock-in is a second concern as the pluggable market allows buyers to pit multiple transceiver manufacturers against each other. On the other hand, CPO consolidates optical functions into the switch package and reduces procurement flexibility.

Manufacturing immaturity is a third barrier. CPO requires precision fiber coupling and thermal isolation at scales that conventional electronics manufacturing is not designed for.

Having said that, these barriers are being actively resolved.

A study by Meta and Broadcom logged over 15 million device hours on CPO switches with zero unserviceable failures, demonstrating a mean time between failures of 2.6 million hours, nearly five times better than the 500,000-hour MTBF of standard 400G pluggable transceivers. Serviceability concerns are also being addressed through modular CPO designs that allow individual optical engine sub-modules to be replaced without affecting the entire switch port count.

CPO creates an architectural problem with lasers. Moving the optical engine onto the GPU or switch package places it directly next to chips that run extremely hot. Lasers are physically sensitive to heat where high temperatures degrade performance and reduce lifespan. Placing a heat-sensitive laser diode next to a GPU that generates enough thermal energy to boil water would result in rapid degradation.

External Laser Source (ELS)

The industry's solution is to use a separate, air-cooled module mounted at the rack faceplate that houses the lasers, known as ELS. The modulator (the component that encodes data onto the light) moves onto the GPU or switch package, but the laser itself remains remote.

But this creates a new problem. The laser is now remote from the optical engines it is feeding. Its light must travel through an additional path of fibers, blind-mate connectors, and splitters before reaching the GPU. Furthermore, a single ELS module must supply light to multiple optical engines simultaneously. A standard CW laser outputs 70 to 100 milliwatts, far too weak to survive the coupling losses and still supply multiple downstream optical engines at adequate power.

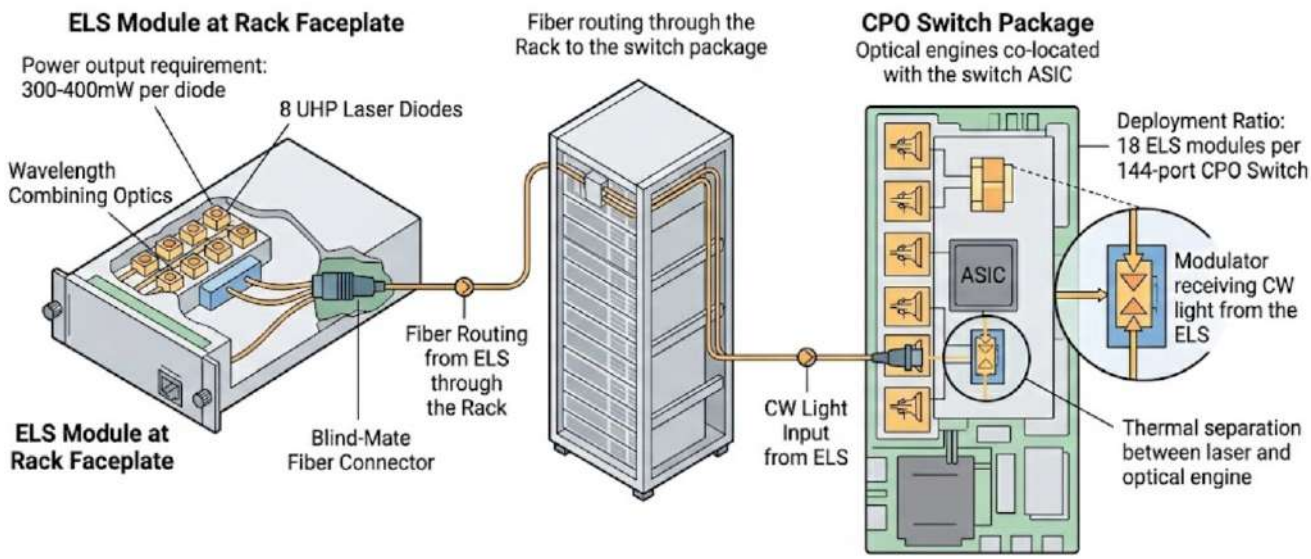
Ultra-High Power (UHP) Lasers

The ELS therefore requires UHP lasers capable of 300 to 400 milliwatts per diode. Manufacturing UHP lasers at this power level is extremely difficult. Ramping a small semiconductor chip to 400 milliwatts creates intense thermal density within the device. If the crystal structure grown during epitaxial manufacturing is not essentially perfect, the laser experiences Catastrophic Optical Damage (COD) where the power becomes so intense that the laser melts its own reflective facets and self-destructs.

Preventing COD requires flawless epitaxial layer structures grown with specialized MOCVD reactors, thick and complex crystal stacks, and manufacturing processes with zero tolerance for defects. The yield challenges for UHP lasers are fundamentally harder than for standard lasers, and the number of companies capable of producing them at scale is currently extremely limited.

An ELS module typically houses 8 UHP laser diodes at different wavelengths plus redundancy components, an optical combining train, blind-mate connectors, and thermal management. A standard 144-port CPO-enabled switch requires approximately 18 ELS modules. Selling one CPO switch creates demand for 18 ELS modules which is a significant multiplier effect that makes ELS a structurally attractive position in the CPO supply chain.

ELS Architecture Diagram



Lumentum

At the time of writing, it is the only supplier currently shipping UHP lasers at competitive yield. The company's advantage derives from decades of experience manufacturing high-reliability lasers for subsea and aerospace applications, where failure tolerance is essentially zero. That accumulated process knowledge translates directly to the yield discipline required for UHP manufacturing. Coherent, the primary competitor, is approximately two years behind Lumentum in UHP yield development and currently purchases EMLs from Lumentum for its own transceiver products.

The economics of ELS are structurally attractive as well. Lumentum captures approximately $260 per ELS module for laser chips alone and $600 to $700 for a full ELS module, compared to a traditional pluggable transceiver where Lumentum captures 5 to 10% of the bill of materials.

Understanding why laser manufacturing is difficult and why so few companies can do it well, requires understanding the epitaxial growth process that creates the crystal structures inside every photonic device.

Epitaxy is not conventional material deposition. In standard deposition, material is sprayed onto a surface and atoms land in whatever configuration they happen to adopt. Epitaxy requires each deposited atom to lock into a specific position extending the perfect crystal structure of the layer beneath it. Think about it, each atom must fall exactly into the correct grid position, or the crystal develops a defect that propagates upward through subsequent layers and eventually destroys the device's performance.

For an EML laser, the wavelength of the emitted light is determined by the thickness and composition of specific quantum well layers within the device stack. These layers must be grown with atom-level uniformity across the entire wafer. For UHP CW lasers, the additional challenge is that the thick, high-power epitaxial stack required to generate 300 to 400 milliwatts without self-destructing must maintain the same uniformity standards over a physically larger die with inherently lower yield than a standard laser die.

Table Summary

Metric	Standard CW laser	UHP laser
Reference use-case	Standard SiPh / datacom CW source	External laser source for CPO
Representative optical output	~50 mW	235–350 mW
Representative chip / cavity geometry	850 × 250 µm die	4 mm cavity public example
Modeled die area index	1.0×	2.5× *(conservative slide assumption)*
Illustrative wafer size	3-inch InP	3-inch InP
Gross dies per wafer	~21,460	~8,584
Modeled yield	85%	60%
Deliverable chips per wafer	~18,240	~5,150
Modeled reactor / process-cost index	1.0×	1.4×
Relative cost per deliverable chip	1.0×	~5.0×

The equipment used for this process is MOCVD (Metal-Organic Chemical Vapor Deposition).

Aixtron

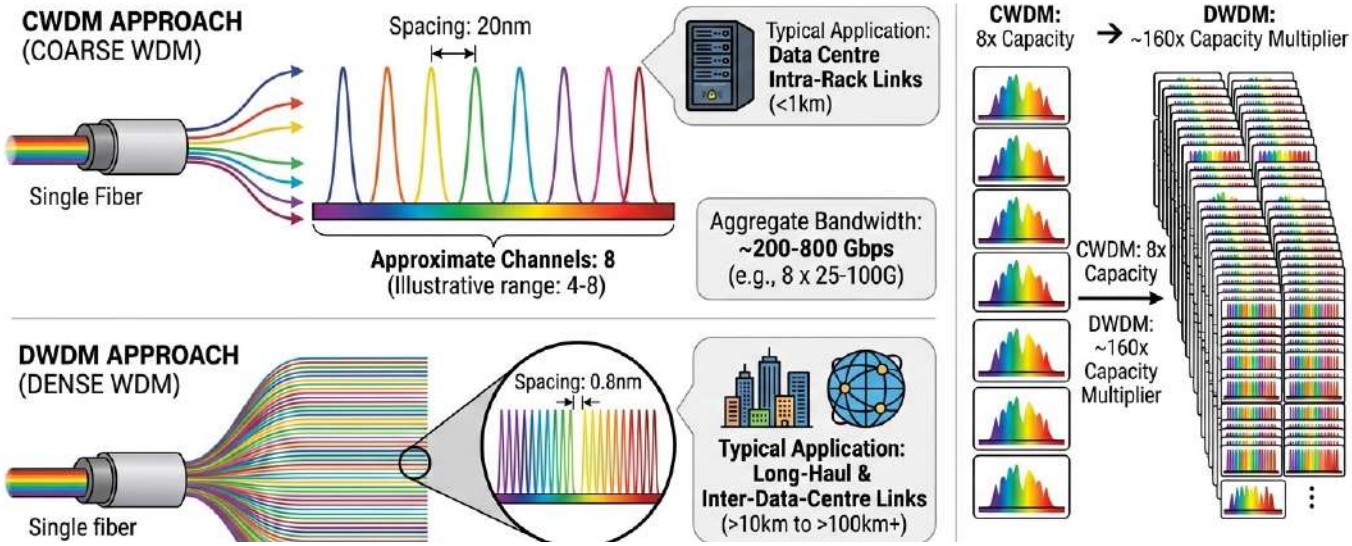
A German equipment company, holds approximately 90% market share in MOCVD reactors for optoelectronics. This near-monopoly position derives from Aixtron's proprietary Planetary Reactor design, which uses a dual-rotation mechanism where it wafers orbit the center of the reaction chamber while simultaneously spinning on their own axes. This creates a gas flow dynamic that achieves near-perfect layer uniformity across the entire wafer surface. Competing designs using simpler single-axis rotation produce poor uniformity at wafer edges, leading to yield loss that makes them economically unviable for precision laser manufacturing.

As the industry transitions from standard CW lasers to UHP CW lasers, the epitaxial intensity of each laser produced increases substantially. UHP lasers require dies two to three times physically larger than standard laser dies, reducing chips per wafer. They require more complex and thicker epitaxial stacks that increase reactor time per wafer. And their inherently lower yield means more wafers must be processed per deliverable chip. Combined, producing the same number of UHP laser chips as a conventional laser fab requires roughly five to ten times as many MOCVD reactors. The CPO transition is therefore a major and durable tailwind for Aixtron.

12. Wavelength Division Multiplexing

Wavelength Division Multiplexing allows multiple independent data streams to share a single optical fiber simultaneously, where each is carried on a different wavelength of light. At the transmit end, a multiplexer combines multiple laser signals at different wavelengths onto one fiber. At the receive end, a demultiplexer separates them back out. Each channel is functionally unaware of the others sharing the same fiber.

CWDM vs DWDM Channel Architecture



CWDM (Coarse WDM)

Uses widely spaced wavelengths of 20nm spacing, requires cheaper components and no temperature control, and supports limited channel counts of typically 4 to 8 channels. It is used in cost-sensitive short-reach applications within data centers.

DWDM (Dense WDM)

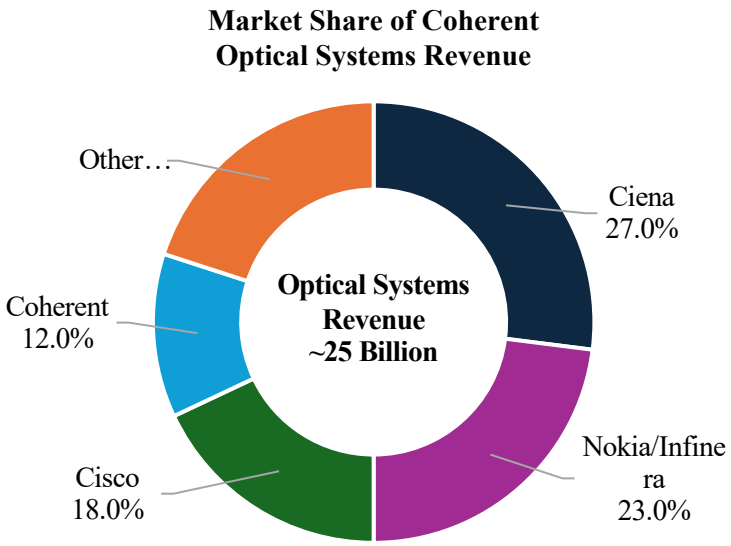
Uses very closely spaced wavelengths of 0.8nm or 0.4nm spacing, enabling 40, 80, or even 160 channels on a single fiber pair, each carrying 100 Gbps or more. DWDM is what allows long-haul fiber routes to carry terabits per second of aggregate traffic on a single fiber pair. It requires precisely temperature-controlled lasers and sophisticated optical amplifiers called EDFAs (Erbium-Doped Fiber Amplifiers) to compensate for accumulated signal loss over hundreds of kilometers.

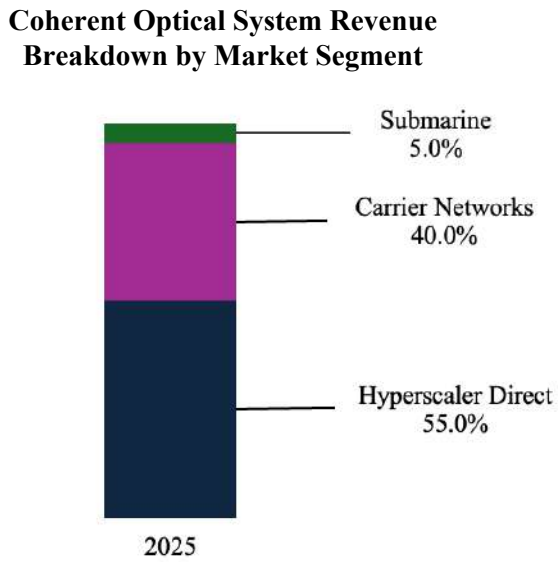
Within data centers, 4-lambda CWDM4 and 8-lambda approaches are standard for connecting racks at 400G and 800G. The shift to 1.6T introduces new 8-lambda and 16-lambda approaches at 200G per lane, directly driving demand for more wavelength channels and more precise laser sources.

For metro, regional, long-haul, and submarine fiber transmission, basic on/off modulation of a laser is grossly inefficient. Coherent optics uses sophisticated modulation schemes to dramatically increase the amount of information carried per unit of optical bandwidth.

A coherent transceiver modulates both the amplitude and the phase of the light wave, and exploits both polarization states of the fiber, encoding multiple bits per symbol. Modern coherent transceivers using PM-64QAM modulation encode 12 bits per symbol per polarization, compared to 1 bit per symbol for basic on/off keying. A single pair of coherent WDM channels on a single fiber pair can carry aggregate terabits per second over thousands of kilometers.
The digital signal processing required to implement coherent modulation and demodulation is extremely demanding. Coherent DSP chips must compensate in real time for chromatic dispersion accumulated over thousands of kilometers, polarization mode dispersion, fiber nonlinearity, and optical noise from amplifiers. The coherent DSP is both the enabler of long-haul high-capacity transmission and one of the primary power consumers in long-haul equipment.

The growth of AI infrastructure is driving significant demand for coherent optical capacity on fiber routes connecting data centers. Hyperscalers building multiple large AI training clusters in different geographic regions require massive optical capacity between those sites for data synchronization, model distribution, and inference traffic routing.





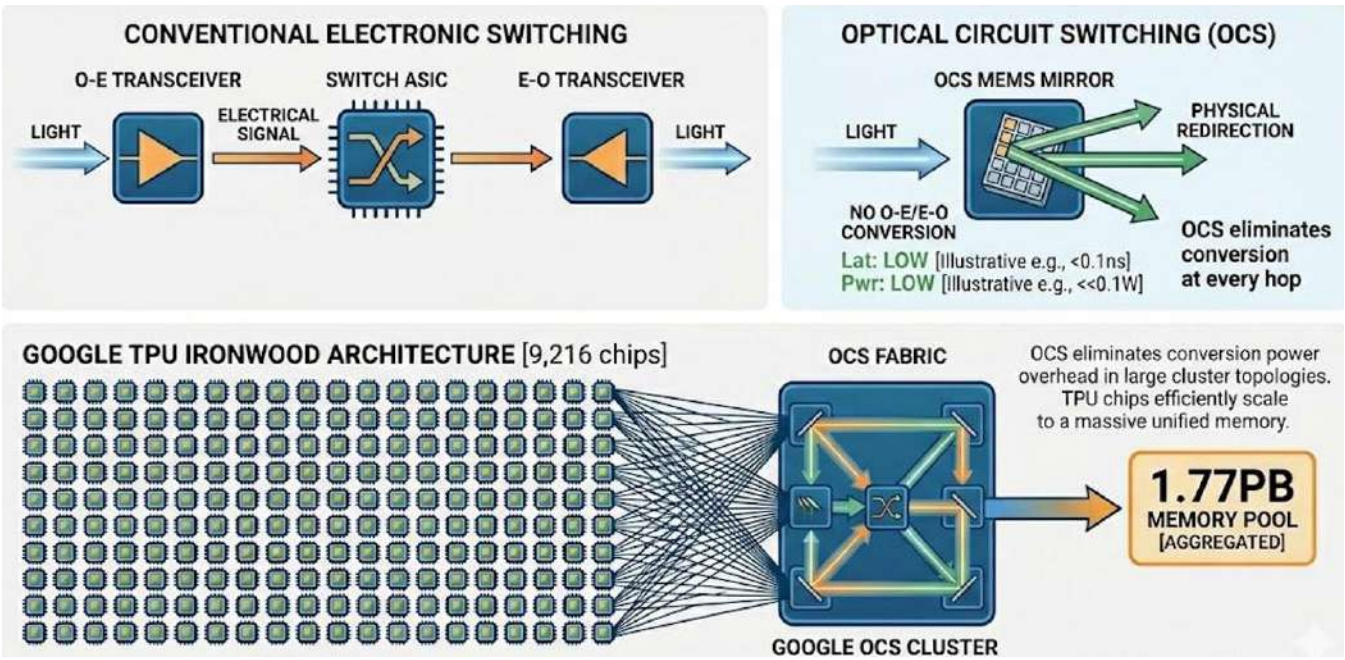
The Coherent Systems Competitive Landscape

Distinct from the data center transceiver market. The primary competitors are Ciena (United States), Nokia (Finland, through its Optical Networks division), Infinera (United States, now acquired by Nokia), and Cisco (through its Acacia coherent module acquisition). Ciena holds the largest market share in carrier coherent optical systems globally and has been the primary beneficiary of hyperscaler-driven capacity expansion.

Nokia's acquisition of Infinera, completed in 2024, created a stronger number-two competitor with complementary technology portfolios. Coherent Corporation (formerly II-VI, rebranded after the acquisition of Viavi's optical business and Coherent Communications) competes at the component and module level, supplying coherent transceiver modules to system integrators and some hyperscalers directly.

Optical Circuit Switches represent an architectural innovation beginning to influence how the largest AI clusters are designed. Rather than routing data through electronic packet switches, which convert light to electrical signals, process them, and convert back to light. OCS switches light paths directly in the optical domain using MEMS mirrors that physically redirect light beams between fibers without any optical-to-electrical conversion.

The advantage of OCS is that it can redirect entire high-bandwidth optical channels between endpoints without incurring the power and latency overhead of electronic switching. Google has pioneered OCS in its TPU supercomputer architecture. Ironwood, Google's TPUv7 generation, interconnects 9,216 chips through an OCS fabric into a unified 1.77 petabyte memory pool which is a memory domain significantly larger than what conventional electronic switching can achieve at comparable power. The OCS fabric allows Google to treat thousands of individual AI accelerators as a single massive processor without the hierarchical switching overhead that conventional InfiniBand or Ethernet topologies require.



Lumentum

Lumentum is currently the primary external supplier for production OCS at hyperscale, supplying MEMS-based optical switches for Google's TPU architecture. OCS deployment is currently limited in scale and supplier base, but multiple hyperscalers are exploring the architecture as a path to scaling AI clusters beyond what conventional networking allows at reasonable power and cost.

15. InfiniBand vs Ethernet

The network protocol war between InfiniBand and Ethernet is a strategically important backdrop for the entire optical layer because it determines which switch ASICs and which networking stacks the optics industry must serve.

Feature	InfiniBand NDR/XDR	Ultra Ethernet
Latency	Very low (~1 µs)	Low (~3 µs)
Throughput	800 Gbps – 1.6 Tbps	Coming Soon: flat 800 Gbps and 1.6 Tbps
Primary switch ASIC supplier	Nvidia	Broadcom, Cisco
Primary transceiver supplier	Coherent, II-VI, Lumentum, Innolight, Hisense	Coherent, II-VI, Lumentum, Innolight, Hisense
Major hyperscaler customers	Microsoft, Meta, Oracle	Microsoft, Meta, Google, Amazon, HPE, Intel

InfiniBand

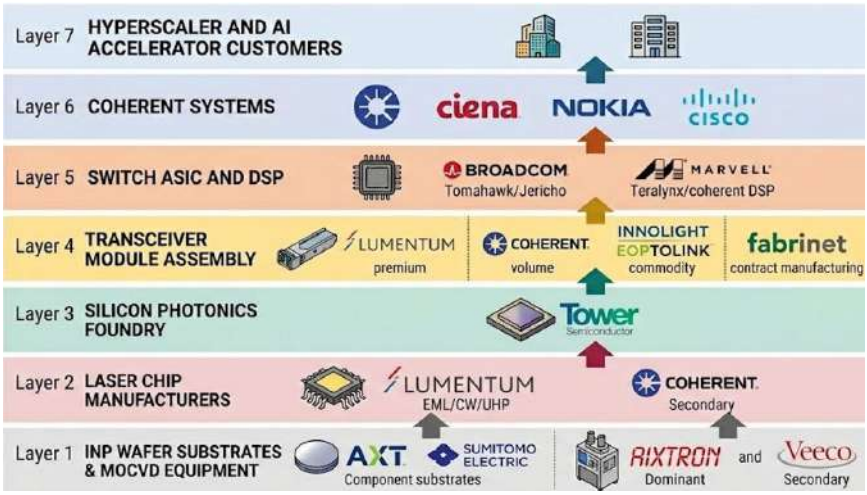
Controlled by Nvidia through its 2020 acquisition of Mellanox, has been the dominant protocol for high-performance AI cluster scale-out networking. InfiniBand's advantages are low latency, high message rate, and a tightly integrated software stack optimized for the collective communication patterns (all-reduce, all-gather) that dominate AI training workloads. Nvidia's NDR InfiniBand at 400 Gbps and the upcoming XDR generation are the standard for H100 and H200 clusters.

Ethernet

Championed primarily by Broadcom whose Tomahawk and Jericho switch ASICs power Ethernet-based AI cluster fabrics. The Ultra Ethernet Consortium, formed in 2023 with members including AMD, Intel, Meta, Microsoft, and others, is developing Ethernet extensions specifically for AI workload communication patterns. The commercial motivation for hyperscalers to support Ethernet alternatives is clear: InfiniBand creates a vendor lock-in dependency on Nvidia that they wish to avoid.

The outcome of this competition matters for the optical supply chain because InfiniBand and Ethernet fabrics use different switch ASIC platforms, drive different transceiver form factor timelines, and create different co-packaging constraints for CPO deployment. Both protocol camps will require optical transceivers and eventually CPO.

Supply Chain Dynamics



Lumentum

Holds the most strategically differentiated position in the optics supply chain for the CPO era. Its UHP laser monopoly for ELS applications, its MEMS OCS position with Google, its EML leadership, and its accumulated subsea-grade reliability process knowledge give it exposure to the highest-value components in the transition. The economics of ELS are particularly attractive: Lumentum captures approximately 40 to 60% of the optical bill of materials per CPO link compared to 5 to 10% in the pluggable era.

Aixtron

The picks-and-shovels play on the entire laser manufacturing industry. Its approximately 90% market share in optoelectronics MOCVD, protected by the Planetary Reactor physics that competitors cannot easily replicate, means that every laser manufactured for every optical link, whether pluggable or CPO, EML or UHP CW, requires Aixtron equipment. The CPO transition multiplies MOCVD demand intensity by five to ten times per UHP laser chip relative to standard laser manufacturing, making the CPO era a structural demand tailwind regardless of which laser companies win the UHP manufacturing race.

Tower Semiconductor

Holds the foundry monopoly for silicon photonics. Its proprietary PDK and 200mm wafer economics give it a structural cost advantage over InP-based alternatives for the applications where SiPho is applicable. As SiPho share grows with the transition to 1.6T and CPO, Tower captures a growing proportion of the optical component manufacturing value.

Corning

Currently the infrastructure layer beneficiary. Every new data ceter, every new fiber route between AI training clusters, and every rack-scale optical interconnect expansion requires more fiber. Corning's manufacturing scale, ultra-low-loss fiber technology, and long-term supply agreements with hyperscalers position it as a durable beneficiary of the AI infrastructure buildout with lower technology risk than component suppliers.

Fabrinet

Its revenue tracks the overall growth of optical component demand without taking positions on which laser architecture or protocol wins. Its customer base includes most major optical component companies, giving it broad exposure to volume growth.

Chinese transceiver assemblers (Innolight, Eoptolink)

Well-positioned in commodity 400G and 800G pluggable modules where manufacturing scale and cost discipline are the primary competitive variables. They are not positioned for the UHP laser and CPO era because those products require process knowledge and yield discipline that commodity assembly operations do not possess. The bifurcation between commodity pluggable assembly (where Chinese manufacturers will likely maintain or grow share) and CPO component manufacturing (where they currently have no credible position) is an important structural dynamic for the optics investment thesis.

Power & Grid Primer Report



Power and Grid Key Terms

Terms	Definitions
Alternating Current (AC)	electric charge that periodically reverses direction and continuously changes magnitude, typically in a sinusoidal wave, rather than flowing in a single direction. It is the standard for power grids because transformers can easily step voltage up for efficient, low-loss, long-distance transmission and step down for safe distribution to residential and industrial users, usually at frequencies of 50 Hz or 60 Hz.
Battery Energy Storage Systems (BESS)	an advanced technology that uses electrochemical batteries, commonly lithium-ion, to store electrical energy, typically from renewable sources or the grid, and release it when needed. It balances electricity supply and demand, enhances grid stability, and provides reliable backup power. **Key components include:** battery cells and modules, battery management systems, power conversion systems, energy management systems and auxiliary systems.
Combined Cycle Gas Turbine (CCGTs)	a highly efficient, low-emission power generation system that combines a gas turbine (Brayton cycle) and a steam turbine (Rankine cycle) to generate electricity, often achieving thermal efficiencies near 60%. It uses waste heat from the gas turbine exhaust to create steam for the steam turbine, maximising power output. Fuel (usually natural gas) is burned to turn a gas turbine. The exhaust heat (around $550C$) passes through a Heat Recovery Steam Generator (HRSG) to produce steam, which drives a secondary steam turbine, increasing total output by over 50% without extra fuel.
Coolant Distribution Unit (CDUs)	a critical liquid-cooling infrastructure component used in high-density data centers, power, and computing environments (such as AI, ML, and HPC). It acts as the "heart" of the liquid cooling system, circulating, regulating, and conditioning coolant to remove heat from IT components, such as high-power CPUs and GPUs. CDUs bridge the facility's main cooling infrastructure with the IT server racks, providing superior thermal transfer capabilities over traditional air-cooling methods.
Cross-Linked Polyethylene (XLPE) Insulated Cables	they are specialised electrical cables designed for power transmission and distribution, utilising a thermoset insulation material created by chemically cross-linking polyethylene molecules into a 3-dimensional network. This process enhances thermal, mechanical, and chemical stability, allowing the cable to operate at higher temperatures (90°C continuously) and withstand higher electrical stresses compared to traditional thermoplastic insulation like PVC.

Power and Grid Key Terms

Terms	Definitions
Data Centers	The critical, high-density energy consumers serving as infrastructure for storing and processing digital data. In power and grid terms, they are massive, 24/7 loads requiring 20–200+ MW, acting as "grid-interactive" assets that provide flexible energy consumption, backup capacity, and bi-directional energy flow opportunities for the electrical grid. Data centers are massive consumers, representing 1.7% of global electricity consumption, with demands projected to rise greatly.
Data Centers as a Grid	Require dedicated substations and high-voltage transmission access to meet their needs.can act as flexible loads, adjusting their power consumption to support grid stability. They are increasingly integrating renewable energy, such as solar and wind, reducing their carbon footprint.
Direct Current (DC) Systems	Electrical networks where charge flows continuously in one direction, maintaining constant voltage polarity and zero frequency. Essential for modern applications, they provide stable power for electronics, battery storage, and solar energy, often overcoming efficiency losses of AC conversion. Electrons flow in a steady, single direction, from the negative terminal to the positive terminal. They are common in batteries, solar cells (photovoltaics), fuel cells, and consumer electronics.
Direct Liquid Cooling (DLC)	A thermal management method that removes heat from high-power electronic components (CPUs, GPUs, servers, or grid power electronics) by circulating a liquid coolant directly through a cold plate physically mounted on the hot components. Unlike air cooling, which moves air across a component, DLC brings liquid, which is up to **3,000x** more effective at conducting heat than air, directly to the heat source, offering far superior efficiency.
Distribution	The final stage in the delivery of electricity, used for carrying energy from the high-voltage transmission system to individual consumers (homes, businesses, and industrial sites). It is the route of the electrical grid, taking power from local substations and distributing it at lower, usable voltages through a network of distribution lines. A complex network of substations, transformers, and feeders (lines) that receive electrical energy from transmission lines and reduce the voltage to levels safe for end-users.
Generation	The process of producing electric energy from primary energy sources (such as fossil fuels, nuclear fuel, or renewable sources). It is the critical first step in the three-part electric grid infrastructure, Generation, Transmission, and Distribution, that delivers electricity to consumers. It involves transforming mechanical energy, kinetic energy, or solar energy into electrical energy using tools like generators (which rotate in a magnetic field) or photovoltaic (PV) cells.

Power and Grid Key Terms

Terms	Definitions
High-Voltage Cables	They are specialised, fully insulated electrical conductors used to transport electricity at high voltage levels, typically 35 kV (35,000 volts) up to 765 kV and higher, between power generation centers, substations, and major industrial users. They are the backbone of the electrical grid's transmission and distribution networks, specifically designed for long-distance power transport with minimal energy losses.
High-Voltage Direct Current (HVDC)	They are specialised, fully insulated electrical conductors used to transport electricity at high voltage levels, typically 35 kV (35,000 volts) up to 765 kV and higher, between power generation centers, substations, and major industrial users. They are the backbone of the electrical grid's transmission and distribution networks, specifically designed for long-distance power transport with minimal energy losses.
Immersion Cooling	A high-efficiency thermal management technique where power electronics, battery cells, or utility hardware are directly submerged in a non-conductive, dielectric liquid. This method replaces traditional air cooling, offering **10-20x** higher thermal conductivity by eliminating thermal resistance between components and the cooling agent. used for Grid-Scale BESS, Uninterruptible Power Supplies (UPS), and power electronics to enhance safety, density, and efficiency.
Interconnection Queue Crisis	The severe backlog of new power generation and storage projects (predominantly wind, solar, and battery) waiting for approval to connect to the electricity grid. As of 2025, this backlog exceeds **2.6 terawatts (TW)** of capacity in the US alone, **2x** the total current operating power capacity, with average wait times for studies exceeding 5 years.
Lithium-Ion Phosphide (LFP)	Subtype of lithium-ion rechargeable battery technology that uses lithium iron phosphate as the cathode material and graphite as the anode. In power and grid applications, LFP is defined by its exceptional thermal stability, long cycle life, high safety, and cost-effectiveness compared to other lithium-ion chemistries (like NMC). It has become the dominant technology for stationary energy storage systems (ESS) and is rapidly gaining market share in electric vehicles.
Long-Duration Energy Storage	Technologies capable of storing and discharging electricity for 10 hours or more, up to several days, weeks, or seasons. Unlike short-term batteries, LDES manages daily, weekly, or seasonal imbalances between renewable energy production and demand, providing stable grids.

Power and Grid Key Terms

Terms	Definitions
Nuclear Energy	It is the clean, baseload electricity generated by harnessing the heat released from controlled nuclear fission reactions, specifically the splitting of uranium-235 atoms within a reactor core. Unlike fossil fuels, it produces no greenhouse gases or air pollutants like sulfur and nitrogen oxides during operation. It is considered the highest-capacity, most reliable source of low-carbon electricity, running 24/7 with a capacity factor typically over 90%.
Phaser Measurement Units (PMUs)	An advanced sensor device used in modern power systems to monitor the electrical grid's health in real-time by measuring synchronised voltage and current phasors. Unlike traditional SCADA systems that provide updates every 2 to 4 seconds, PMUs deliver high-speed, timestamped data, 30 to 120x per second, allowing for the observation of dynamic, fast-moving phenomena. The key to their function is time synchronisation, using GPS to ensure measurements from geographically dispersed locations are aligned, allowing operators to see a "live" picture of the entire grid.
Photovoltaic (PV) Cells	An electronic device that directly converts sunlight into electricity (direct current, or DC) through the photovoltaic (PV) effect. These cells, generally made of semiconducting materials like silicon, are the building blocks of solar panels, which are interconnected to form arrays that can provide power to local devices or feed directly into the utility grid.
Power Distribution Units (DPUs)	A device likened to a sophisticated, heavy-duty power strip, that receives electrical power (AC) from a single source (UPS or grid) and distributes it to multiple IT devices within data center racks. They are crucial for bridging power gaps, managing power consumption, and improving uptime in high-density environments. It can be rack-mounted horizontally (1U/2U) or vertically (0U), allowing for space-efficient installation in server cabinets. Provides power consumption monitoring, energy reporting, and remote, real-time access to manage load.
Power Usage Effectiveness (PUE)	Measures data center energy efficiency by calculating the ratio of total facility power to the power used specifically by IT equipment. A PUE of **1.0** represents perfect efficiency, where all energy powers IT, with lower ratios indicating better efficiency. The ideal ratio is 1.0; however, the average data center operates **~1.58.** It helps identify energy waste, benchmark performance over time, and reduce operational costs. PUE = Total Facility Power / IT Equipment Power.

Power and Grid Key Terms

Terms	Definitions
Reactive Power Compensation Equipment	Electrical devices, specifically shunt reactors, used to absorb excess inductive reactive power, counteract the capacitive effect of long transmission lines, and stabilise voltage levels. These devices are essential for maintaining the balance between reactive power production and consumption, particularly during low-load conditions or when using extensive underground cable networks.
Real-Door Heat Exchange	A high-efficiency, liquid-cooled, or air-assisted cooling solution that replaces the standard rear door of an IT equipment rack. It intercepts hot exhaust air directly at the source, cooling it via a liquid-filled coil (using chilled water or a coolant) before it enters the data center room, essentially rendering the rack thermally neutral. This technology operates as a "bridge" between traditional air cooling and direct-to-chip (D2C) liquid cooling, enabling high-density computing (AI/ HPC) while significantly improving power efficiency and lowering PUE.
Renewable Energy	Electricity generated from self-replenishing natural sources, solar, wind, hydro, geothermal, and biomass, that is integrated into the electrical grid. It displaces conventional fossil fuels, offering a clean, sustainable power supply that often fluctuates, requiring advanced grid management.
Small Module Reactors (SMR)	Advanced nuclear fission reactors that are physically smaller than traditional nuclear power plants, generally producing 300 megawatts electric (MWe) or less per unit (around one-third of a conventional reactor's capacity). They are designed to be "modular," meaning major components are factory-assembled, transportable, and installed at a site as a complete unit.
Static Synchronous Compensators (SVCs)	Flexible Alternating Current Transmission System (FACTS) device that acts as a shunt-connected reactive power compensation technology. It uses solid-state power electronics, typically Voltage Source Converters (VSCs) based on Insulated Gate Bipolar Transistors (IGBTs), to generate, absorb, and control reactive power to regulate the grid voltage and improve power transfer capability.
Substation	A critical, high-voltage component of an electrical power grid that serves as a junction point to transform voltage levels, switch circuits, and provide protection to the system. It is the essential link between generation, transmission, and distribution systems, acting as a transfer point that steps up voltage for efficient long-distance transmission or steps it down for safe residential and commercial use. It consists of power transformers, circuit breakers, switches, busbars, and protection devices.

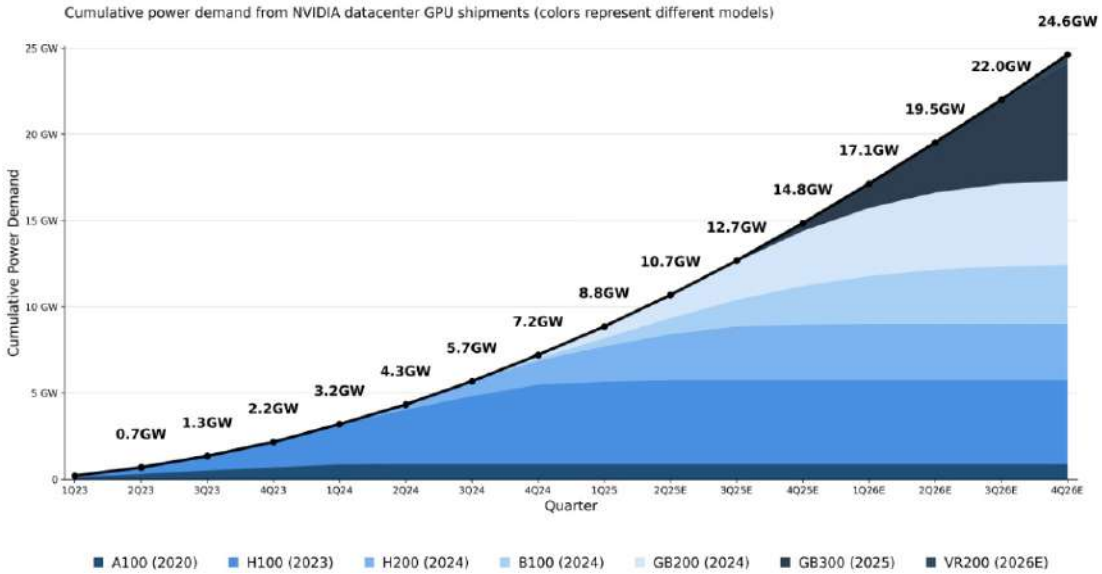
Power and Grid Key Terms

Terms	Definitions
Switchgear	Centralised, metallic-enclosed assembly of electrical disconnect switches, fuses, circuit breakers, and relays used in power systems to control, protect, and isolate electrical equipment. It acts as the "nervous system" or "traffic cop" of the power grid, directing electricity flow during normal operation and instantly disconnecting faulty circuits to prevent damage and widespread outages. Defined by its ability to manage electrical energy from generation to distribution and consumption. It is not a single device but a collection of interconnected components, e.g., housed in metal structures.
Switch Model Power Supply (SMPS)	It is known as a switching power supply or switcher. It is an electronic power converter that uses a high-frequency switching regulator to efficiently convert electrical power from one voltage level to another, typically transforming AC mains power into stable, regulated DC voltage. Unlike linear power supplies that dissipate excess energy as heat, an SMPS operates by rapidly toggling a transistor (MOSFET) on and off (typically 20 kHz to 500 kHz or higher), allowing for efficiencies often exceeding 85–95%.
Transformers	A static electrical device that transfers AC electrical energy between circuits through electromagnetic induction (EMI), primarily for the purpose of stepping up (increasing) or stepping down (decreasing) voltage levels. It is essential for efficient, long-distance power transmission and safe distribution, operating at the same frequency.
Transmission	It is the bulk, high-voltage movement of electricity from power generation plants to substations, acting as the "backbone" of an electric grid. It is designed to transport large amounts of energy over long distances (often hundreds of miles) with minimal energy loss. It consists of transmission lines, step-up and step-down transformers, substations and high-voltage direct current transmissions.
Uninterrupted Power Supply (UPS) Systems	An electrical apparatus providing instant, high-quality backup power to critical loads during grid failures or voltage fluctuations. Utilizing stored energy (batteries/flywheels), it ensures continuous operation and protects sensitive equipment from outages, sags, and noise, acting as a crucial bridge until generators start or systems safely shut down.
Voltage Regulator Modules (VRM)	A specialized buck DC-to-DC converter designed to step down a higher voltage (typically 12V or 48V) to a lower, precise, and highly stable voltage required by high-performance semiconductor components like CPUs, GPUs, and ASICs. It serves as a critical "gatekeeper" of power integrity at the Point-of-Load (PoL).

A conventional enterprise server rack draws 5 to 10 kilowatts. A rack of eight Nvidia H100 GPUs draws 60 to 100 kilowatts, representing ten times the density. A full H100 server pod of 32 racks draws approximately 3 megawatts. A hyperscale AI training cluster of 100,000 H100s draws approximately 500 megawatts of IT load, rising to 650 to 750 megawatts with cooling and facility overhead. That is approximately the output of a mid-sized nuclear power plant, consumed by a single training cluster.

The next generation of Nvidia's Blackwell B200 GPU has a thermal design power of 1,000 watts. That is 43% more than the H100. At 100,000 GPU scale, a Blackwell cluster approaches 700 megawatts of IT load before cooling.

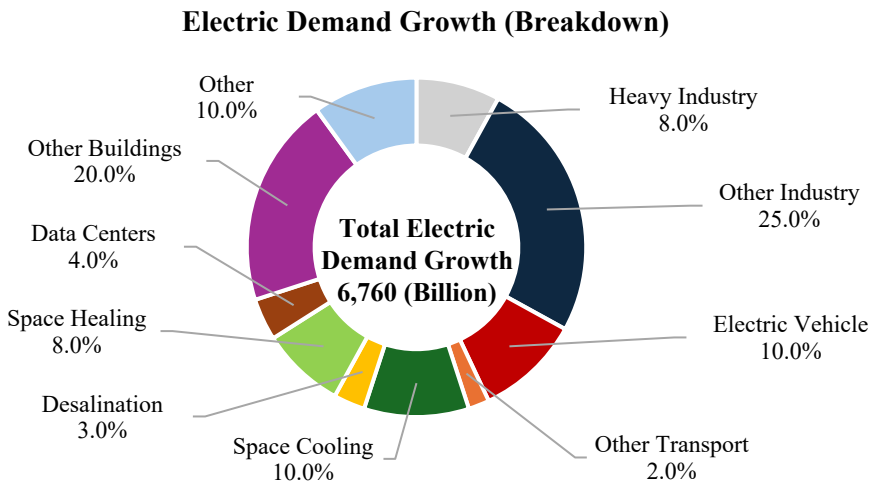
AI Power Demand



The aggregate demand growth is equally striking. US data center electricity consumption was approximately 200 terawatt-hours in 2023.

The IEA projects it could reach 500 to 600 terawatt-hours by 2030 under a moderate AI growth scenario, and significantly more under aggressive scenarios. To put this in context, total US electricity generation in 2023 was approximately 4,200 terawatt-hours. Data centers are moving from approximately 5% of US electricity consumption toward a potential 12 to 15%. This is a demand addition equivalent to adding a new California to the grid in less than a decade.

Approximately 60% of the energy consumed by data centers today comes from fossil fuels, and companies are increasingly forced to choose between AI capabilities and environmental commitments. The electricity generation mix that will serve AI infrastructure through the 2030s is one of the most consequential and contested questions in energy policy.



Source: https://www.goldmansachs.com/what-we-do/goldman-sachs-global-institute/articles/smart-demand-management-can-forestall-the-ai-energy-crisis

Understanding the basics is necessary context for evaluating the bottlenecks described in subsequent sections.

Generation

Produces electricity from primary energy sources by burning fossil fuels, nuclear fission, or harvesting renewable energy from wind and solar. Generation output is measured in watts and must be matched to consumption in real time. The grid has very limited storage capacity, which means every kilowatt consumed must simultaneously be produced somewhere on the network.

Transmission

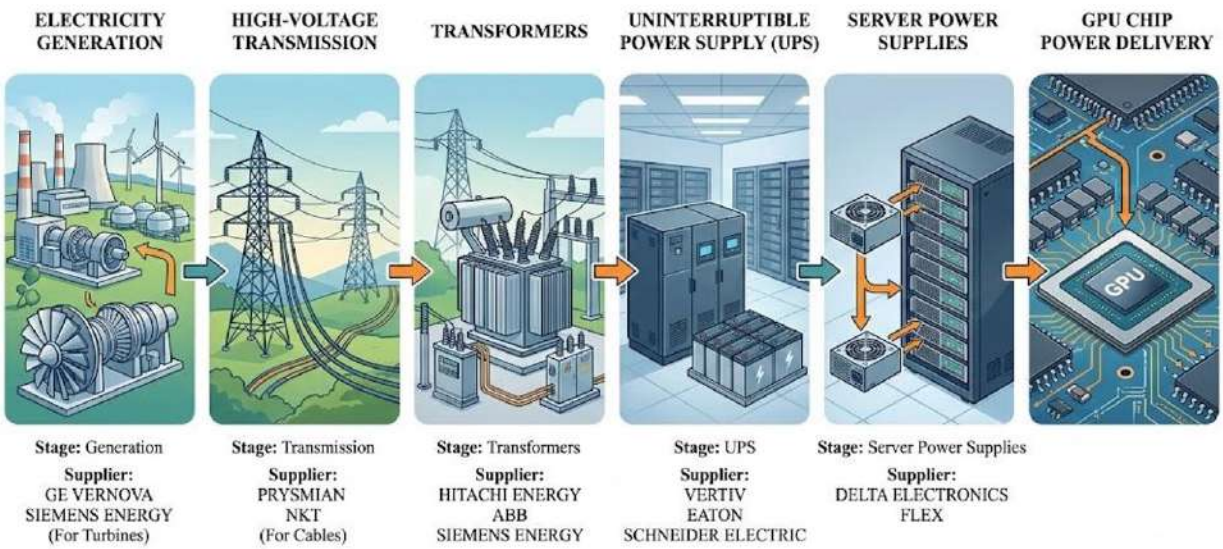
Carries electricity from generation sites (which are often remote from population centers) to regional distribution networks at very high voltages of 115 to 765 kilovolts AC. High voltage reduces current for a given power level, which reduces resistive losses in the transmission conductors (power loss equals current squared times resistance, so halving current reduces losses fourfold). High-voltage direct current (HVDC) transmission is used for very long distances and submarine links where AC losses become prohibitive.

Substations

Step voltage down from transmission levels to distribution levels using transformers. A transmission substation might step 345 kilovolts down to 69 kilovolts for regional distribution. A distribution substation steps 69 kilovolts down to 12 kilovolts for local neighborhood distribution. A final service transformer steps 12 kilovolts down to the 120 or 240 volts that reaches homes and businesses.

Distribution

Carries electricity through the local network of poles and underground cables to end consumers.



For a Data Center, the Full Power Delivery Chain:

1. Electricity arrives at the data center from the high-voltage transmission network, which carries power at between 115 and 765 kilovolts over long distances from generation sources.
2. A transmission substation at or near the data center campus steps this down to a distribution voltage of 34 to 69 kilovolts.
3. A dedicated on-site substation steps it down again to the medium voltage range of 12 to 35 kilovolts that is practical for distribution within the facility.
4. From there, on-site medium voltage switchgear routes and isolates power across the campus before it passes through the UPS and generator systems that buffer the load against grid disturbances and provide backup during outages.
5. Power distribution units then step voltage down further and distribute it to individual server racks, where each server's internal power supply converts the incoming alternating current to the direct current rails the components require.
6. Finally, voltage regulator modules mounted directly on the server motherboard perform the last conversion, stepping down to the 0.7 to 1.0 volts at which the GPU itself operates.

The fundamental investment thesis in power infrastructure is a severe mismatch between the speed of AI demand growth and the speed at which power infrastructure can respond. Understanding why supply is so slow to respond requires understanding three distinct bottlenecks.

The Interconnection Queue Problem

In the United States, any new facility wanting to connect to the high-voltage transmission grid must join an interconnection queue managed by regional transmission organizations (RTOs) such as PJM (serving the Mid-Atlantic and Midwest), MISO (serving the broader Midwest and South), ERCOT (serving Texas), and CAISO (serving California). The queue requires the RTO to conduct interconnection studies determining whether the grid can physically accommodate the new load without destabilizing voltage and frequency across the network.

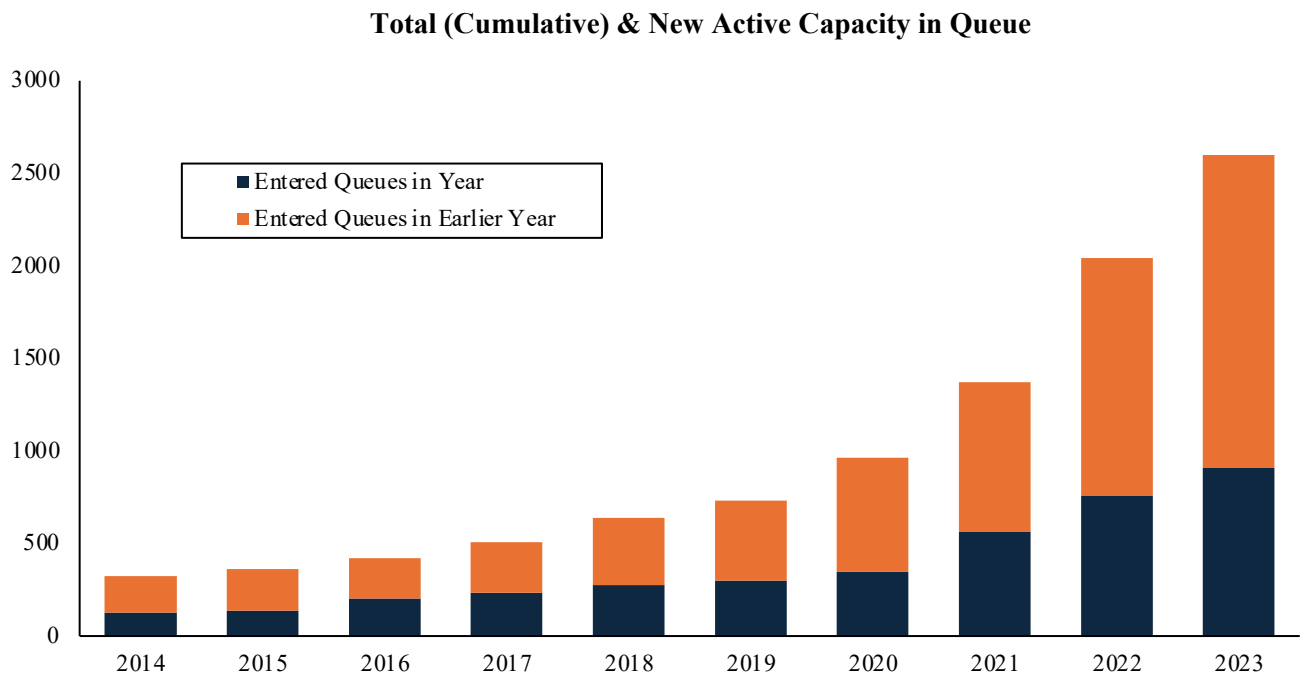
The scale of queue backlogs is staggering. PJM's interconnection queue currently contains projects representing over 300 gigawatts of new load and generation which is more than the existing total generating capacity of its service territory. Average interconnection study timelines have grown from two years in 2008 to five years or more in congested markets today. A data center developer who secures a site, raises capital, and begins construction may wait five years for a grid connection. FERC Order 2023, issued in 2023 to reform the interconnection process, is expected to improve timelines but not resolve the fundamental capacity constraint.

Permitting Timelines

New high-voltage transmission lines in the United States face a permitting process involving environmental impact assessment, rights-of-way acquisition, and multi-jurisdictional regulatory approval that routinely takes seven to ten years from proposal to energization. A transmission line needed to carry power from a new generation source to a data center cluster may take longer to permit and build than the data center it is designed to serve. This is not a temporary dysfunction. It is a structural feature of US infrastructure permitting that bipartisan reform efforts have repeatedly failed to resolve.

Equipment Manufacturing Constraints

The physical equipment required to upgrade the grid, including the transformers, switchgear and high-voltage cables, has manufacturing lead times that have extended dramatically under surging demand. Large power transformers, the most critical single piece of equipment in grid expansion, currently have lead times of 18 to 36 months from order to delivery, up from four to eight weeks a decade ago. This bottleneck is examined in detail in Section 4.



Source: Lawrence Berkeley National Laboratory

A power transformer converts electrical power from one voltage level to another using electromagnetic induction between two coils of wire wound around a common iron core. The manufacturing challenge is the combination of massive physical scale, precise engineering tolerances, and specialized materials that make transformer production fundamentally difficult to expand quickly.

A large transmission-class transformer weighs 200 to 800 tons, contains hundreds of kilometers of copper windings, and is filled with thousands of liters of specialized insulating oil. Each unit is custom-engineered to specific voltage, impedance, and capacity requirements. There is no such thing as a standard, off-the-shelf transmission transformer. Every order triggers a new engineering design, custom core steel cutting, and hand assembly by workers with skills developed over years.

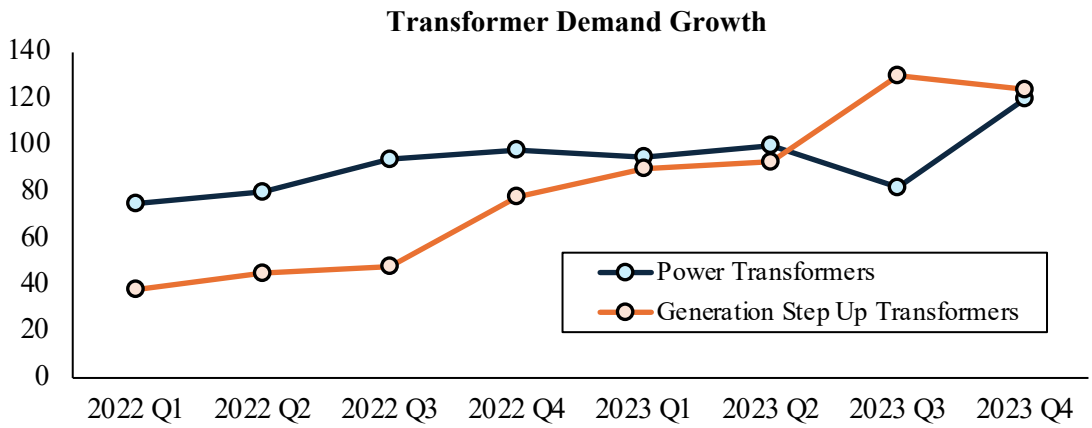
Why Lead Times have Exploded

US transformer manufacturing capacity was structurally underinvested for decades. Between the 1990s and 2015, grid expansion in most developed markets was slow, transformer demand was moderate, and domestic US manufacturing capacity contracted as imports from lower-cost producers filled the market. The combination of AI data center buildout, renewable energy project deployment (both solar farms and offshore wind require significant substation infrastructure), electric vehicle charging infrastructure investment, and general grid modernization programs has converged simultaneously to create a demand shock for which no supplier was prepared.

Current lead times for large power transformers (defined as transformers rated above 100 MVA) are 18 to 36 months from order to delivery in most markets. Data center substation transformers at medium voltage, typically custom units in the 50 to 200 MVA range, have stretched from four weeks historically to 18 to 24 months. The Congressional Research Service has designated large power transformer supply as a national security concern, noting that the United States imports a significant proportion of its large transformer fleet from South Korea, Mexico, and Europe.

The Supply Chain

1. Hitachi Energy is the global leader in transformer manufacturing with facilities across Europe, Asia, and the Americas. Its order backlog has extended to multi-year terms.
2. Siemens Energy is the second major global player, manufacturing across Germany and internationally.
3. GE Vernova (spun out of General Electric in 2024) manufactures transformers in the United States, benefiting from the domestic manufacturing premium that US buyers are increasingly willing to pay to reduce supply chain risk. Eaton and SPX Transformer Solutions are the largest purely domestic US transformer manufacturers.
4. WEG (Brazil) has been expanding US transformer manufacturing capacity.
5. ABB retains a position in medium-voltage distribution transformers.



The Investment Thesis

Transformer manufacturers operate in a market where demand now consistently exceeds supply, where products are custom-engineered (limiting competitive pressure on pricing), and where capacity expansion requires multi-year investment in factory floor space, specialized tooling, and skilled workforce development. This is a structural rather than cyclical demand uplift as AI data centers will require transformer upgrades for the foreseeable future regardless of the interest rate cycle. Companies with existing production capacity, established customer relationships, and US domestic manufacturing credentials are well positioned for a multi-year order book expansion.

Source: Wood Mackanzie

Beyond transformers, grid expansion requires a broader set of high-voltage equipment, each with its own supply chain dynamics and key suppliers.

High-Voltage Cables

Carry power from generation sites and substations to data center campuses. For data centers on large campuses requiring dedicated grid feeds, high-voltage underground cable is often required rather than overhead lines. The global market for high-voltage cables is dominated by Prysmian (Italy), Nexans (France), and NKT (Denmark).

Offshore wind and subsea interconnect demand has consumed much of the high-voltage cable manufacturing capacity in Europe, creating lead time extensions that also affect onshore data center power infrastructure. Prysmian has been one of the clearest beneficiaries of both the renewable energy buildout and the data center power investment cycle.

Switchgear isolates and routes power circuits within substations and data center power plants. High-voltage switchgear is supplied by ABB, Siemens Energy, GE Vernova, Schneider Electric, and Eaton. The switchgear market has similar supply constraints to transformers given that the same suppliers serve both markets and face the same demand surge across the same timescales.

Reactive Power Compensation Equipment

Maintains voltage stability on transmission networks carrying large, variable loads. AI data centers with fluctuating GPU utilization are significant sources of load variability that can affect grid voltage quality. Hitachi Energy and Siemens Energy are the dominant suppliers.

Grid Monitoring & Control

The sensors, protection relays, and control systems that manage power flow across the grid is a growing segment as utilities add intelligence to manage more complex, variable demand patterns.

GE Vernova's Grid Solutions division, Siemens Energy, and Schweitzer Engineering Laboratories are key players. Software-driven grid management platforms from companies like Itron and Landis+Gyr are increasingly relevant as utilities need to model and manage AI-driven load growth in their planning.

Once power enters the data center fence, it passes through a series of conversion and conditioning steps before reaching the GPU. Each step involves efficiency losses and represents a distinct supply chain.

The Incoming Substation

Steps medium-voltage utility power (typically 13.8 kilovolts or 34.5 kilovolts) down to the distribution voltage used within the facility. A hyperscale AI campus may have multiple dedicated substations on-site. The medium-voltage substation equipment is supplied by the same companies as utility-grade equipment. They are ABB, Siemens Energy, Eaton, Schneider Electric.

UPS Systems (Uninterruptible Power Supply)

Provide a buffer between the utility grid and the critical IT load. Data center cannot tolerate even millisecond power interruptions as a voltage sag lasting a few cycles will crash servers and corrupt data. The UPS system isolates the IT load from utility disturbances and provides battery-backed power during the seconds or minutes required for diesel generators to reach operating speed.

Traditional double-conversion UPS systems rectify incoming AC to DC, charge the battery bank, then invert back to clean AC. The IT load always runs from the inverter, completely isolated from grid disturbances where the best power quality but with 3 to 6% conversion overhead. Modern Eco-mode or delta-conversion UPS systems pass utility power directly to the load when grid quality is acceptable, with the inverter only providing correction, achieving efficiencies above 99% in normal operation.

Battery technology for data center UPS is transitioning from traditional VRLA (valve-regulated lead acid) to lithium-ion. Lithium-ion batteries offer higher energy density, longer cycle life of 10 to 15 years versus 4 to 6 years for VRLA, faster charging, remote monitoring capability, and smaller footprint. Vertiv, Eaton, and Schneider Electric are the dominant UPS system suppliers for data centers.

Generators

These are the last line of defense when the grid fails entirely. A large hyperscale data center has dozens of diesel generators, each 2 to 4 megawatts, staged in redundant configurations. On-site diesel tanks typically provide 24 to 48 hours of run time at full load. Extended outages require fuel delivery coordination. The transition toward gas turbine generators running on natural gas or hydrogen, and toward fuel cell systems from companies like Bloom Energy, is accelerating driven by emissions commitments and the efficiency advantages of natural gas.

Power Distribution Units (PDUs)

Step voltage from the UPS output down to the 208V or 480V that server racks receive. Within each server, a Switch Mode Power Supply (SMPS) converts incoming AC to the DC voltage rails needed by server components. Modern data center servers increasingly use 48-volt distribution internally — higher voltage means lower current at the same power, reducing resistive losses across the many meter of copper in a server chassis.

Voltage Regulator Modules (VRMs)

Takes the 12V or 48V server bus and step it down to the 0.7 to 1.0 volts that modern CPUs and GPUs operate at, with sub-millivolt precision. An H100 drawing 700 watts at approximately 1 volt requires 700 amperes of current delivered to the chip with extreme accuracy.

VRM ICs are supplied by Monolithic Power Systems, Renesas, and Infineon, all of which are benefiting from AI server proliferation multiplying the VRM content per rack.

Battery Energy Storage Systems (BESS) are one of the fastest-growing categories in the power infrastructure landscape, serving multiple roles simultaneously in the AI-driven electricity system. Understanding BESS requires understanding the distinct applications it serves, because the technology requirements and economics differ significantly across those applications.

Grid-scale BESS

At the utility scale, BESS installations of 100 megawatts to multiple gigawatts are deployed to provide two primary services. Frequency regulation involves responding within milliseconds to deviations in grid frequency (which must be maintained at 50 or 60 Hz) caused by sudden changes in generation or load. Batteries excel at this because they can inject or absorb power almost instantaneously. This is far faster than any thermal generator.

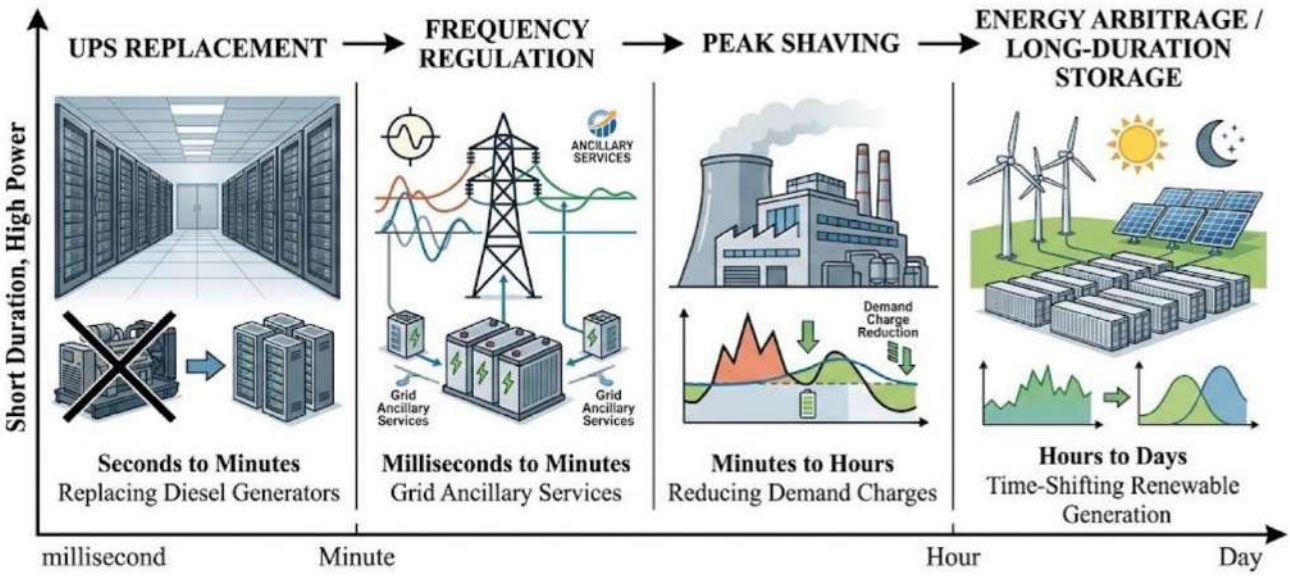
Energy arbitrage involves charging during periods of low electricity prices (typically overnight or during periods of high renewable generation) and discharging during peak demand periods when prices are high. AI data centers with predictable base load patterns create relatively stable demand signals that interact with variable renewable generation, creating arbitrage opportunities for co-located BESS.

Data Center BESS

At the data center level, BESS serves three roles.

1. First, it replaces or supplements diesel generators for backup power, providing cleaner, faster-responding energy storage with lower maintenance requirements and no fuel logistics challenges.
2. Second, it performs peak shaving — absorbing the difference between base load and peak GPU utilization spikes to reduce demand charges on utility bills, which at hyperscale can save tens of millions of dollars annually.
3. Third, it enables demand response participation, where the data center agrees to reduce grid consumption during grid stress events in exchange for payments from the utility or RTO.

BESS Application Spectrum



The Technology Landscape

Lithium iron phosphate (LFP) chemistry has emerged as the dominant battery technology for grid-scale and data center BESS applications. LFP offers lower energy density than the lithium nickel manganese cobalt (NMC) chemistry used in electric vehicles. However, it provides superior thermal stability (significantly lower fire risk), longer cycle life of 4,000 to 6,000 cycles versus 1,000 to 2,000 for NMC, and lower cost at scale. For stationary applications where energy density matters less than longevity and safety, LFP's trade-offs are highly favorable.

- CATL (Contemporary Amperex Technology, China) is the world's largest manufacturer of LFP batteries and supplies a significant proportion of global grid-scale BESS.
- BYD (China) is the second-largest.
- LG Energy Solution and Samsung SDI (both South Korea) are the leading non-Chinese suppliers.

The dominance of Chinese manufacturers in BESS supply creates a supply chain concentration risk analogous to the semiconductor equipment concentration in the Netherlands and Japan where hyperscalers and utilities deploying BESS at scale are increasingly seeking to qualify non-Chinese supply.

EnerSys, Fluence (a joint venture between Siemens and AES), and Powin are the leading system integrators for grid-scale BESS in Western markets, packaging cells from cell manufacturers into complete, grid-connected storage systems with power electronics and control software.

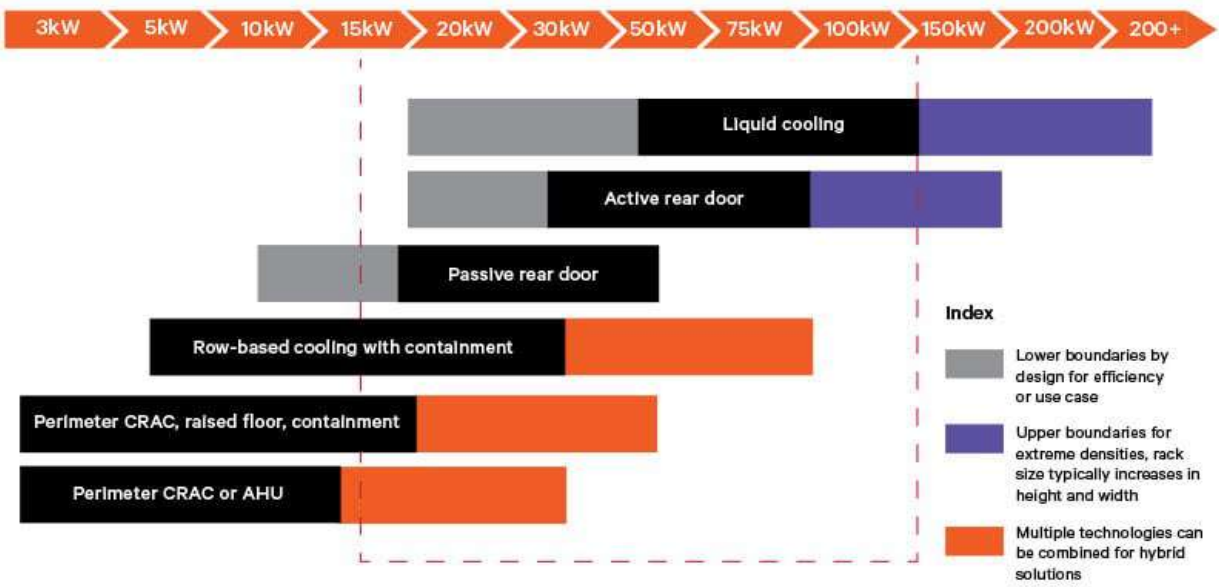
Long-Duration Energy Storage

The two-to-four-hour discharge duration of standard LFP BESS is sufficient for frequency regulation, peak shaving, and backup applications, but is insufficient to fully time-shift renewable generation across multi-day weather patterns. Long-duration energy storage technologies (that are capable of 8 to 100 hours of discharge) are the missing piece that would allow solar and wind generation to fully displace fossil fuel baseload. Several chemistries are in development. Flow batteries (vanadium redox, iron-air) scale energy capacity independently of power rating by increasing tank size, making them potentially cost-effective at long duration.

ESS Inc. and Form Energy (iron-air chemistry) are the most advanced Western long-duration storage developers. Commercialization at grid scale remains several years away for most technologies.

Every watt consumed by a GPU must be removed as heat. Cooling is the direct consequence of power consumption, and it accounts for 30 to 50% of total data center energy use. The metric that captures this relationship is PUE (Power Usage Effectiveness) which is the total facility power divided by IT equipment power. A PUE of 1.0 would mean 100% of power goes to computing with zero overhead. Traditional enterprise data centers run at PUE of 1.5 to 2.0. Leading hyperscale facilities with advanced cooling achieve PUE of 1.1 to 1.2.

IT Power Trends



Why AI is Breaking Air Cooling

Conventional data center cooling uses chilled air circulated through raised floors and server racks. Hot aisle and cold aisle containment prevents hot and cold air from mixing, improving efficiency. This approach has served the industry for decades and remains viable for CPU-dominated workloads at rack densities of 5 to 15 kilowatts.

However, a rack of H100 servers draws 60 to 100 kilowatts and a rack of GB200 NVL72 systems draws 120 kilowatts. The heat flux per square meter of floor space from these racks exceeds what air cooling can remove efficiently.

The Cooling Technology Spectrum

Rear-door heat exchangers (RDHx) attach to the back of server racks and use chilled water to absorb heat from the exhaust air before it reaches the room. This doubles effective cooling capacity per rack without requiring liquid cooling inside the servers and can be retrofitted to existing air-cooled facilities. RDHx is an effective bridge technology for facilities upgrading from CPU to GPU workloads without full reconstruction.

Direct Liquid Cooling (DLC) circulates chilled water through cold plates attached directly to CPUs and GPUs on the server motherboard. Water has approximately 1,000 times the volumetric heat capacity of air, enabling efficient cooling of racks at 60 to 100-plus kilowatt densities. DLC is increasingly standard for new AI server deployments and is required for GB200 and beyond. Vertiv, Asetek, Aavid (part of Boyd Corporation), and Coolcool are key suppliers of DLC systems.

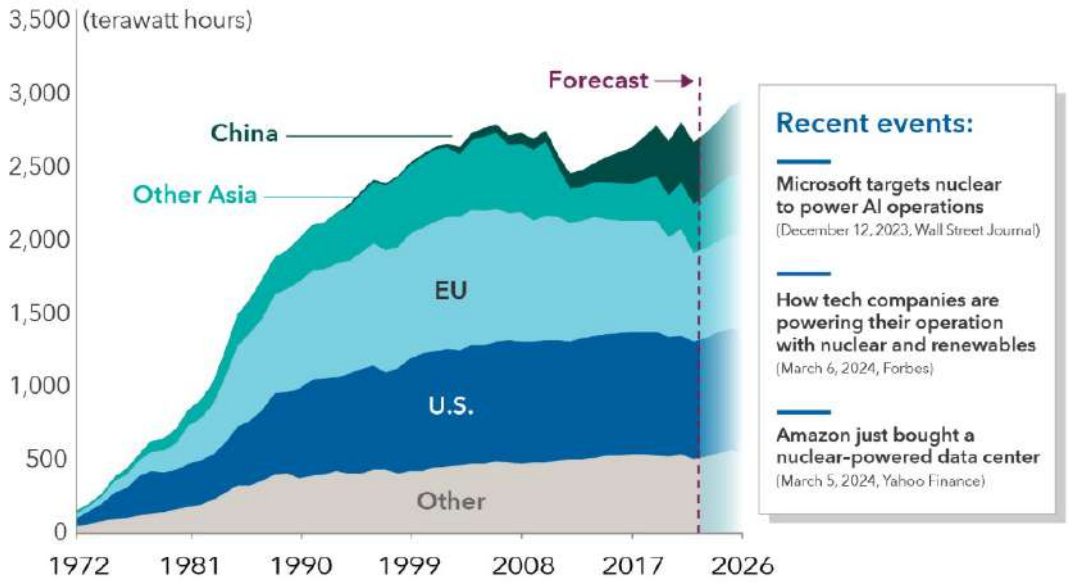
Immersion cooling submerges servers directly in a dielectric fluid. Single-phase immersion keeps the fluid liquid as the fluid is circulated through an external heat exchanger. Two-phase immersion allows the fluid to boil at the chip surface, with the vapor condensing on a cooling coil above. This phase dramatically increases heat transfer efficiency. Bute its drawbacks are higher upfront infrastructure cost, the engineering challenge of maintaining and servicing servers submerged in fluid tanks, and the need for purpose-built facilities rather than retrofits.

Vertiv and Schneider Electric are the dominant suppliers of data center cooling infrastructure across all categories. Both companies have guided to significant revenue growth from liquid cooling adoption and are benefiting from the forced infrastructure upgrade cycle that AI GPU density is driving.

Source: https://www.vertiv.com/en-asia/about/news-and-insights/articles/blog-posts/liquid-cooling-technology-towards-the-data-center-of-the-future/

Hyperscalers have a specific electricity requirement that makes nuclear uniquely attractive. They need large quantities of firm, 24/7, carbon-free power. Solar and wind are carbon-free but they do not produce power when the sun is not shining, or when the wind is not blowing. Natural gas is firm but not carbon-free. Nuclear is the only electricity source that is simultaneously firm, carbon-free, and scalable to the multi-hundred-megawatt blocks that large AI data center campuses require.

Expansion of Nuclear Power by Region



The Nuclear Renaissance Driven by Hyperscalers

- In 2023 and 2024, the four largest hyperscalers all made significant nuclear commitments:
- Microsoft signed a 20-year power purchase agreement to purchase electricity from the restarted Three Mile Island Unit 1 in Pennsylvania, now renamed the Crane Clean Energy Center. The plant had been shut down in 2019; Constellation Energy restarted it specifically for this contract. Microsoft pays a premium for the certainty of firm, carbon-free baseload power.
- Google signed a series of agreements with Kairos Power, a small modular reactor developer, to purchase power from commercial SMR deployments beginning in the early 2030s.
- Amazon acquired a data center campus adjacent to a nuclear power plant in Pennsylvania, establishing a physical co-location model where the data center takes power directly from the plant's dedicated interconnect.
- Oracle's Larry Ellison has publicly discussed designing new data center campuses around three small nuclear reactors as the dedicated power source.

Small Modular Reactors

The most strategically significant nuclear development for AI infrastructure is the emergence of Small Modular Reactors (SMRs). These are factory-built nuclear reactor designs rated at 50 to 300 megawatts, small enough to be sited on data center campuses or co-located with industrial facilities. SMRs promise faster construction timelines (5 to 7 years versus 15-plus years for conventional large-scale nuclear), lower upfront capital (factory manufacturing with standardized designs), and flexible siting options.

- NuScale Power was the first SMR design to receive NRC design approval in the United States.
- TerraPower (backed by Bill Gates) is developing a sodium-cooled fast reactor. X-energy is developing a high-temperature gas-cooled design.
- Kairos Power uses molten fluoride salt coolant. Several of these companies have received commitments from hyperscalers as anchor customers.

The critical caveat is that no SMR has yet achieved commercial operation in the United States. Timelines have repeatedly slipped, costs have increased above initial projections, and regulatory processes remain complex. NuScale's first planned deployment, called the Carbon Free Power Project, was cancelled in 2023 when projected electricity costs exceeded initial estimates. The hyperscaler commitments represent options on future nuclear capacity rather than near-term power solutions.

Sources: Capital Group, International Energy Agency (IEA)

If nuclear is the preferred destination, natural gas is how the industry gets there in the interim. Natural gas plants can be permitted in two to three years, built in an additional two to three years, and operated flexibly to match data center load patterns. They produce significantly fewer emissions than coal and no direct particulate pollution. For a data center that needs firm power now rather than after a decade of SMR development, natural gas is the default answer.

Combined Cycle Gas Turbines (CCGTs)

CCGTs are the most efficient gas generation technology, pairing a gas turbine with a steam turbine that recovers heat from the gas turbine exhaust. CCGTs achieve thermal efficiencies of 55 to 65%. They convert over half the chemical energy of natural gas into electricity, compared to 35 to 40% for older simple-cycle designs.

GE Vernova, Siemens Energy, and Mitsubishi Power are the three global gas turbine manufacturers, effectively an oligopoly in the large-frame gas turbine market. All three have multi-year backlogs driven by the combination of data center, power demand and the retirement of older coal and nuclear plants that gas is filling.

On-Site Natural Gas Generation

An increasingly attractive option for data centers seeking to reduce grid dependency. Rather than relying entirely on utility power through a potentially constrained grid interconnection, some hyperscalers are building private on-site gas generation, what has been called a "shadow grid." Big tech data centre developers are building private natural gas shadow grid power plants to sidestep strained grids. These on-site gas plants provide guaranteed power availability independent of grid interconnection queues and can be built faster than utility-scale generation requiring transmission infrastructure.

Bloom Energy's solid oxide fuel cells are being deployed at several hyperscale data center campuses, providing high-efficiency, low-emission on-site natural gas generation with greater reliability than combustion-based generators. Solid oxide fuel cells convert natural gas to electricity electrochemically rather than through combustion, achieving 60-plus percent efficiency with extremely low NOx emissions.

LNG and gas supply security is an important dimension of the natural gas bridge. Gas-fired generation requires a reliable supply of natural gas, which in many markets means exposure to pipeline constraints and gas price volatility. Regions without domestic gas production or pipeline access must import liquefied natural gas (LNG), adding supply chain complexity and cost. The geopolitical dimensions of LNG supply, particularly the US-Europe LNG trade that developed following Russia's invasion of Ukraine, are relevant context for data center power planning in Europe and Asia.

Renewable energy, solar and wind, is the cheapest form of new electricity generation in most markets and is the foundation of hyperscaler carbon neutrality commitments.

The Cost Revolution

The cost of solar photovoltaic electricity has fallen by approximately 90% over the past decade and continues declining. Utility-scale solar in favorable locations now produces electricity at below $20 per megawatt-hour which is cheaper than any fossil fuel alternative. Onshore wind has experienced similar cost reductions. This cost trajectory has driven enormous deployment where the United States added more solar capacity in 2023 than in any previous year, and global solar additions set records for the third consecutive year.

The Intermittency Problem

Despite this progress, solar and wind cannot alone solve the AI power problem because they produce electricity only when meteorological conditions permit. As mentioned briefly, solar produces nothing at night and reduced output in cloudy conditions and wind produces nothing during calm periods. A data center running 24 hours a day, 365 days a year.

The Hybrid Reality

In practice, hyperscalers use renewable energy purchases (through power purchase agreements or direct ownership) to meet carbon accounting targets and source low-cost electricity during daylight hours, combined with firm generation (gas or nuclear) to ensure 24/7 power availability. BESS (discussed in Section 7) plays a growing role in smoothing the gap between renewable production and data center demand at timescales of hours. Long-duration storage (8 to 100 hours) would substantially reduce the amount of firm generation required, but commercial long-duration storage at the required scale is not yet available.

Regulated electric utilities with significant data center load growth in their service territories represent an underappreciated investment category at the intersection of technology infrastructure and energy infrastructure.

The Rate-Base Mechanism

Regulated utilities in the United States operate under a rate-of-return regulatory model. The utility is permitted to earn a regulated return (typically 9 to 11%) on its rate base, which is the value of the capital it has invested in infrastructure serving customers.

When a utility builds a new substation, transmission upgrade, or generation investment to serve growing data center load, that capital investment is added to the rate base, increasing the asset base on which the regulated return is earned. Data center load growth is a capital investment catalyst that expands the earnings base for years or decades.

Virginia: The Most Concentrated Data Centre Utility Market

Dominion Energy serves Northern Virginia, which is the largest data center market in the world by installed capacity. Fairfax and Loudoun Counties alone contain more than 100 data center representing tens of gigawatts of power demand. Dominion's capital expenditure program has been driven substantially by data center load growth, and its regulated earnings have expanded accordingly. Understanding Dominion's rate case outcomes and data center pipeline disclosures is essential for anyone analyzing the company as an AI infrastructure proxy.

Georgia Power and Duke Energy are the other two US utilities most exposed to concentrated data center load growth, with Meta, Microsoft, Google, and Amazon all operating or building large campuses in their service territories. AEP (American Electric Power) serves significant data center concentrations in Ohio and Texas.

International Utility Exposure

In Europe, data center concentration in Ireland (serving the EU market with favorable English-language business environment and tax structure), the Netherlands, and the Nordic countries (with abundant hydroelectric power and natural cooling from cold climates) creates specific utility exposure. Elia (Belgium/Germany), RTE (France), and National Grid (UK) are managing data center load growth that is reshaping their capacity planning assumptions.

The Risk: Load Growth that Arrives Faster than Infrastructure

The inverse of the opportunity is the risk that data center load growth arrives before the utility has completed the infrastructure upgrades required to serve it reliably. A utility that cannot connect new data centers due to substation constraints or transformer lead times loses load growth revenue while still carrying the capital cost of partially completed infrastructure. Managing the timing mismatch between data center development timelines and utility infrastructure timelines is the primary operational challenge for utilities in high-density data center markets.

Humanoid Robots Primer Report



Real Application Key Terms

Terms	Definitions
Actuators	They are the "muscles" and "joints," acting as specialised components that convert electrical, hydraulic, or pneumatic energy into physical movement. They empower robots to perform tasks like walking and lifting by providing torque, speed, and force through motors, gears, and sensors.
Cameras	Specialised, high-definition optical sensors and imaging systems that act as the robot's "eyes," enabling real-time environmental perception, object recognition, navigation, and manipulation. These systems integrate with AI to process spatial data, allowing robots to understand their surroundings and interact safely within dynamic, human-centric environments.
Coreless Micro Motors	A specialised type of DC permanent magnet servo motor designed without an iron core in the rotor. They act as high-precision, low-inertia actuators primarily used in dexterous hands, fingers, and robotic eyes. Their unique "hollow-cup" design allows for incredibly fast response times (3-5 times faster than conventional motors) and superior energy efficiency (often exceeding 80% efficiency).
Cycloidal Drives	A high-performance compact gear transmission mechanism utilised in humanoid robots, collaborative robots (cobots), and exoskeletons to convert high-speed, low-torque rotation from a motor into low-speed, high-torque output with minimal backlash. It is characterised by its "wobbly" motion principle, offering superior torque density and shock resistance compared to traditional gears, making it ideal for robotic joints that require high payload capacity and durability.
Dextrous General Manipulation	The capability of a multi-fingered end-effector (robotic hand) to perform a wide variety of complex, high-precision, and versatile object manipulation tasks, similar to a human, across unstructured environments, with minimal reprogramming required for unseen objects or different tasks.
Encoder	A high-precision sensor, typically integrated into joint actuators, that converts mechanical movement (rotational angles or linear displacement) into electrical signals (digital or analog). Acting as the "eyes" of the robot's motion control system, it provides real-time feedback on the exact position, speed, and direction of joints.
Exteroceptive Sensing	The use of sensors to gather information about the external environment, distinct from proprioceptive sensors that monitor the robot’s internal state (e.g., motor speed, battery level, joint angles). These sensors enable the robot to perceive its surroundings, detect objects, avoid obstacles, and interact with the physical world, functioning similarly to human senses such as sight, hearing, and touch.

Real Application Key Terms

Terms	Definitions
Field Oriented Control (FOC)	A sophisticated motor control technique used in high-performance robotics to manage brushless DC (BLDC) and induction motors. In humanoid robots, FOC is critical for achieving smooth, efficient, and precise joint movement by controlling the magnetic field within the motors. It converts 3-phase AC currents into a rotating reference system aligned with the rotor's magnetic field. It treats AC motors like simple, linear DC motors, allowing independent, precise control over torque and magnetic flux.
Frameless Torque Motors	A high-torque, compact, permanent-magnet brushless servo motors designed without a housing, shaft, or bearings. Consisting only of a stator and rotor, they are directly integrated into humanoid robot joints, providing high power density, superior torque-to-inertia ratios, and direct-drive capabilities, essential for agile movement and compact, lightweight design. They consist of 2 core components: the stator (windings) and rotor (magnets). They lack a dedicated casing, shaft, or bearings, allowing them to be integrated directly into the structure of a robot actuator.
Harmonic Drives	A strain wave gear, a specialised, high-precision, low-backlash, and compact gear transmission mechanism that serves as the core "joint" technology in humanoid robots. harmonic drives convert high-speed, low-torque output from electric motors into low-speed, high-torque motion, allowing robots to replicate human-like movements with exceptional dexterity, smoothness, stability.
Humanoid Robots	They are designed to operate in environments built for humans (stairs, doors, using tools) without requiring costly redesigns of factories or homes. They use AI (specifically large vision and language models) to perceive, reason, and act. Actuators ("Muscles"): High-torque electric motors at joints. Sensors ("Senses"): Cameras (vision), LiDAR (mapping), and IMUs (balance). Control System ("Brain"): AI that processes data to walk, manipulate objects, and interact with humans.
Impedance Control	A control strategy that manages the dynamic relationship between a robot's motion (position/velocity) and the forces it exerts on its environment, making the robot behave like a programmable mass-spring-damper system. It enables compliant interaction, safety, and adaptability to uncertain environments.
Light Detection and Ranging (LIDAR)	An active, 3D laser-based perception system that allows robots to "see" and map their environment by emitting laser pulses and measuring the time it takes for them to reflect back. It acts as a core sensory system for navigation, obstacle avoidance, and mapping, providing dense 3D point cloud data, a real-time 3D model of surroundings.

Real Application Key Terms

Terms	Definitions
Model Predictive Control (MPC)	An advanced, model-based control strategy that calculates optimal robot actions by predicting future behavior over a finite time horizon while adhering to physical constraints (e.g., joint limits, obstacle avoidance). It is a receding horizon strategy, meaning it solves an optimisation problem at each time step, applies only the first input, and then repeats the process with updated feedback. It is an iterative, optimal, and feedback-based control strategy. It consists of 3 main components: a predication model, a cost function, and an optimal algorithm.
Motion Control Units (MCU)	They are specialised, high-performance embedded microcontrollers or System-on-Chips (SoCs) that function as the robot's "cerebellum". They are responsible for managing real-time, low-level motion tasks, such as balance, motor coordination, joint torque, and position control, by processing sensor feedback and executing complex, high-speed algorithms within milliseconds. MCUs act as the "little brain," performing essential, real-time control, including walking gait generation, balance maintenance, and force control. MCUs operate on very high-frequency feedback loops (often in the kilohertz range) to instantly adjust motor speeds and positions in response to environmental changes (e.g., uneven ground, external forces).
Original Equipment Manufacturer (OEM)	Semiconductor companies design, brand, and sell their own chips or specialised production equipment, retaining full intellectual property (IP) rights. These firms, such as top equipment suppliers Applied Materials and ASML, are vital for developing advanced manufacturing tools for 3D NAND and AI applications.
Perception System	A comprehensive, multi-layered, and intelligent assembly of hardware sensors and software algorithms designed to enable the robot to sense, interpret, and understand its internal state and external environment in real-time. It functions as the robot's sensory organs and nervous system, allowing it to mimic human-like awareness to interact with surroundings, recognise objects, and operate in human-centric, unstructured environments.
Proprioceptive Sensing	Enables robots to measure their internal states, such as motor speed, joint angles, and torque, monitoring how they move and function. Unlike exteroceptive sensors that perceive the environment, these internal sensors (encoders, IMUs) are essential for balance, kinematic modeling, and, when combined with machine learning, enabling robots to interpret contact and adapt to complex, unstructured terrains without cameras.
Quadra Flow Cooling Systems (QFCS)	Advantech's QFCS is an advanced, compact, and highly efficient thermal management solution designed for edge computing and industrial applications. A high-performance, low-noise, and compact heatsink designed for high-computing applications.

Real Application Key Terms

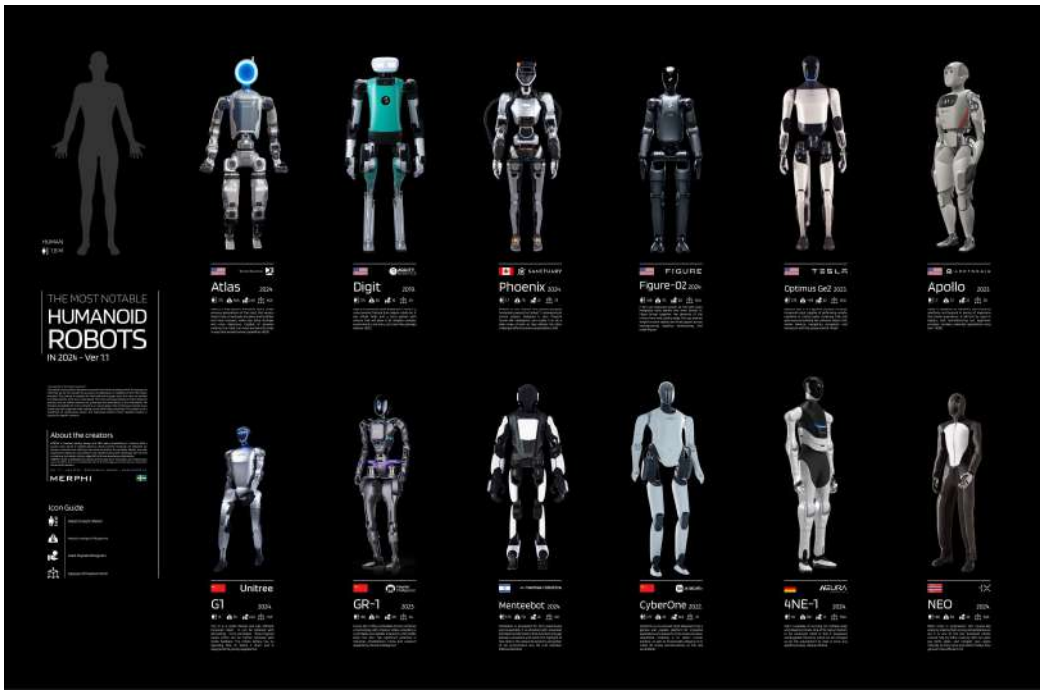
Terms	Definitions
QFCS Efficiency	Features 2-phase heat exchange pipes and a mixed-fin structure for maximum thermal dissipation. **Performance:** Capable of maintaining 100% performance without throttling at 60°C (e.g., in 45W TDP scenarios). **Design:** Uses a "vertical airflow" design and direct contact copper block with the CPU. **Features:** Low noise (under 38dB in certain models), tool-free installation, and thin, lightweight construction.
Radar	Measures the speed and distance of surrounding vehicles. AI processes this sensor data to predict the behavior of others and send instructions to steering, braking, and acceleration actuators.Ranging from Level 0 (no automation) to Level 5 (full automation in all conditions). Most current "full" systems are actually Level 3 or 4, operating within specific "geofenced" areas.
Robotics-as-a-Service (RaaS)	An emerging business model where humanoid robots, bipedal, human-like machines, are provided to customers not as a one-time capital purchase, but as a flexible, subscription-based, or pay-per-use service. In this model, the robotics developer retains ownership of the hardware, delivering the robot alongside its AI software, maintenance, and continuous upgrades for a recurring fee. This model enables companies to treat high-cost humanoid robotic labour as an operational expense (OpEx) rather than a heavy capital expense (CapEx), allowing for easier deployment, faster scaling, and reduced operational risk.
Self-Driving Cars	Vehicles capable of sensing their environment and navigating without human involvement. They use a "sensor fusion" approach—combining data from multiple sources to create a real-time, 3D map of their surroundings
Software-Defined Vehicles (SDV)	A vehicle whose features, functions, and performance are primarily managed, updated, and enhanced through software rather than hardware. SDVs use centralised computing, modular architecture, and over-the-air (OTA) updates to continuously upgrade vehicle capabilities throughout their lifespan.
Tactile Interaction	The process of sensing and responding to physical contact, such as pressure, force, vibration, or texture, between a robot and its environment or a human. It enables robots to detect touch, interpret tactile data for tasks like manipulation, and facilitate safe, intuitive human-robot communication.
Teleoperations	The remote control and operation of a robotic system by a human from a distance, typically through a master-slave configuration using communication links. It allows operators to manipulate environments in real-time using interfaces like joysticks, VR headsets, or exoskeletons, often providing "telepresence" through sensory feedback.

Real Application Key Terms

Terms	Definitions
Vision Language-Action Models (VLA)	A specific, advanced subset of VLMs designed for physical robots, are unified, end-to-end AI frameworks that map visual observations and language instructions directly to executable motor commands (actions). They process synchronised data from cameras (RGB/Depth), microphone (speech), and proprioceptive sensors (gripper pose, joint angles). Unlike traditional, modular robotics (separate modules for seeing, planning, and acting), VLAs integrate these processes into a single neural network architecture, often transformer-based. They directly output action tokens or continuous motor commands (e.g., joint velocities) rather than merely selecting from pre-programmed skills.

For decades, robots were capable machines with limited intelligence. They could weld, paint, and assemble with extraordinary precision in structured environments, but they could not adapt to unexpected situations, understand natural language instructions, or operate in the unstructured chaos of the real world. What has changed in the past three years to make physical AI a credible near-term deployment story is the convergence of three shifts simultaneously.

Cool Robots



The AI Shift

The most important change is the least obvious to someone watching a robot video. The physical hardware of humanoid robots such as the actuators, the frame, the sensors has existed in roughly its current form for a decade.

What has changed is the intelligence. Large language models and their multimodal descendants have given robots the ability to interpret natural language instructions, reason about novel situations, and generalize learned behaviors to new contexts. A robot trained on a warehouse task no longer needs to be reprogrammed from scratch for a slightly different warehouse layout. Vision-Language-Action models, discussed in detail in Section 4, represent a qualitative shift in robot capability that is directly analogous to what GPT-3 represented for natural language processing in 2020.

The Cost Shift – the Moore's Law Moment for Robotics

Unitree's G1 humanoid is available for $16,000. In 2023, comparable capability cost $500,000. That is a 97% cost reduction in approximately two years which has a similar trajectory that tracks closely to the early cost curves of smartphones and solar panels.

The primary driver is China's domestic actuator and motor supply chain reaching scale, combined with Unitree's highly integrated local sourcing model where 90% of vendors sit within a few hours of its Hangzhou factory. At $16,000 a unit, they are approaching the territory where a Robot-as-a-Service subscription at $500 per month can compete with warehouse labor costs.

Labor Scarcity

The demographic reality of most developed economies is a shrinking working-age population and chronic labor shortages in exactly the sectors, such as logistics, light manufacturing, elder care, agriculture, where humanoid robots are initially most deployable. This is not a temporary economic cycle phenomenon. Japan, South Korea, Germany, and the United States all face decades of declining labor force participation in these sectors regardless of economic conditions. The humanoid robot addresses a structural demand that grows more acute over time.

Source: https://www.reddit.com/user/HosSsSsSsSsSs/

The obvious objection to humanoid robots is, why does it have to be human-shaped?

Any robot designed for a specific task will outperform a humanoid at that task. Industrial robotic arms are faster, stronger, and more precise at repetitive manufacturing tasks than any humanoid. Autonomous forklifts are better at moving pallets. Robotic vacuum cleaners are better at cleaning floors. Why build a general-purpose machine in the shape of a human?

It is due to built environment.

The world has been designed for the human body. For example, door handles are at human hand height, stairs have human-scale risers, car pedals are at human leg distance and shelving is at human reach height. A bolt pattern on a BMW body cell was designed to be torqued by a human wrist holding a specific tool. Retrofitting any of this for a non-humanoid robot requires rebuilding the environment which is a new conveyance tracks, modified fixtures, custom tooling. Slip a humanoid robot into that environment and it operates with the same tools, in the same spaces, on the same fixtures the human workforce already uses. There is no guardrails to re-weld, no need for any new conveyance track and no retrofitting.

This is the economic argument for the humanoid form. The cost of adapting the robot to the environment is zero, because the environment was already built for this body shape. The alternative that one may suggest is adapting the environment to the robot. However, this has a cost that dwarfs the cost of the robot itself at scale.

The secondary argument is generality.

A single humanoid robot can, in principle, perform every physical task a human worker can perform. That includes tasks across different environments, different employers, and different contexts. That same unit that works a warehouse day shift can theoretically operate a different facility's night shift.

The third argument is the teleoperation bridge.

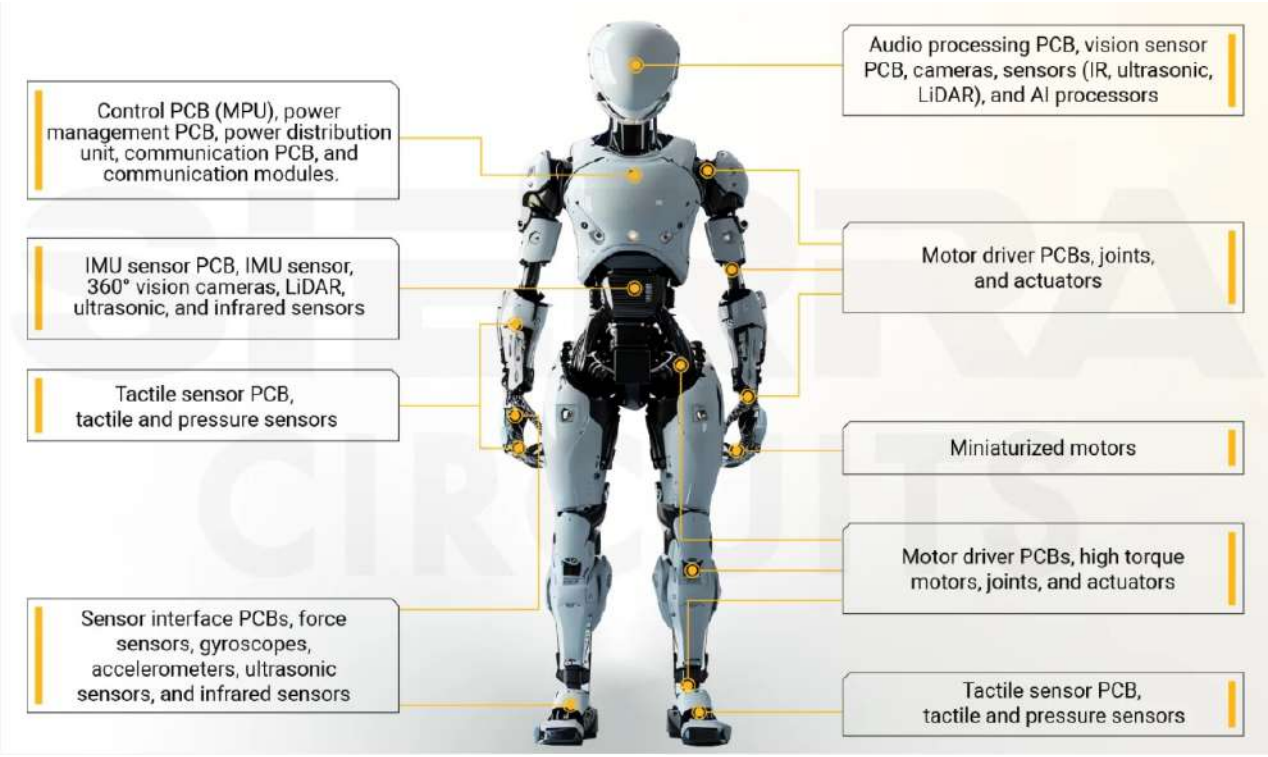
Even before full autonomy is achieved, a humanoid robot operated remotely by a human worker in a lower-cost geography can perform physical labor in a developed market at dramatically reduced cost. One teleoperation supervisor in a lower-cost geography can many robots simultaneously and steps in when the AI brain encounters situations it cannot handle.

This so-called hybrid model (part autonomous, part remotely supervised) means the economic case does not require waiting for artificial general intelligence. It requires only that remote operation latency falls below approximately 40 milliseconds, which 5G infrastructure can already deliver.

At the component level, a humanoid robot is a system of seven interdependent subsystems. Understanding each subsystem independently is essential because the investment opportunities, the supply chain concentrations, and the competitive dynamics differ substantially across them.

Like a human body, the robot has senses (cameras, LiDAR, depth sensors, torque sensors), a brain (an AI chip running VLA models), a nervous system (motion control units translating brain instructions into joint commands), muscles (actuators that move the joints), a skeleton (the structural frame), a bloodstream (the battery and power management system), and a communications layer (the wireless stack enabling teleoperation and fleet coordination).

Anatomy of a Human Robot



The cost structure is counterintuitive for investors accustomed to AI infrastructure themes. In a data center, compute dominates the bill of materials while in a humanoid robot, it is the actuators and gearboxes. These are the mechanical components that make joints move and accounts for approximately 40 to 45% of the total bill of materials. AI compute chips account for perhaps 10 to 15%.

This means the cost reduction path for humanoid robots runs primarily through mechanical engineering and manufacturing scale rather than chip advancement, which has important implications for which companies capture value as volumes scale.

Source: https://www.protoexpress.com/blog/humanoid-robotics-pcb-design-and-advanced-manufacturing-techniques/

The most significant AI development specifically relevant to robotics is the emergence of Vision-Language-Action models, or VLAs.

Earlier generations of robots ran on hand-coded motion sequences. This might include explicit instructions for every possible situation the robot might encounter. This approach works well for highly structured, repetitive environments (a car assembly line that never changes) but fails completely when the environment varies unexpectedly (a warehouse where boxes arrive in different configurations, orientations, and sizes). Reprogramming a robot for each new task or environment required specialist engineers and weeks of work.

VLA models collapse the three separate processing stages, known as perceive, plan, act, into a single multimodal model that takes raw sensor inputs (camera images, depth data, proprioceptive feedback from the joints) and natural language instructions, and outputs motor commands directly. In simple terms, you can tell the robot what to do in ordinary language, show it the environment through its cameras, and it figures out the joint-by-joint movements needed to accomplish the task without explicit programming.

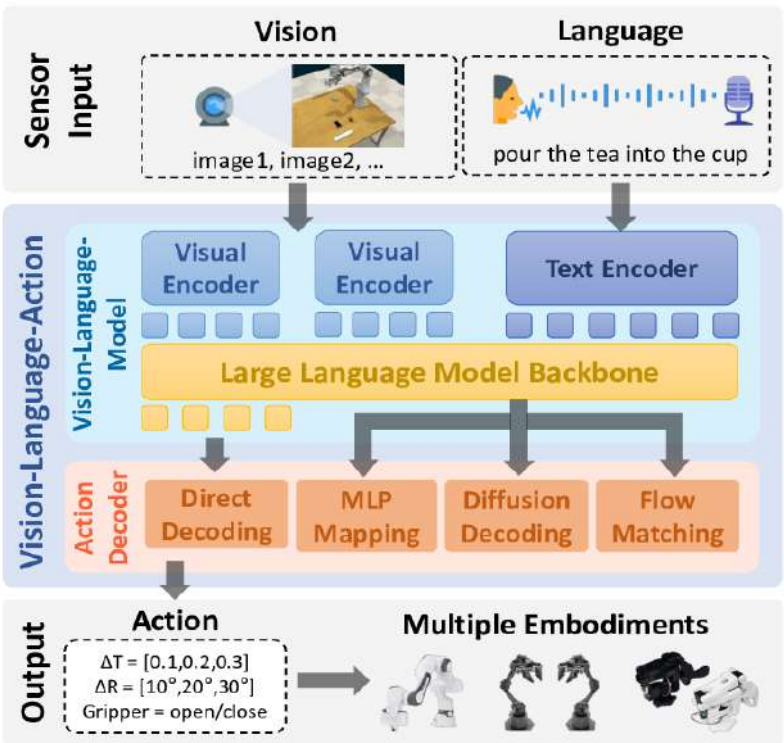
Figure AI's Helix model, currently running on BMW pilot production lines, exemplifies the architecture the industry is converging on. Helix runs two parallel loops that are similar to Daniel Kahneman “Dual-System Foundation Model”.

- A slow System 2 reasoning loop at 7 Hz that handles scene understanding and task planning
- A fast System 1 visuomotor policy at 200 Hz that handles moment-to-moment balance and precise manipulation.
- The slow loop tells the robot what to do and the fast loop executes it in real time with sub-50 millisecond latency.
- This dual-loop architecture mirrors the cognitive structure of human decision-making such as deliberate reasoning for novel situations and fast reflexes for learned motions.

Nvidia's GR00T N1, released in early 2025, is the first open foundation model specifically designed for humanoid-scale VLA applications. It is bundled with Isaac Sim, Nvidia's physics-based robot training simulator, and can be fine-tuned on partner fleet data from Agility Robotics' Digit, Apollo, and Figure robots.

The practical implication is that a robot OEM that logs its operational data in ROS (Robot Operating System) bag files can now fine-tune a foundation model from Nvidia rather than building one from scratch, collapsing AI development cycles from twelve months to potentially four weeks.

VLA Model Architecture



Source: https://www.researchgate.net/figure/Architecture-and-pipeline-of-VLA-models-A-VLA-model-comprises-a-VLM-and-an-action_fig1_397595915

The data flywheel is the most important long-term competitive dynamic in VLA development. Whoever owns the largest fleet of deployed robots accumulates operational data at a rate that compounds with fleet size. This data is used to fine-tune the VLA model, which improves performance, which enables more deployments, which generates more data. Tesla understands this dynamic from its Autopilot development where the company with the most vehicles on the road accumulates training data at a rate that purpose-built robotics companies cannot match. The same logic applies to humanoid robots, which is why both Tesla Optimus and Unitree are racing to deploy fleets rather than simply sell individual units.

Actuators are the robot's muscles. They convert electrical energy into the mechanical motion that moves joints.

Rotary Actuators

They spin to create torque. They are mechanically simpler and currently more cost-effective, which is why they dominate today's humanoid designs. Unitree's H1 uses a rotary-only joint configuration. Rotary systems require reduction gearboxes to convert the motor's high-speed, low-torque output into the low-speed, high-torque output useful for moving a limb.

Gearbox architectures compete across different joint requirements.

1. Harmonic drives (also called strain-wave gearboxes) offer very high precision and zero backlash, as such the joint moves exactly where commanded with no slop, making them preferred for precise manipulation tasks. They are also the most expensive gearbox type and have limited torque capacity. Harmonic Drive Systems of Japan holds the original IP and manufacturing lead, with LeaderDrive of China as the primary lower-cost challenger.

2. Cycloidal drives sit between the two in cost and are extremely durable, making them preferred for rugged industrial applications and heavier-duty joints. Japan's Nabtesco dominates industrial cycloidal drives, while Shuanghuan Driveline and Sumitomo Heavy are significant players. Spinea, owned by Timken, produces ultra-compact cycloidals that fit humanoid joint geometries.

Linear Actuators

They push and pull rather than rotate, offering significant performance advantages over rotary systems for specific joint types. Tesla's Optimus is approximately half rotary and half linear, which contributes substantially to its cost premium over Unitree's rotary-only H1. Linear actuators offer zero-power holding (the screw's self-locking action maintains position without continuous power draw, unlike rotary actuators that must fight gravity continuously), lower heat generation (no constant power burn against gravity reduces motor temperature, extending continuous operating time from approximately one hour for comparable rotary setups to two to three hours), and higher load capacity per unit weight.

Summary Table

Actuator Type	Precision / Backlash	Torque Density	Cost	Durability / Cycle Life	Primary Application	Dominant Supplier	Chinese Challenger
Harmonic Gear (Rotary)	Excellent (Near Zero)	High	High	Moderate	Wrists, Shoulders, Small Joints	HD Systems (Japan)	LeaderDrive
Planetary Gear (Rotary)	Good (Low)	Moderate-High	Moderate	High	Hips, Shoulders, High-Load Joints	Schaeffler / Wittenstein	ZD Leader
Cycloidal Gear (Rotary)	Moderate-Good	Very High	Moderate-High	Very High	Main Joint / Heavy Lift	Nabtesco (Japan)	Shuanghuan
Ball Screw (Linear)	Good-High	Moderate	Moderate	Moderate	Initial Prototyping, Mid-Load Joints	NSK / THK (Japan)	HGT / PMI
Roller Screw (Linear)	Excellent (High)	Very High	Very High	Highest	Knee, Ankle, High-Load Actuation	SKF (Sweden)	Shanghai Beite

6. Actuation

Planetary roller screws are the premium linear actuator component, lasting approximately five times longer than ball screws under comparable load cycles. They are one of the most supply-constrained components in the humanoid supply chain. SKF of Sweden is a major supplier. Shanghai Beite Technology announced a 1.85-billion-yuan investment in 2024 specifically to build planetary roller screw production capacity, illustrating the strategic importance China's supply chain is placing on this component.

The motors inside the actuators are important as well. Frameless torque motors (a ring of rare-earth magnets against electrified copper windings, bolted directly into a joint without a separate housing) are used by Optimus across 28 joints for lifting assistance. They require neodymium-iron-boron (NdFeB) permanent magnets, placing rare earth magnet suppliers including MP Materials (US), JL Mag (Hong Kong), and Lynas Rare Earths (Australia) in the supply chain.

At the fingertip level, where space constraints require miniaturization, coreless micro motors are used. Their supply is dominated by specialist private companies, with Chinese challengers Moons' Industries and Shenzhen Zhaowei emerging as lower-cost alternatives.

A robot's sensors are its connection to the physical world. The sensor suite determines what the robot can perceive, and perception quality directly limits what the robot can safely and accurately do.

Cameras

Cameras provide color, texture, and motion information. Most humanoid designs use stereoscopic RGB-D (colour plus depth) cameras for close manipulation tasks. The binocular arrangement allows the robot to estimate depth from parallax. Tesla's Autopilot heritage has led it to bet on a pure camera array for Optimus, trusting VLA models to infer depth from motion parallax without dedicated depth sensors. Most other designs blend camera types with additional sensing modalities.

LiDAR (Light Detection and Ranging)

LIDAR uses pulsed laser light to measure precise distances to surfaces, building a 3D point cloud of the environment. LiDAR is particularly valuable for SLAM (Simultaneous Localization and Mapping). The SLAM algorithm allows a robot to build a map of its environment while simultaneously tracking its own position within that map. Structured-light and time-of-flight depth modules provide similar information at lower cost and shorter range, used for shelf-level manipulation tasks.

IMUs (Inertial Measurement Units)

IMUs combine accelerometers and gyroscopes to track the robot's orientation and acceleration continuously. They are critical for balance and gait stability as the robot needs to know whether it is tilting, accelerating, or rotating at all times to maintain upright posture.

Encoders

Report the exact rotational or linear position of each joint, providing the proprioceptive feedback that tells the motion control system whether the commanded joint movement actually occurred. Dual encoders with active thermal compensation are used in demanding applications where precise position tracking must be maintained across temperature variations and mechanical wear.

Torque & Force Sensors

Allow the robot to detect contact and measure the forces it is applying. Novanta and Sensata are key suppliers. These sensors are what allow a humanoid robot to handle an egg gently, basically knowing how hard it is gripping and adjusting the grip force in real time, rather than simply moving to a commanded position without regard for what it encounters.

The computing that fuses all of these sensor streams into a coherent world model must operate under 50 milliseconds which is the approximate latency of human reflexes. At higher latency the robot appears and behaves clumsily, stumbling over unexpected objects and misjudging distances.

The AI Compute Layer

This is where Nvidia's dominance from the data center extends into the physical robot. The Jetson Orin NX, a palm-sized system-on-module delivering 275 trillion operations per second (TOPS), serves as the robot's frontal cortex. It runs the VLA inference, processing sensor fusion, and handling scene understanding. The forthcoming Jetson Thor triples this compute capacity. At approximately $60 per unit, the Jetson represents outstanding compute-per-dollar for edge inference.

Nvidia's competitive moat in robotics compute is not primarily the chip itself, it is also the software ecosystem around it. Isaac Sim provides a physics-accurate virtual environment where robots can be trained across thousands of parallel scenarios before deployment.

- GR00T provides the foundation model.
- The CUDA programming environment that the entire AI developer community knows connects data center training to edge deployment.
- Switching to a non-Nvidia chip for robotics compute means abandoning the entire Isaac/GR00T/CUDA toolchain, which is a multi-year re-engineering cost that most robotics companies are unwilling to pay.

But rival compute options exist. They include AMD's Versal, Intel's range of edge AI modules, Horizon Robotics' Journey 6 (a Chinese alternative with strong cost positioning), and Cambricon. For Chinese OEMs deploying at scale in the domestic market and seeking to reduce US chip dependency, Horizon Robotics is the increasingly credible alternative. Outside China, Nvidia's software moat keeps most developers on Jetson.

The Motion Control Layer

Sits one level below the AI chip and handles the real-time execution of joint commands. While the VLA model reasons about what to do at 7 to 200 Hz, the motion control units (MCUs) execute those instructions at over 2,000 samples per second, translating the high-level command "swing left arm forward" into thousands of tiny torque adjustments per actuator while simultaneously maintaining whole-body balance.

This requires automotive-grade microcontrollers positioned physically close to the actuators rather than sharing space with the hotter, higher-power AI chip. These chips run on mature 28nm nodes and are relatively inexpensive, but they are safety-critical and must meet automotive reliability standards. Texas Instruments, Infineon, STMicroelectronics, and Renesas are the dominant suppliers.

Unitree's G1 achieves approximately 2 to 3 hours of operation per charge under normal working conditions. However, real-world industrial deployments require longer continuous operation where the economics of a Robot-as-a-Service model depend on the robot running multiple shifts, not requiring a two-hour recharge after every two-hour work period. The combination of motor heat buildup (which triggers thermal throttling and forced cooling pauses) and energy density limitations in current battery cells is the primary operational constraint on humanoid robot deployment at scale.

The relevant metric is watt-hours per meter of locomotion which is the robot's miles-per-gallon equivalent. Efficiency gains will come from the following directions

- Drivetrain optimization (frameless torque motors and roller screw linear actuators that run cooler and require less continuous power to hold position), smarter gait algorithms that minimize unnecessary energy expenditure in stance and locomotion, and battery cell advances that increase energy density.
- High-C-rate battery cells, cells capable of both rapid charging and sustained high-current discharge during peak movement, are the specific battery requirement that humanoid robots impose. Samsung SDI is trialing 46-millimetre-format cylindrical cells for humanoid applications. CATL's leadership in pack modularity is strategically important as the industry moves toward standardized battery formats that enable hot-swap battery trays (analogous to DJI's drone battery ecosystem) rather than full recharge cycles.

Autonomous charging infrastructure also becomes critical at fleet scale. A humanoid robot that cannot autonomously return to its docking station at the end of a task cycle cannot operate unsupervised. Docking station design, whether inductive charging pads, motorized floor connectors, or hot-swap battery bays, is an emerging supply chain opportunity for companies including Amphenol (high-cycle connectors), Advantech (robotic charging bay controllers), and CATL (modular pack formats).

10. Teleoperation

The most underappreciated near-term deployment model for humanoid robots is not full autonomy but teleoperation. Teleoperation is remote human control of the robot through a low-latency communication link and haptic interface.

The model works as follows. A human operator in a lower-cost geography wears a haptic glove and views a real-time video feed from the robot's cameras. The operator controls the robot's movements with sub-50 millisecond round-trip latency which is the threshold at which the line between remote pilot and local muscle memory begins to blur. When the robot encounters a situation its autonomous AI cannot handle, the operator steps in. As the operator's interventions are logged and used to fine-tune the VLA model, the frequency of required interventions decreases over time. The robot gradually becomes more autonomous as it accumulates real-world training data from the supervised episodes.

This creates a labor arbitrage dynamic that is similar to the call-center offshoring wave of the early 2000s. One teleoperation supervisor managing four to five robots simultaneously, working from a geography with lower labor costs, can perform physical work in a developed market at a fraction of the direct labor cost. The technology that enables this is not new. For instance, remote surgery has used teleoperation for decades. However, the cost of the hardware and the quality of the communications infrastructure have only recently reached the thresholds required for industrial-scale deployment.

Teleoperation Visualization



The communications infrastructure requirements are specific.

- Round-trip latency below 40 milliseconds, sufficient bandwidth for real-time video streams from multiple camera angles, and reliability sufficient for industrial environments.
- Private 5G networks within factory or warehouse environments can achieve the required latency, together with public 5G with edge compute relays can approach it.
- The networking companies most relevant to this layer include Cisco (industrial IoT switches with embedded security, WiFi-6 with sub-3 millisecond latency), Semtech (LoRa for long-range low-power telemetry in large facilities), and Silicon Labs (ultra-low-power Bluetooth and WiFi-6 transceivers for intra-robot subsystem communication).

Source: https://www.htx.gov.sg/whats-happening/all-news---events/all-news/2025/htxplains-teleoperation-in-robotics

The humanoid robot OEM competitive landscape divides into three clusters, Scale-Chasers, Cost-Disrupters, and Legacy Springboards, that have fundamentally different financial profiles, competitive strategies, and timelines.



The Scale-Chasers

Building toward mass production volumes and the economics that come with scale.

- Tesla Optimus is the Western archetype. Musk has stated targets of several thousand units in 2025, scaling to 50,000 to 100,000 in 2026, with a stretch goal of half a million by 2027 and a target average selling price of $20,000 to $30,000. Tesla's competitive advantages are vertical integration (it designs its own actuators, inference chips, and trains on data from its vehicle fleet), manufacturing scale from its EV business, and the data flywheel from Optimus units working in Tesla's own factories.
- Figure AI, with its BMW pilot deployment using the Helix VLA model, and Agility Robotics (Digit, deployed by Amazon in warehouse operations and GXO Logistics in fulfilment centres), also sit in this cluster.
- UBTech's Walker S1 rounds out the scale-chasers with Hong Kong-listed shares giving public market investors direct access.

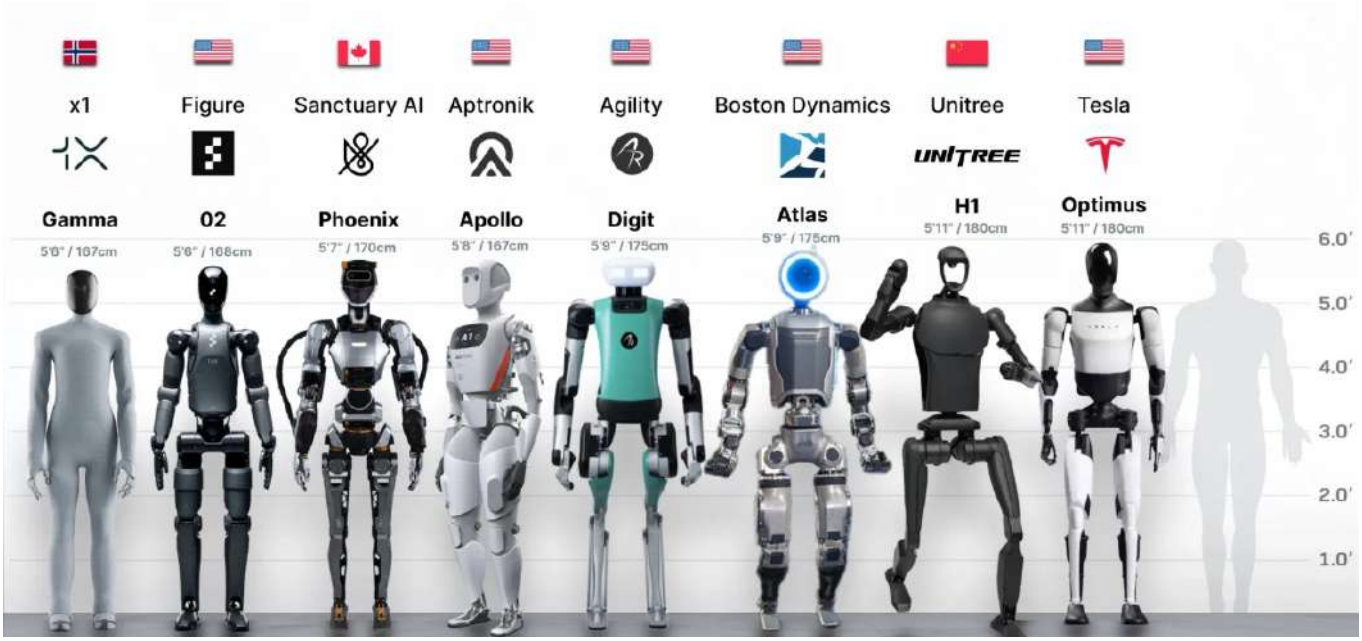
The Cost-Disrupters

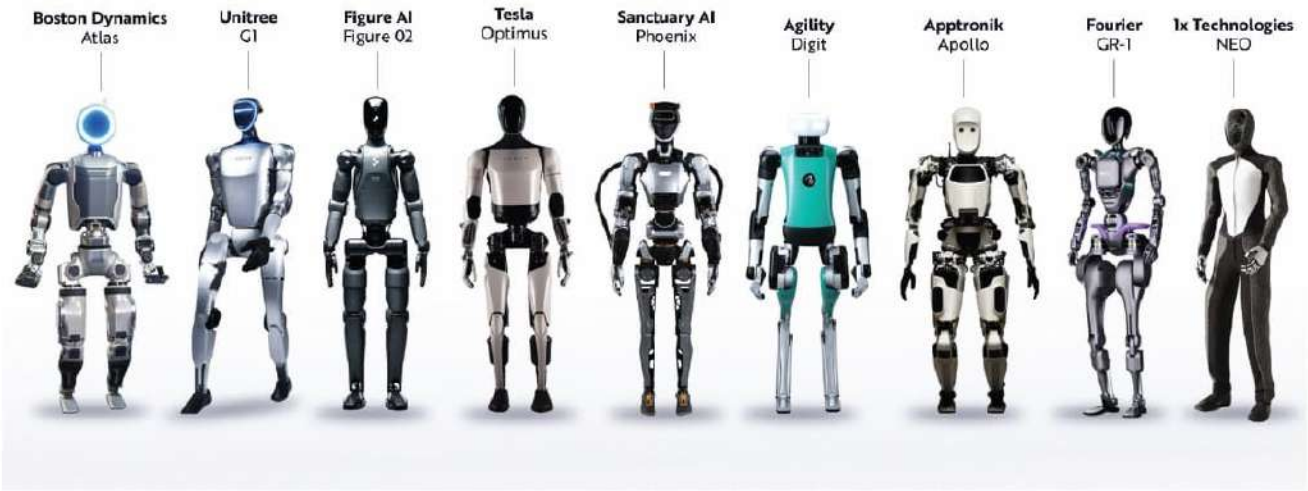
Compress average selling prices and challenge incumbents on economics rather than capability.

- Unitree is the archetype, its G1 at $16,000 and H1 at comparable pricing represent a cost structure that Western OEMs cannot currently approach.
- XPeng's Iron robot (developed with Sanctuary AI, visible in the image), Xiaomi's CyberBot, and EngineAI's PM01 (priced at $13,700, famously shown in videos with internet celebrity iShowSpeed performing backflips in mainland China) also compete here.

The cost-disrupter thesis is that reaching consumer price points first creates a self-reinforcing data flywheel and brand position analogous to what Tesla achieved in EVs.

Cool Robots





The Legacy Springboards

Have deep robotics IP and engineering heritage but longer development cycles.

Boston Dynamics, now owned by Hyundai following SoftBank's partial divestiture, brings the most credible locomotion IP in the industry, accumulated over decades of DARPA-funded research. The all-electric Atlas unveiled in 2024 is the first Boston Dynamics design built explicitly for commercial industrial deployment rather than research demonstration. Hyundai's manufacturing scale and automotive engineering expertise addresses the commercialization gap that limited Boston Dynamics under Alphabet and SoftBank ownership.

Apptronik's Apollo, deployed in Mercedes-Benz Group facilities, and Rainbow Robotics' HUBO (a South Korean company with Samsung investment) also sit in this cluster.

Honorable mentions include companies with indirect but significant exposure such as Toyota and Honda, which have decades of humanoid research (ASIMO was Honda's bipedal robot demonstrator from 2000) but have been outpaced by faster-moving startups.

Source: https://www.voronoiapp.com/technology/The-Current-Generation-of-Humanoid-Robots-2025-4656

The deployment will be gradual, moving up a ladder from structured industrial environments to unstructured service environments to consumer homes. Monitoring progress against five specific conditions tells investors how far along the deployment curve the industry is.

Condition 1: Autonomous Power Management

Today's robots still require planned charging cycles and suffer thermal throttling during extended operation. The tipping point for continuous industrial deployment requires three-plus hours of sustained operation at full task load without thermal shutdown, plus autonomous return to charging infrastructure without operator intervention.

High-C-rate battery cells, smarter battery management firmware, GaN-based charging electronics, and thermal optimization through linear actuator adoption are all contributing to this. Samsung SDI's 46-format cell trials and CATL's modular pack development are the key milestones to monitor.

Condition 2: Sub-15 Millisecond Teleoperation Latency

At 40 milliseconds round-trip, teleoperation is feasible. At 15 milliseconds, the line between remote pilot and local muscle memory effectively disappears. The operator's actions feel immediate and natural. Achieving this at scale requires private 5G networks with edge compute relays at the facility level. Every 10 milliseconds shaved off round-trip latency unlocks a measurable productivity improvement in the teleoperated stage of deployment.

Condition 3: Dextrous General Manipulation

The current killer application for humanoid robots is pick and place in structured environments where they take items from known positions and putting them in other known positions. The next capability threshold is genuine dexterous manipulation in unstructured environments: handling objects of unknown shape, weight, and fragility; opening containers; assembling components with sub-millimeter tolerances. Google's Gemini Robotics framework classifies the requirement as interactive (responds to voice and environment changes), dextrous (sub-millimeter hand-eye coordination), and general (works across different tasks without reprogramming).

Condition 4: Safety Standards and Regulatory Frameworks

There are currently no safety standards for a humanoid robot operating in a public environment. This is why early deployments are confined to private facilities behind access controls and light curtains. The regulatory framework will be reactive rather than proactive where standards will emerge from accumulated deployment data in controlled environments. The ISO 13849 functional safety standard and UL certification processes will extend to humanoid robots as deployments mature.

Condition 5: Sub-Two-Year Payback Economics

The ideal cost of a robot should be approximately two years of labor cost for a replaced worker. At current developed-market warehouse wages of approximately $40,000 to $50,000 per year, that implies a robot (or RaaS subscription) cost of $80,000 to $100,000 total over two years. Unitree's G1 at $16,000 hardware cost plus service and maintenance is approaching this threshold for lower-complexity tasks. The next leg of economics comes from amortizing tooling costs and standardizing BOMs as production volumes scale.